



Chapter V

Polymer Materials

General Introduction to Polymers

Basic Classification of Materials

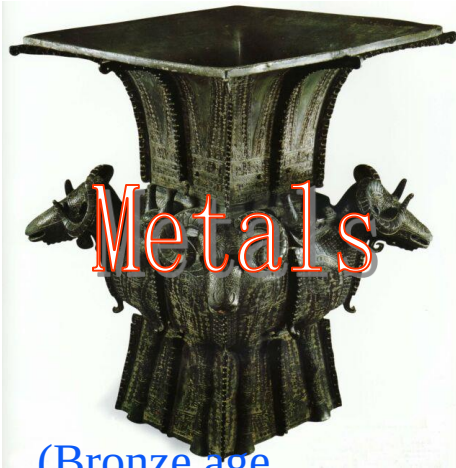
Classification by compositions

Ceramics



(New stone age,
5100 years ago)

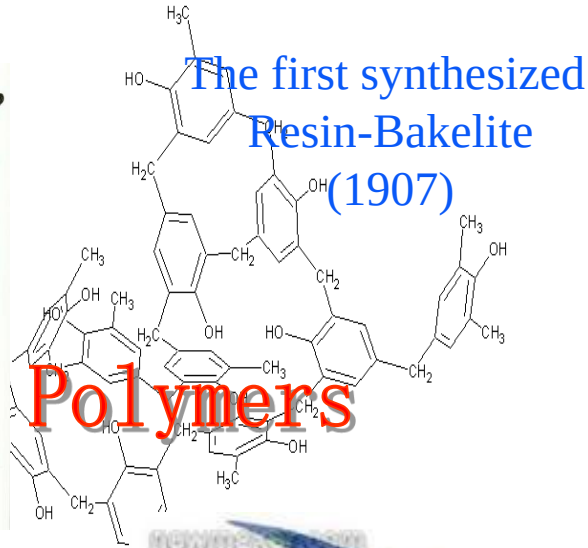
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Metals

(Bronze age,
Shang dynasty)

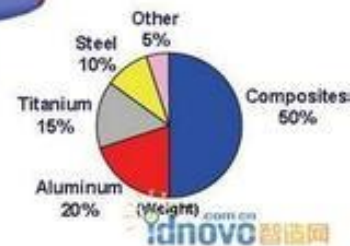
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The first synthesized Resin-Bakelite (1907)

Polymers

Composites

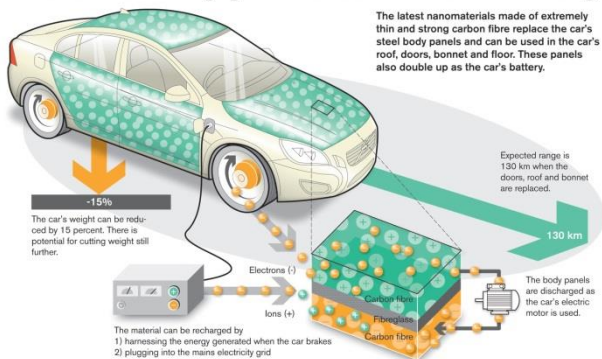




Polymers In Our Life



The car's body panels serve as a battery



Do you know the history of polymer science?



Cotton from plants is a natural polymer and is used to make clothing



Silk from silk worms is used to make cloth



Wood is a natural polymer used for paper



Natural latex from rubber tree is a polymer



Rubber can be used to make tires



or rubber bands!



Crustacean shells are made of chitin, a natural polymer



"Carbs" like spaghetti are natural polymers

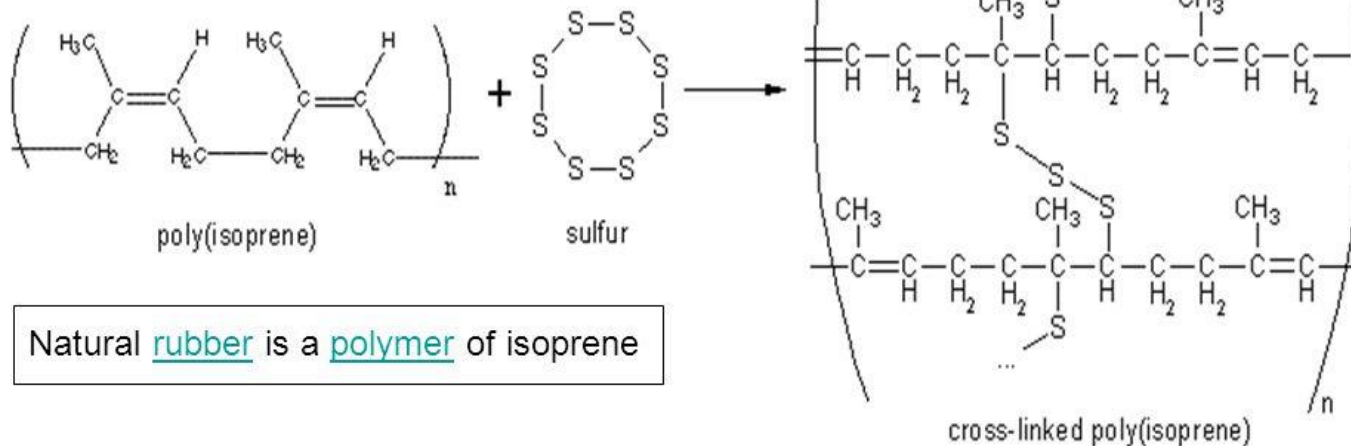


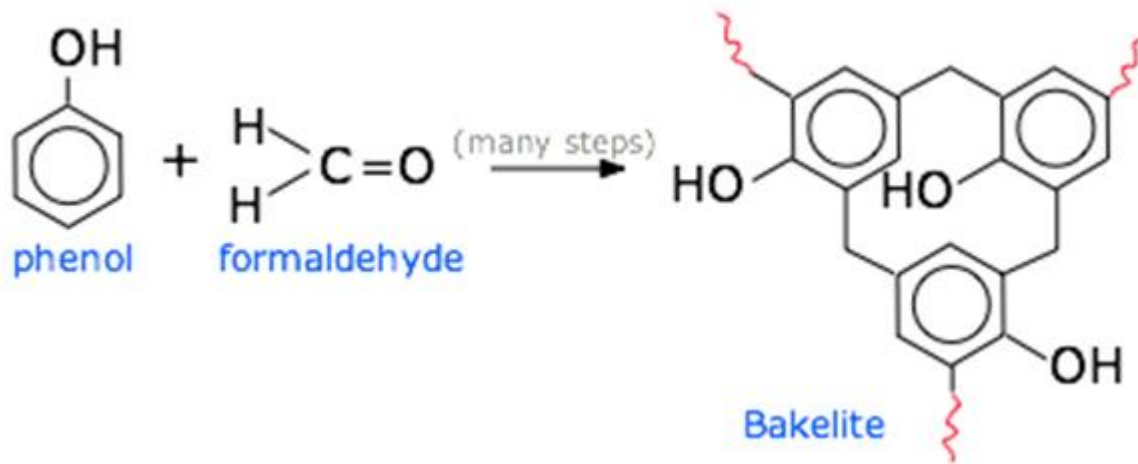
Proteins from eggs and other foods are also natural polymers





GOODYEAR

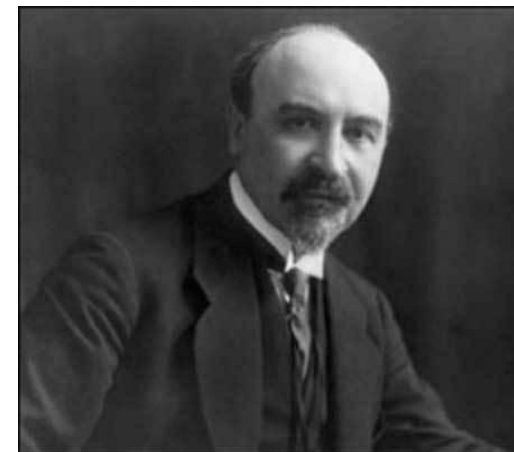




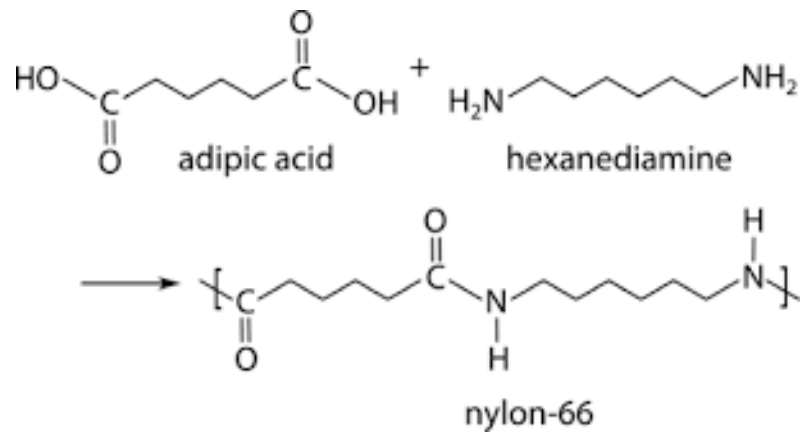
19世纪大师级化学家Baeyer



Earlier, bodies of telephone sets were made using **Bakelite**.



Baekeland, Leo Hendrik





Origins of Polymer Science



1920, Hermann Staudinger made his macromolecular hypothesis. (1953 Nobel prize)



Hermann Staudinger (1881 – 1965) was a German chemist who demonstrated the existence of macromolecules, which he characterized as polymers.

1929---Staudinger said, polymerization is a process that micromolecules combine each other by covalent bond then form long chain macromolecules. The solution property of macromolecular is the behavior of long chain macromolecular “molecular colloid”. It’s different to the micelle state combined with micromolecules.

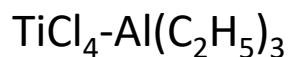
Then Staudinger founded the theory of macromolecule that contest the theory of colloid, he earned the Nobel Prize in chemistry 1953 for **“the discovery of chains formed macromolecule”**

Origins of Polymer Science



In 1950s' Ziegler and Natta invented the coordination polymerization and orientation polymerization. (1963 Nobel Prize)

Ziegler-Natta catalyst:

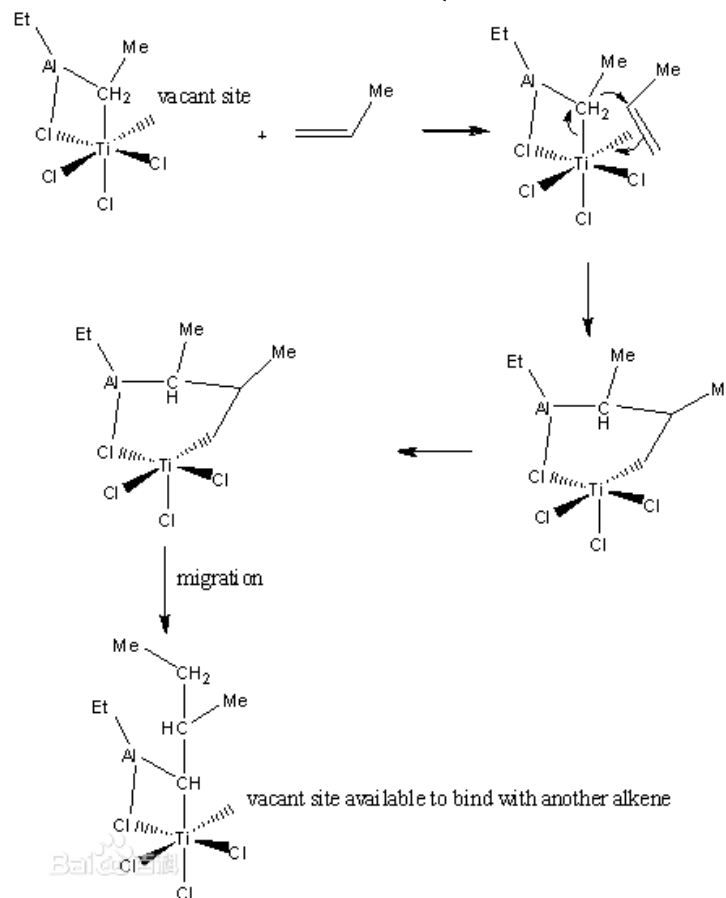


齐格勒, K.

Karl Ziegler
1989—1973



Giulio Natta,
1903—1979



“the discovery of organometallic mixed catalysts for the polymerization of olefins” .

Origins of Polymer Science



Paul Flory contributed to the polymerization mechanism, solution thermodynamics and a variety of polymer properties. (1974 Nobel prize)



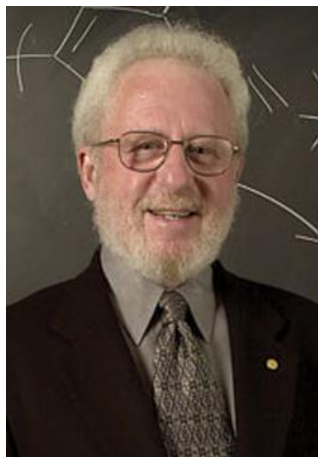
Paul John Flory (1910 – 1985) , American chemist

Earned the Nobel Prize in chemistry 1974 for "for his fundamental achievements, both theoretical and experimental, in the physical chemistry of macromolecules".

Origins of Polymer Science



Conjugated Polymers (2000 Nobel Prize)



Alan J. Heeger (艾伦·黑格)
University of California, Santa Barbara,
CA, USA



Alan MacDiarmid (麦克德尔米德)
University of Pennsylvania,
Philadelphia, PA, USA



Hideki Shirakawa (白川英树)
University of Tsukuba, Tokyo,
Japan

The Nobel Prize in Chemistry 2000 was awarded jointly to Alan J. Heeger, Alan G. MacDiarmid and Hideki Shirakawa

"for the discovery and development of conductive polymers".



5.1 Definition

Polymer – is a large molecule consisting of a number of repeating units with molecular weight typically several thousand or higher

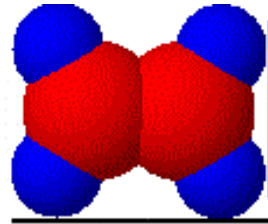
Monomer – is the smaller molecule(s) that are used to prepare a polymer

Oligomer – is a molecule consisting of reaction of several repeat units of a monomer but not large enough to be consider a polymer (dimer , trimer, tetramer, . . .)

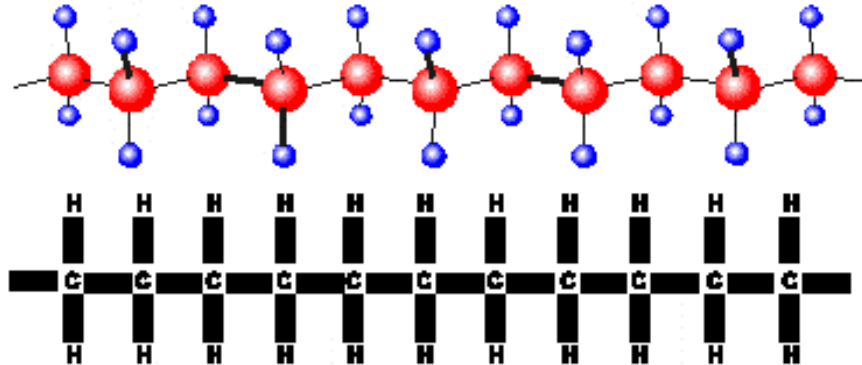
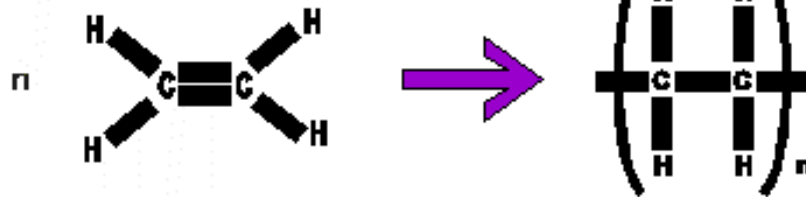
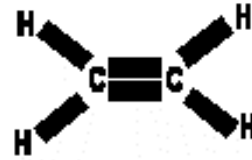
Degree of polymerization - number of repeating units



A monomer and a polymer



a monomer ethene



a polymer

poly(ethene)



5.2 Types of polymers and polymerizations

5.2.1 Types of polymers

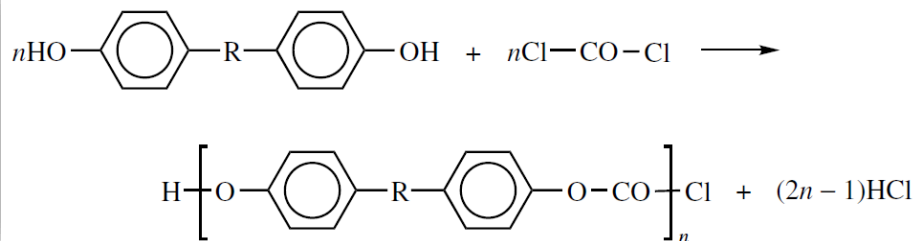
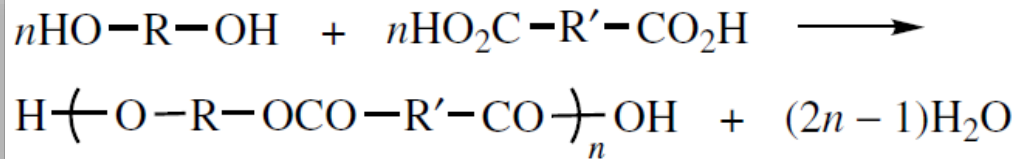
1. Carothers Classification

Classified by Carothers, 1929

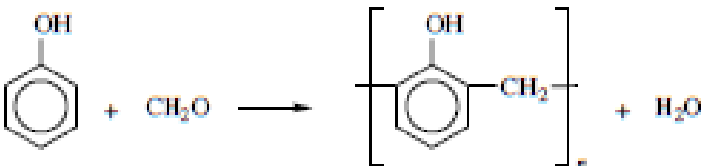
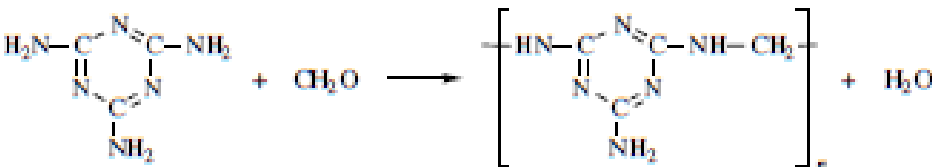
- Condensation polymers
- Addition polymers

Condensation polymers

The polymers are formed from polyfunctional monomers by the various condensation reactions of organic chemistry with the elimination of some small molecule such as water.





Type	Characteristic Linkage	Polymerization Reaction
Polyamide	$-\text{NH}-\text{CO}-$	$\text{H}_2\text{N}-\text{R}-\text{NH}_2 + \text{HO}_2\text{C}-\text{R}'-\text{CO}_2\text{H} \longrightarrow \text{H} \left(\text{NH}-\text{R}-\text{NHCO}-\text{R}'-\text{CO} \right)_n \text{OH} + \text{H}_2\text{O}$ $\text{H}_2\text{N}-\text{R}-\text{NH}_2 + \text{ClCO}-\text{R}'-\text{COCl} \longrightarrow \text{H} \left(\text{NH}-\text{R}-\text{NHCO}-\text{R}'-\text{CO} \right)_n \text{Cl} + \text{HCl}$ $\text{H}_2\text{N}-\text{R}-\text{CO}_2\text{H} \longrightarrow \text{H} \left(\text{NH}-\text{R}-\text{CO} \right)_n \text{OH} + \text{H}_2\text{O}$
Protein, wool, silk	$-\text{NH}-\text{CO}-$	Naturally occurring polypeptide polymers; degradable to mixtures of different amino acids. $\text{H} \left(\text{NH}-\text{R}-\text{CONH}-\text{R}'-\text{CO} \right)_n \text{OH} + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{N}-\text{R}-\text{CO}_2\text{H} + \text{H}_2\text{N}-\text{R}'-\text{CO}_2\text{H}$
Polyester	$-\text{CO}-\text{O}-$	$\text{HO}-\text{R}-\text{OH} + \text{HO}_2\text{C}-\text{R}'-\text{CO}_2\text{H} \longrightarrow \text{H} \left(\text{O}-\text{R}-\text{OCO}-\text{R}'-\text{CO} \right)_n \text{OH} + \text{H}_2\text{O}$ $\text{HO}-\text{R}-\text{OH} + \text{R}''\text{O}_2\text{C}-\text{R}'-\text{CO}_2\text{R}'' \longrightarrow \text{H} \left(\text{O}-\text{R}-\text{OCO}-\text{R}'-\text{CO} \right)_n \text{OH} + \text{R}''\text{OH}$ $\text{HO}-\text{R}-\text{CO}_2\text{H} \longrightarrow \text{H} \left(\text{O}-\text{R}-\text{CO} \right)_n \text{OH} + \text{H}_2\text{O}$
Polyurethane	$-\text{O}-\text{CO}-\text{NH}-$	$\text{HO}-\text{R}-\text{OH} + \text{OCN}-\text{R}'-\text{NCO} \longrightarrow \left(\text{O}-\text{R}-\text{OCO}-\text{NH}-\text{R}'-\text{NH}-\text{CO} \right)_n$
Polysiloxane	$-\text{Si}-\text{O}-$	$\text{Cl}-\text{SiR}_2-\text{Cl} \xrightarrow[\text{-HCl}]{\text{H}_2\text{O}} \text{HO}-\text{SiR}_2-\text{OH} \longrightarrow \text{H} \left(\text{O}-\text{SiR}_2 \right)_n \text{OH} + \text{H}_2\text{O}$
Phenol-formaldehyde	$-\text{Ar}-\text{CH}_2-$	
Urea-formaldehyde	$-\text{NH}-\text{CH}_2-$	$\text{H}_2\text{N}-\text{CO}-\text{NH}_2 + \text{CH}_2\text{O} \longrightarrow \left(\text{HN}-\text{CO}-\text{NH}-\text{CH}_2 \right)_n + \text{H}_2\text{O}$
Melamine-formaldehyde	$-\text{NH}-\text{CH}_2-$	
Polysulfide	$-\text{S}_n-$	$\text{Cl}-\text{R}-\text{Cl} + \text{Na}_2\text{S}_n \longrightarrow \left(\text{S}_n-\text{R} \right)_n + \text{NaCl}$
Polyacetal	$-\text{O}-\underset{\text{R}}{\text{CH}}-\text{O}-$	$\text{R}-\text{CHO} + \text{HO}-\text{R}'-\text{OH} \longrightarrow \left(\text{O}-\text{R}'-\text{OCHR} \right)_n + \text{H}_2\text{O}$

Addition polymers

Formed from monomers without the loss of a small molecule. Unlike condensation polymers, the repeating unit of an addition polymer has the same composition as the monomer.

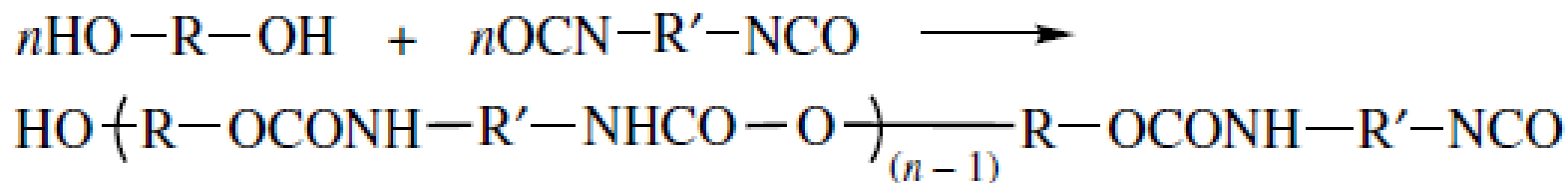
The major addition polymers are those formed by polymerization of monomers containing the carbon–carbon double bond.



The development of polymer science with the study of new polymerization processes and polymers showed that the original classification by Carothers was not entirely adequate and left much to be desired.

For example:

Polyurethanes (聚亚胺酯):



Using Carothers' original classification, one would classify the polyurethanes as addition polymers, since the polymer has the same elemental composition as the sum of the monomers. However, the polyurethanes are **structurally** much more similar to the condensation polymers than to the addition polymers.

2. Classification from the structural considerations

Condensation polymer:

1. Synthesis involved elimination of small molecules; or
2. Contains functional groups as part of polymer chain; or
3. Repeating unit lacks certain atoms in the monomer which it can be degraded.

Addition polymers:

Not fulfill any of the three requirements.



Structure consideration: condensation polymers with repeating units are joined by functional groups.



R: aliphatic or aromatic groups

Z: functional units —OCO— , —NHCO— , —S— , —O— , —OCOO— , $\text{—SO}_2\text{—}$

5.2.2 Types of polymerizations - mechanism

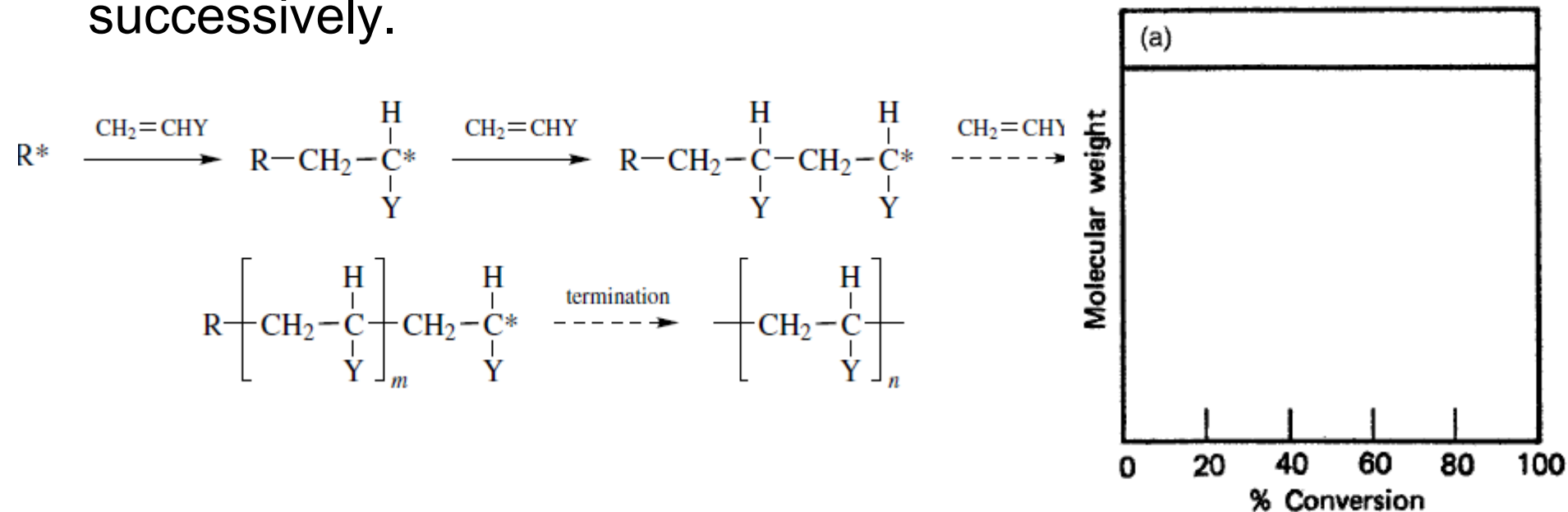
Mechanism of how polymer molecules are built up. (Flory)

Step & chain polymerization:

- Identities of the species that can react with each other.
- Molecular size with the extent of conversion.

a. Chain polymerization

- Initiator involved to generate reactive center.
- Chain growth by monomers reacting with the reactive center successively.



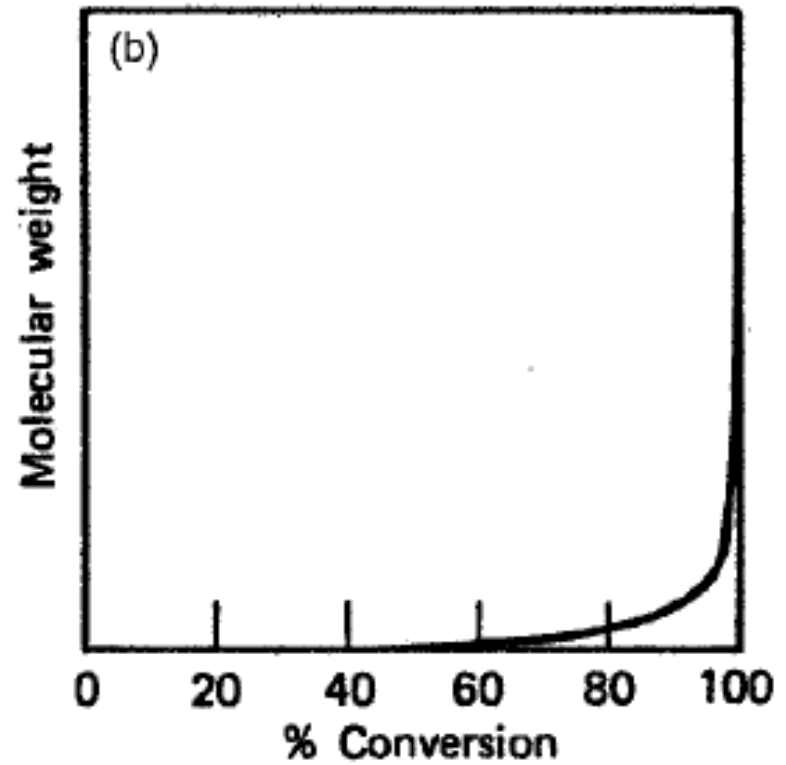
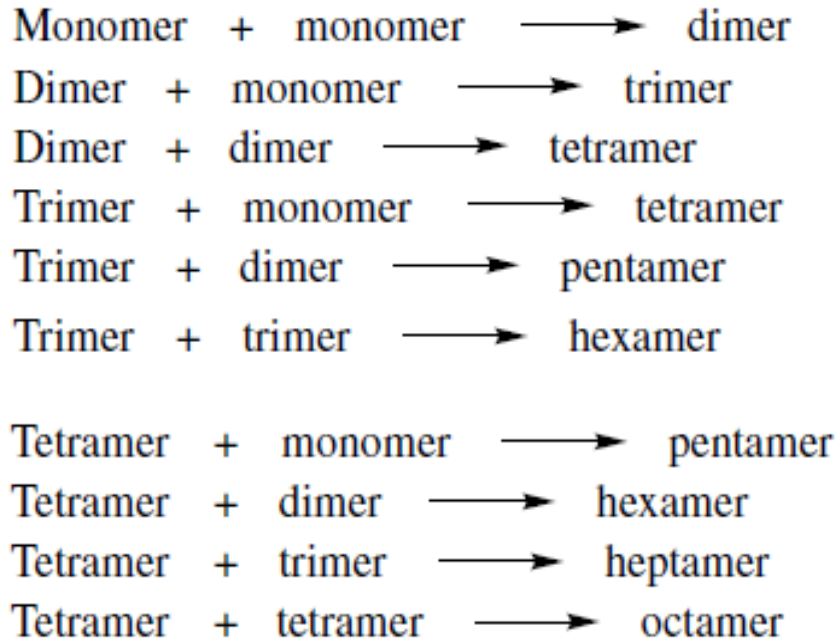
Reaction mixture: monomer, high molecular weight polymer, initiators.

Initiation step is much slower than propagation step, certain termination step.



b. Step polymerization

Reaction occurs between any of the different sized species present in the reaction systems.

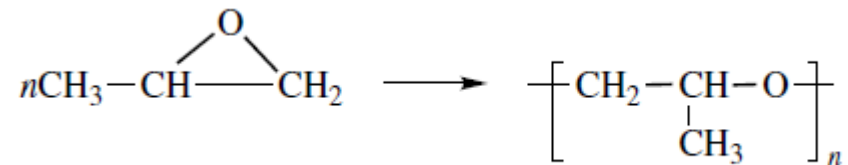
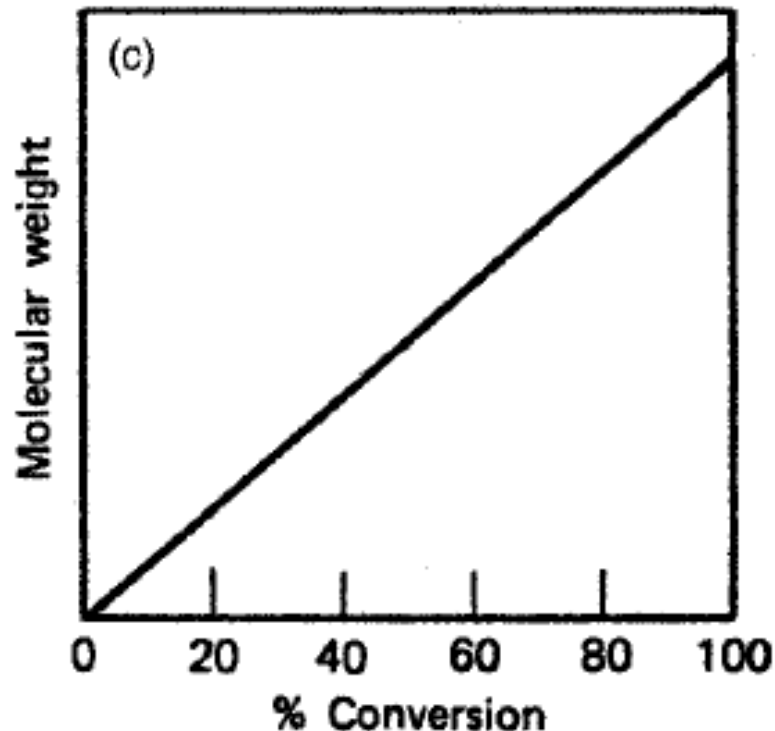




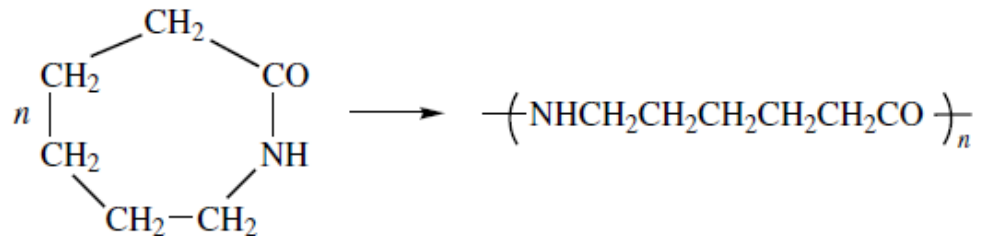
c. Nonterminating chain polymerization

Fast initiation, no termination step, living growth.

The ring-opening polymerizations of cyclic monomers such as propylene oxide:



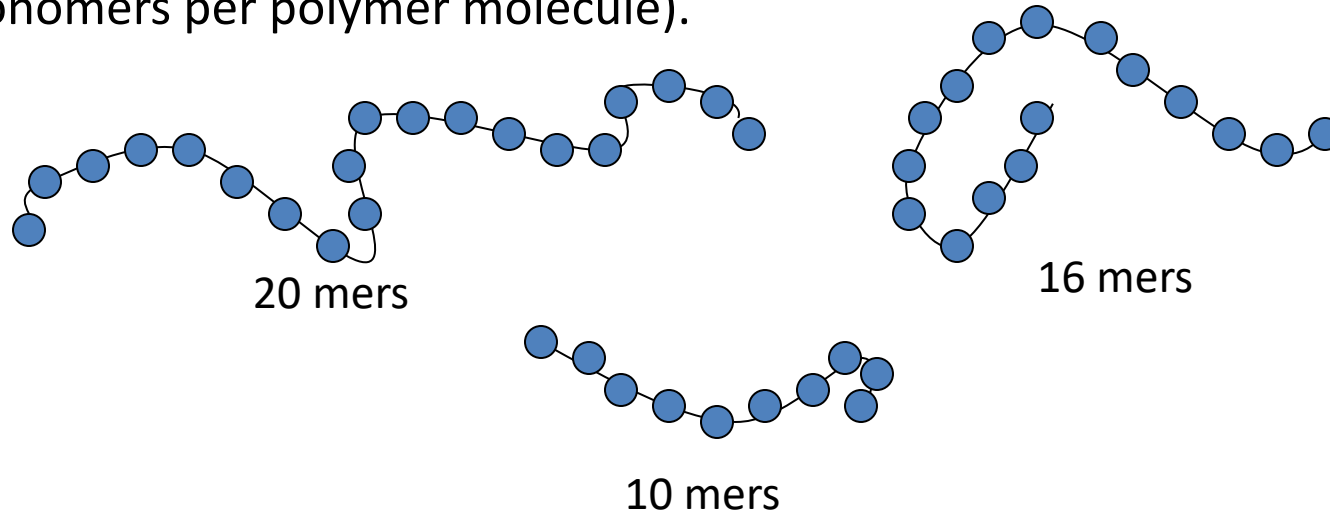
or ϵ -caprolactam (ϵ -己内酰胺):





5.3 Molecular weight

Not only are there different structures (molecular arrangements) but there can also be a distribution of molecular weights (i.e. number of monomers per polymer molecule).



$$\text{Average molecular weight} = \frac{20 + 16 + 10}{3} M_{\text{monomer}} = 15.3 M_{\text{monomer}}$$

This is what is called number average molecular weight.



Number-average molecular weight:

- $\overline{M}_n$ is determined by experimental methods that count the number of polymer molecules in a sample of the polymer.
- The methods for measuring $\overline{M}_n$ are those that measure the colligative properties of solutions.
vapor pressure lowering (vapor pressure osmometry), freezing point depression (cryoscopy), boiling point elevation (ebulliometry), and osmotic pressure (membrane osmometry).
- The colligative properties are the same for small and large molecules when comparing solutions at the same molal (or mole fraction) concentration.
For example, a 1-molal solution of a polymer of molecular weight 10^5 has the same vapor pressure, freezing point, boiling point, and osmotic pressure as a 1-molal solution of a polymer of molecular weight 10^3 or a 1-molal solution of a small molecule such as hexane.



➤ Number-average molecular weight:

$$\overline{M}_n = \frac{w}{\sum N_x} = \frac{\sum N_x M_x}{\sum N_x} = \frac{\sum N_x M_x}{N_1 + N_2 + \dots + N_x} = \frac{\sum N_x M_x}{N}$$

$$\overline{M}_n = \sum \frac{N_x}{N} M_x = \sum \underline{N_x} M_x$$

$\overline{M}_n$: number-average molecular weight;

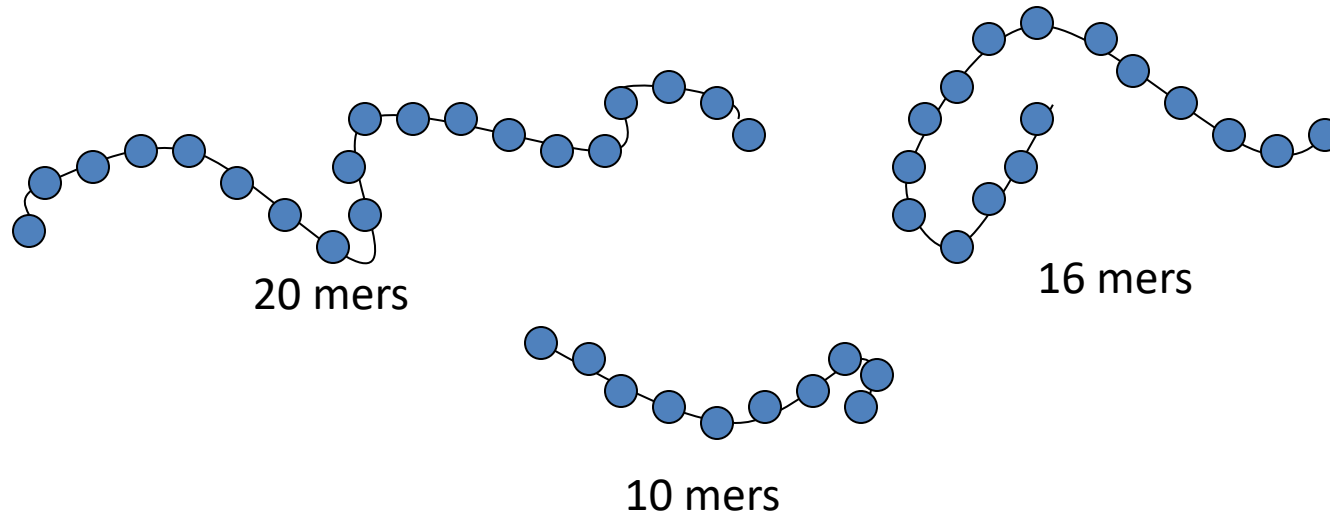
w: total mass of the polymer;

N_x : the number of moles of molecules having a molecular weight of M_x

$\underline{N_x}$: the mole fraction (or the number-fraction) of molecules of size M_x



Not only are there different structures (molecular arrangements) but there can also be a distribution of molecular weights (i.e. number of monomers per polymer molecule).



$$\text{Average molecular weight} = \left(20 \times \frac{20}{46} + 16 \times \frac{16}{46} + 10 \times \frac{10}{46} \right) M_{\text{monomer}} = 16.4 M_{\text{monomer}}$$

This is what is called weight average molecular weight.



➤ **Weight-average molecular weight:**

$$\overline{M}_w = \sum w_x M_x \quad w_x = \frac{c_x}{c}$$

$$\overline{M}_w = \frac{\sum c_x M_x}{c}$$

w_x : the weight fraction of the molecule whose weight is M_x ;

c_x : the weight of M_x molecules;

c : the total weight of all the polymer molecules.

$$c_x = N_x M_x$$

$$c = \sum c_x = \sum N_x M_x$$



$$\overline{M}_w = \frac{\sum c_x M_x}{c} = \frac{\sum N_x M_x^2}{\sum N_x M_x}$$



Weight-average molecular weight:

- Light scattering by polymer solutions, unlike colligative properties, is greater for larger-sized molecules than for smaller-sized molecules.
- Since the amount of light scattered by a polymer solution increases with molecular weight, this method becomes more accurate for higher polymer molecular weights.

There is no upper limit to the molecular weight that can be accurately measured except the limit imposed by insolubility of the polymer. The lower limit of M_w by the light scattering method is close to 5000–10,000. Below this molecular weight, the amount of scattered light is too small to measure accurately.



➤ Viscosity-average molecular weight:

$$\overline{M}_v = \left[\sum w_x M_x^a \right]^{1/a} = \left[\frac{\sum N_x M_x^{a+1}}{\sum N_x M_x} \right]^{1/a}$$

$$a = 0.5 - 0.9$$

- a is a constant.
- The viscosity- and weight-average molecular weights are equal when a is unity.
- M_v is less than M_w for most polymers, since a is usually in the range 0.5–0.9.
- However, M_v is much closer to M_w than M_n , usually within 20% of M_w .
- The value of a is dependent on the hydrodynamic volume of the polymer, the effective volume of the solvated polymer molecule in solution, and varies with polymer, solvent, and temperature.



Question: A polymer had been mixed with 10mol with a molecular weight of 10^4 , and 5mol with a molecular weight of 10^5 , calculate their average molecular weight (the a is 0.6).

$$\overline{M}_n = \sum \frac{N_x}{N} M_x = \sum \frac{N_x}{N} M_x$$

$$\overline{M}_w = \frac{\sum c_x M_x}{c} = \frac{\sum N_x M_x^2}{\sum N_x M_x}$$

$$\overline{M}_v = \left[\sum w_x M_x^a \right]^{1/a} = \left[\frac{\sum N_x M_x^{a+1}}{\sum N_x M_x} \right]^{1/a}$$



Answer:

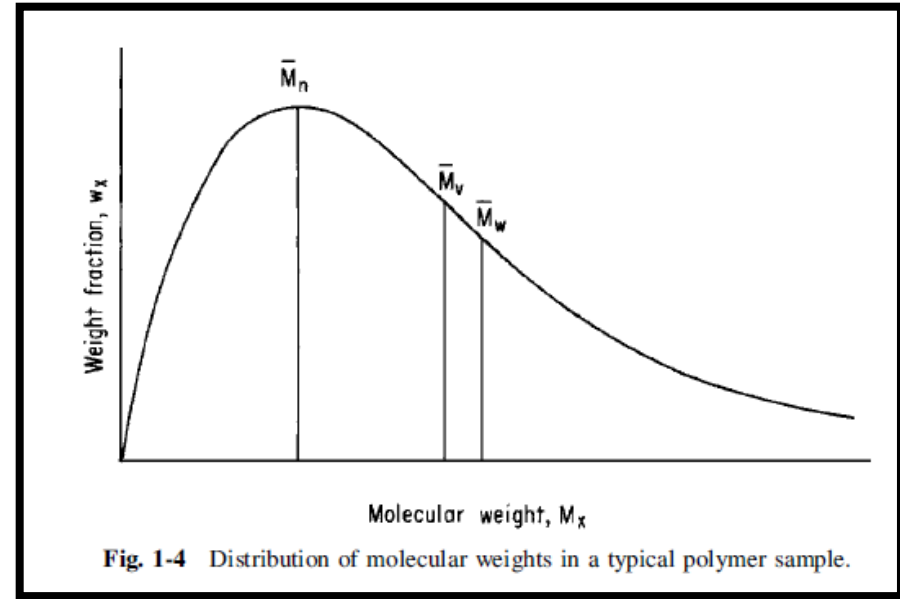
$$\overline{M}_n = \frac{\sum NiMi}{\sum Ni} = \frac{10 \times 10^4 + 5 \times 10^5}{10 + 5} = 40000$$

$$\overline{M}_w = \frac{\sum NiMi^2}{\sum NiMi} = \frac{10 \times (10^4)^2 + 5 \times (10^5)^2}{10 \times 10^4 + 5 \times 10^5} = 85000$$

$$\overline{M}_v = \left(\frac{10 \times (10^4)^{0.6+1} + 5 \times (10^5)^{0.6+1}}{10 \times 10^4 + 5 \times 10^5} \right)^{1/0.6} \approx 80000$$

Polydispersity index

- M_v is commonly used as a close approximation of M_w , since the two are usually quite close (within 10–20%).
- Thus in most instances, one is concerned with the M_n and M_w of a polymer sample. M_n is biased toward the lower-molecular-weight fractions, while M_w is biased toward the higher-molecular weight fractions.



- The ratio of the two average molecular weights M_w/M_n depends on the breadth of the distribution curve and is often useful as a measure of the polydispersity in a polymer.



Polydispersity index (PDI)

$$PDI = M_w/M_n$$

$$\frac{M_w}{M_n} = 1 \quad \text{monodisperse}$$

Polymer sample consisting of molecules all of which have the same chain lengths.

$$\frac{M_w}{M_n} > 1 \quad \text{polydisperse}$$

Polymer sample consisting of molecules with the variety of chain lengths.



- The weight-average molecular weight is a much better indicator of the properties to be expected in a polymer. the utility of M_n resides primarily in its use to obtain an indication of polydispersity in a sample by measuring the ratio M_w/M_n .
- Since samples with the same average molecular weight may possess different molecular weight distributions, information regarding the distribution allows the proper choice of a polymer for optimum performance.



Discussion:

1. What is the relationship among polymer, monomer, repeating unit, degree of polymerization and molecular weight?
3. How to understand the molecular weight of polymers?
4. Practice: calculates the number-average and weight-average molecular weight

Practice: calculates the number-average and weight-average molecular weight

Components	Weight Fractions	Average relative molecular weight
A	0.1	12000
B	0.19	21000
C	0.24	35000
D	0.18	49000
E	0.11	75000
F	0.08	102000
G	0.06	122000
H	0.04	146000

Average molecular weight

Assumption: the total mass of the polymer is 100 g

M_x	w_x	$w_x \times M_x$	c_x / g	$N_x = c_x / M_x$	$N_x / \sum N_x$	$N_x \cdot M_x / \sum N_x$
12000	0.1					
21000	0.19					
35000	0.24					
49000	0.18					
75000	0.11					
102000	0.08					
122000	0.06					
146000	0.04					

$$\overline{M}_w = \sum w_x M_x$$

$$\overline{M}_n = \sum \frac{N_x}{N} M_x$$

Average molecular weight

Assumption: the total mass of the polymer is 100 g

M_x	w_x	$w_x \times M_x$	c_x / g	$N_x = c_x / M_x$	$N_x / \sum N_x$	$N_x \cdot M_x / \sum N_x$
12000	0.1	1200	10	0.00083	0.2694	3233
21000	0.19	3990	19	0.00090	0.2925	6143
35000	0.24	8400	24	0.00069	0.2217	7760
49000	0.18	8820	18	0.00037	0.1188	5820
75000	0.11	8250	11	0.00015	0.0474	3557
102000	0.08	8160	8	0.00008	0.0254	2587
122000	0.06	7320	6	0.00005	0.0159	1940
146000	0.04	5840	4	0.00003	0.0089	1293

Total
51980

$$\overline{M}_w = \sum w_x M_x$$

$$\overline{M}_n = \sum \frac{N_x}{N} M_x$$

Total 32333

Calculate number average molecular weight, weight average molecular weight and the polydispersity index (PDI) for a polymer sample that contains the following mixture.

	Mole fraction	Molecular weight
1	0.1	1000000
2	0.2	700000
3	0.4	500000
4	0.2	250000
5	0.1	100000

Assumption: the total N of the polymer is 1 mol

$$\overline{M}_n = \sum \underline{N}_x M_x$$

$$\overline{M}_w = \sum w_x M_x$$

M_x	$\underline{N}_x$	N_x	$\underline{N}_x \times M_x$	$c_x = N_x \times M_x$	$w_x = \frac{c_x}{C}$	$w_x M_x$
1000000	0.1	0.1	100000	100000	0.2	200000
700000	0.2	0.2	140000	140000	0.28	196000
500000	0.4	0.4	200000	200000	0.4	200000
250000	0.2	0.2	50000	50000	0.1	25000
100000	0.1	0.1	10000	10000	0.02	2000
Total 500000				Total 623000		

$$PDI = \frac{M_w}{M_n} \approx 1.246$$



5.4 Nomenclature of polymer

1. Nomenclature based on monomer source

- Polymers synthesized from a single monomer as in addition and ring-opening polymerizations. Such polymers are named by adding the name of the monomer onto the prefix “poly” without a space or hyphen.
- When the monomer has a substituted parent name or a multiworded name or an abnormally long name, parentheses are placed around its name following the prefix “poly.”

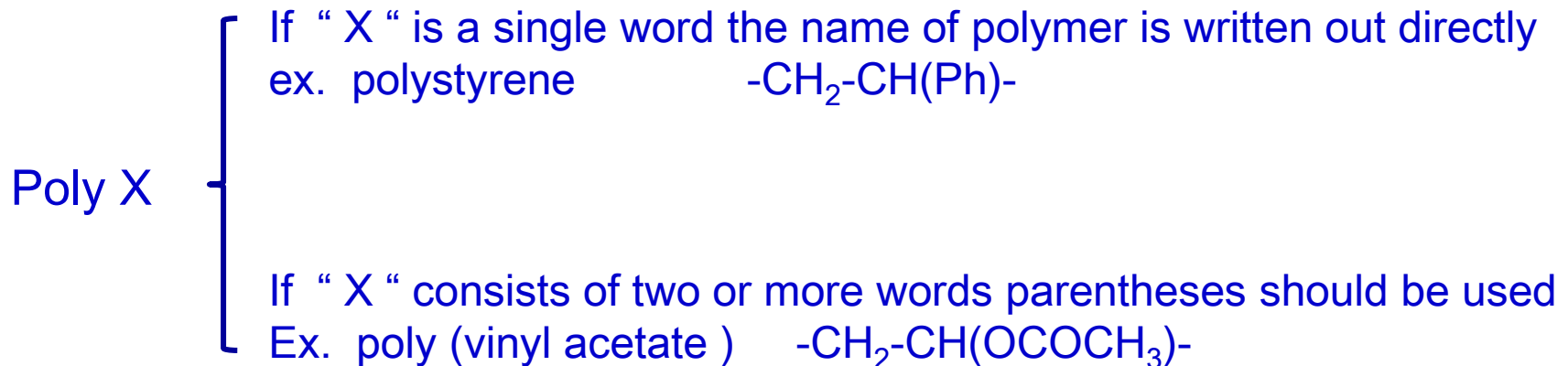




TABLE 1-2 Typical Addition Polymers

Polymer	Monomer	Repeating Unit
Polyethylene	$\text{CH}_2=\text{CH}_2$	$-\text{CH}_2-\text{CH}_2-$
Polyisobutylene	$\text{CH}_2=\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}$	$-\text{CH}_2-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-$
Polyacrylonitrile	$\text{CH}_2=\text{CH}-\text{CN}$	$-\text{CH}_2-\underset{\text{CN}}{\text{CH}}-$
Poly(vinyl chloride)	$\text{CH}_2=\text{CH}-\text{Cl}$	$-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-$
Polystyrene	$\text{CH}_2=\text{CH}-\phi$	$-\text{CH}_2-\underset{\phi}{\text{CH}}-$
Poly(methyl methacrylate)	$\text{CH}_2=\underset{\text{CO}_2\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}$	$-\text{CH}_2-\underset{\text{CO}_2\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-$
Poly(vinyl acetate)	$\text{CH}_2=\text{CH}-\text{OCOCH}_3$	$-\text{CH}_2-\underset{\text{OCOCH}_3}{\text{CH}}-$
Poly(vinylidene chloride)	$\text{CH}_2=\underset{\text{Cl}}{\overset{\text{Cl}}{\text{C}}}$	$-\text{CH}_2-\underset{\text{Cl}}{\overset{\text{Cl}}{\text{C}}}-$
Polytetrafluoroethylene	$\text{F} \quad \text{F}$ $\text{C}=\text{C}$ $\text{F} \quad \text{F}$	$-\underset{\text{F}}{\overset{\text{F}}{\text{C}}}-\underset{\text{F}}{\overset{\text{F}}{\text{C}}}-$
Polyisoprene (natural rubber)	$\text{CH}_2=\underset{\text{CH}_3}{\text{C}}-\text{CH}=\text{CH}_2$	$-\text{CH}_2-\underset{\text{CH}_3}{\text{C}}=\text{CH}-\text{CH}_2-$

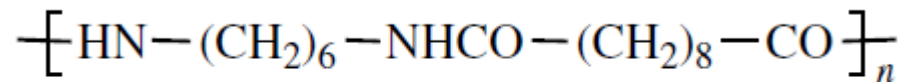


2. Nomenclature based on structure (Non-IUPAC)

International Union of Pure and Applied Chemistry (IUPAC)

- A number of the more common condensation polymers synthesized from two different monomers have been named by a semisystematic, structure-based nomenclature system other than the more recent IUPAC system.
- The name of the polymer is obtained by following the prefix poly without a space or hyphen with parentheses enclosing the name of the structural grouping attached to the parent compound.

Poly + “(the name of the structural group attaching to the parent compound)”



Poly(hexamethylene sebacamide)

the polymer from hexamethylene diamine and sebacic acid is considered as the substituted amide derivative of the compound sebacic acid.



3. IUPAC Structure-Based Nomenclature System

Selection of a preferred **constitutional repeating unit (CRU)**.

- CRU: smallest possible repeating unit; bivalent unit for single strand.
- Naming: poly (name of CRU)
- CRU name is according to the IUPAC rules for small molecules.

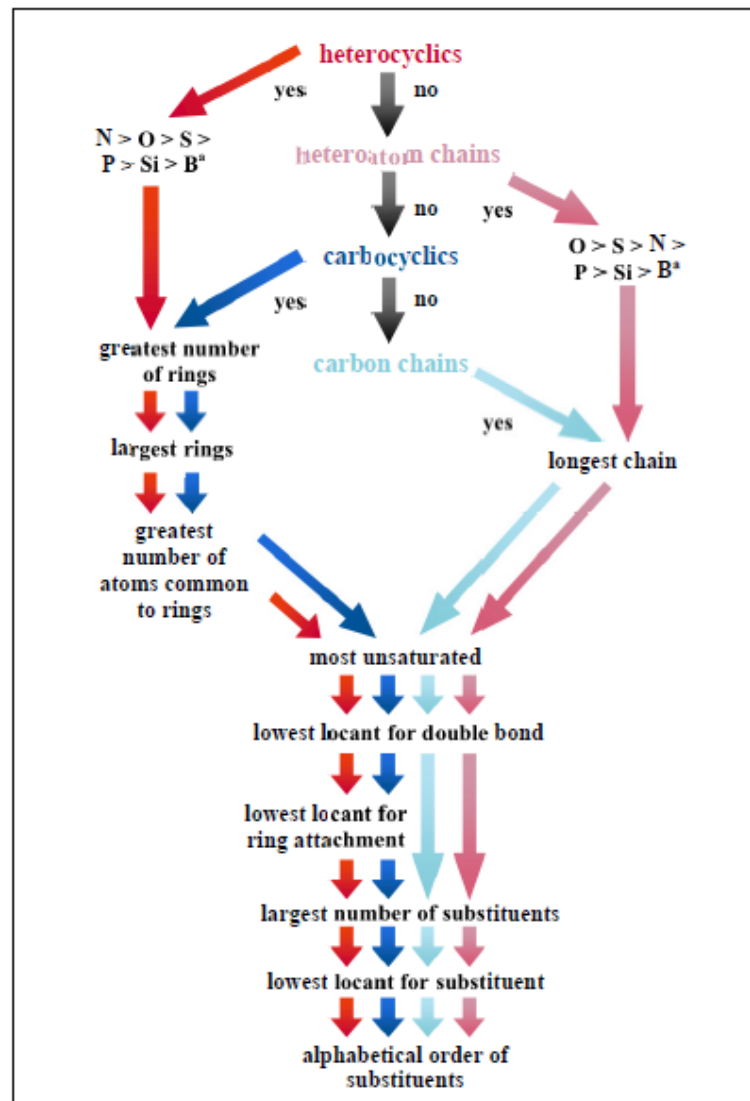
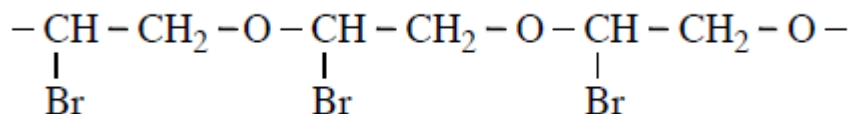


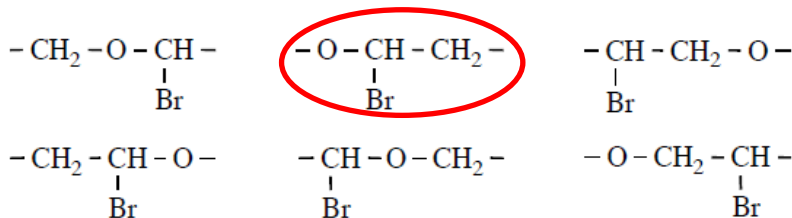
Figure 1 The order of subunit seniority. The senior subunit is at the top centre. Subunits of lower seniority are found by following the arrows. The type of subunit, be it a **heterocycle**, a **heteroatom chain**, a **carbocycle**, or a **carbon chain**, determines the colour of the arrow to follow. * Other heteroatoms may be placed in these orders as indicated by their positions in the periodic table.⁸



(i) a large enough part of the polymer chain is drawn to show the structural repetition, e.g.,



(ii) the smallest repeating portion is a CRU, so all such possibilities are identified. In this case:



The preferred CRU:

oxy(1-bromoethane-1,2-diyl)

The polymer:

poly[oxy(1-bromoethane- 1,2-diyl)]

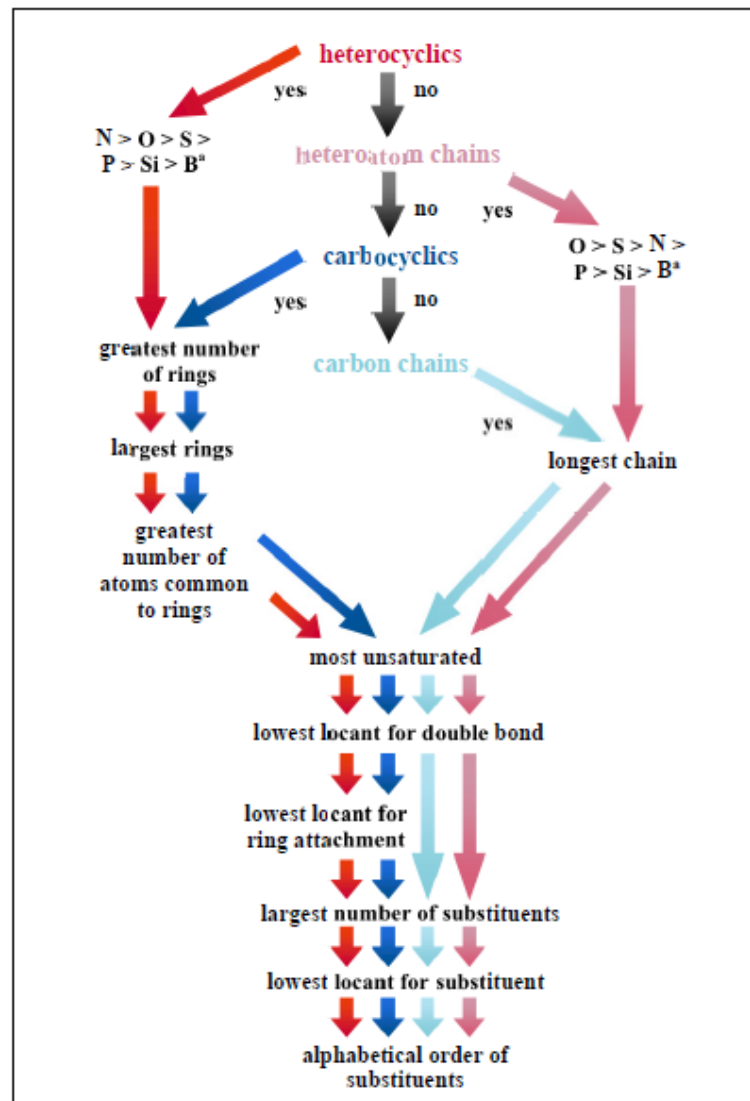


Figure 1 The order of subunit seniority. The senior subunit is at the top centre. Subunits of lower seniority are found by following the arrows. The type of subunit, be it a **heterocycle**, a **heteroatom chain**, a **carbocycle**, or a **carbon chain**, determines the colour of the arrow to follow. * Other heteroatoms may be placed in these orders as indicated by their positions in the periodic table.⁸



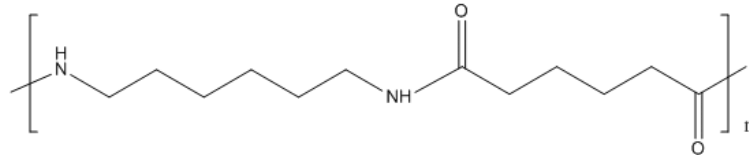
3. Nomenclature based on trade name

Nylon – unsubstituted, nonbranched polyamides.

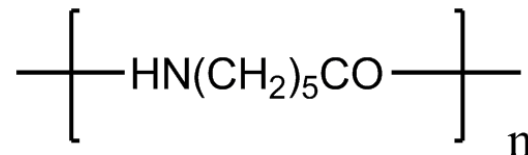
- Nylon – m, n
m: the number of methylene groups in the diamine portion
n: the number of carbon atoms in the diacyl portion.

Nylon-6,6: poly(hexamethylene adipamide)

Nylon-6,10: poly(hexamethylene sebacamide)



- Nylon – n
From single monomers with n, the number of carbon atoms in the repeating units.





4. Copolymers Units named by sources/structures, then combine.

Table 1 – Qualifiers for copolymers.⁷

Copolymer	Qualifier	Example
unspecified	<i>co</i> (C)	poly(styrene- <i>co</i> -isoprene)
statistical	<i>stat</i> (C)	poly[isoprene- <i>stat</i> -(methyl methacrylate)]
random	<i>ran</i> (C)	poly[(methyl methacrylate)- <i>ran</i> -(butyl acrylate)]
alternating	<i>alt</i> (C)	poly[styrene- <i>alt</i> -(maleic anhydride)]
periodic	<i>per</i> (C)	poly[styrene- <i>per</i> -isoprene- <i>per</i> -(4-vinylpyridine)]
block	<i>block</i> (C)	poly(buta-1,3-diene)- <i>block</i> -poly(ethene- <i>co</i> -propene)
graft ^a	<i>graft</i> (C)	polystyrene- <i>graft</i> -poly(ethylene oxide)

^a The first name is that of the main chain.

Table 2 – Qualifiers for non-linear (co)polymers and polymer assemblies.⁵

(Co)polymer	Qualifier	Example
blend	<i>blend</i> (C)	poly(3-hexylthiophene)- <i>blend</i> -polystyrene
comb	<i>comb</i> (C)	polystyrene- <i>comb</i> -polyisoprene
complex	<i>compl</i> (C)	poly(2,3-dihydrothieno[3,4- <i>b</i>][1,4]dioxine)- <i>compl</i> -poly(vinylbenzenesulfonic acid) ^a
cyclic	<i>cyclo</i> (P)	<i>cyclo</i> -polystyrene- <i>graft</i> -polyethylene
branch	<i>branch</i> (P)	<i>branch</i> -poly[(1,4-divinylbenzene)- <i>stat</i> -styrene]
network	<i>net</i> (C or P)	<i>net</i> -poly(phenol- <i>co</i> -formaldehyde)
interpenetrating network	<i>ipn</i> (C)	(<i>net</i> -polystyrene)- <i>ipn</i> -[<i>net</i> -poly(methyl acrylate)]
semi-interpenetrating network	<i>sipn</i> (C)	(<i>net</i> -polystyrene)- <i>sipn</i> -polyisoprene
star	<i>star</i> (P)	<i>star</i> -polyisoprene

^a In accordance with IUPAC organic nomenclature, square brackets enclose locants that refer to the numbering of the components of the fused ring.

Statistical copolymer:

They can exhibit a short-range order with preferred sequences, and thus are different from a random mixing, but possess no order in the chemical composition over the long range.

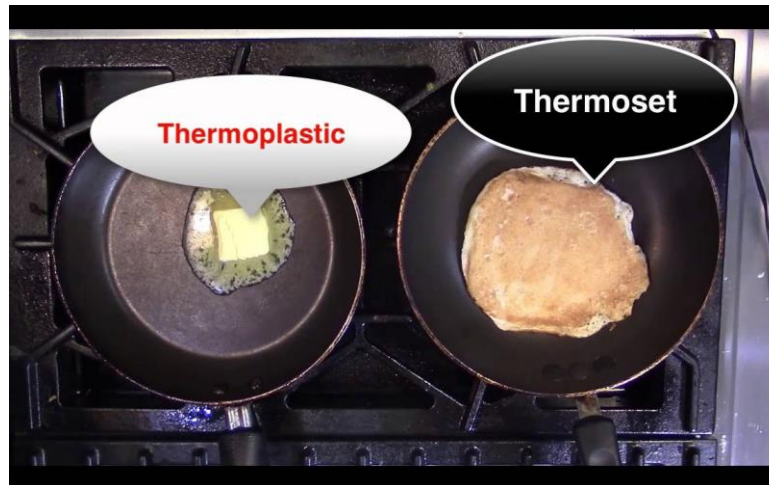


5.5 Classification

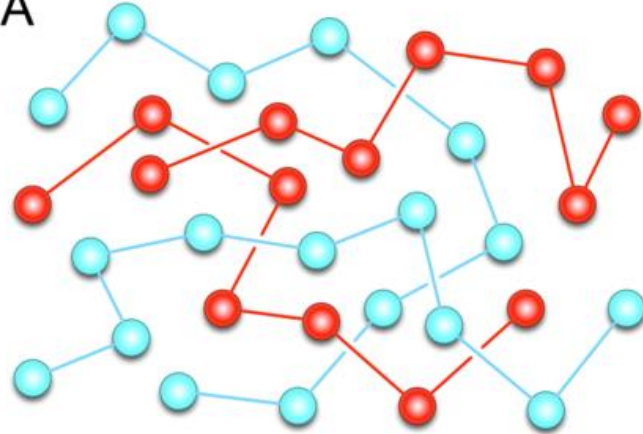
1. Classification by Thermal Behavior

Thermoplastics - materials become fluid and processible upon heating, allowing them to be transformed into desired shapes that are stabilized by cooling. Example: PE, PVC, PMMA, PS

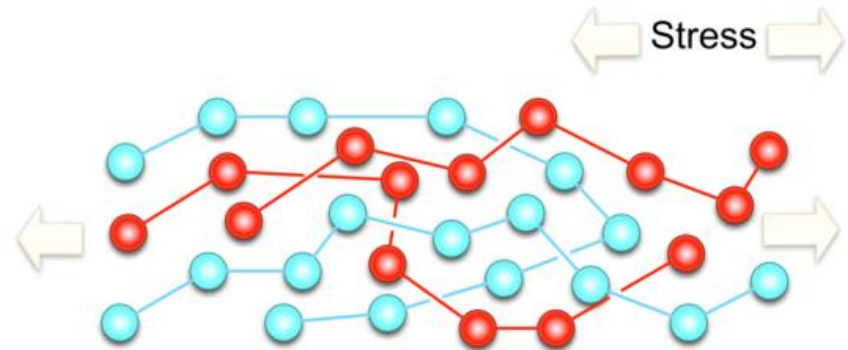
Thermosets - initial mixture of reactive, low molar mass compounds reacts upon heating in the mold to form an insoluble, infusible network. Example: polyester



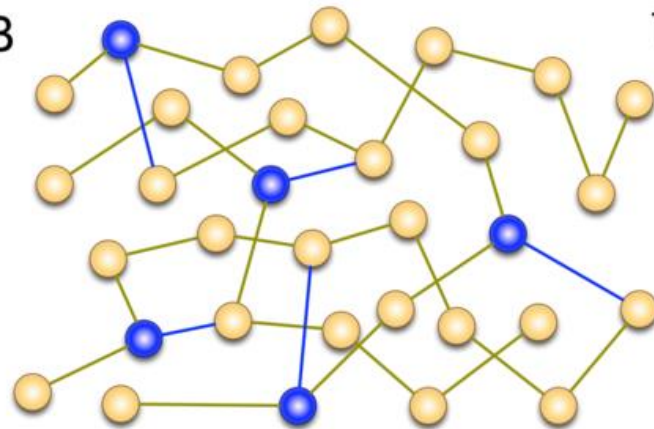
A



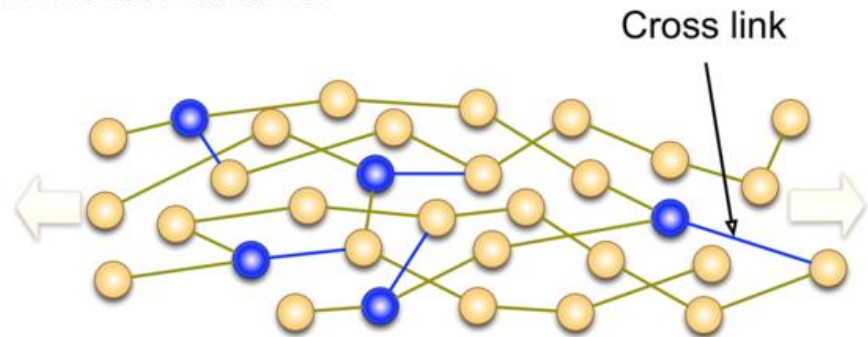
Thermoplastic Elastomer



B



Thermoset Elastomer





Thermoplastics

- The molecules have weak, straight-chain bonds between them that can be broken by heating.
- Thermoplastics are prepolymerized by the manufacturer, and don't require polymerization again during processing.
- They are elastic and flexible in nature.
- They dissolve in organic solvents.
- On heating, they will soften and ultimately melt.
- The melting point of thermoplastics is lower than their degradation point.
- Thermoplastics are in the form of solid pellets before use.
- In solid state, their structure consists of hard crystalline regions together with elastic amorphous regions.



Thermosets

- They have strong chemical bonds between molecules, including cross-linking, due to which they do not separate on heating.
- Thermosets have to undergo a two-stage polymerization during the processing stage.
- They are hard and brittle.
- They do not dissolve in organic solvents.
- On heating, thermoset polymers will char, not melt.
- Their degradation point is lower than their melting point.
- Thermosets are in the liquid state before processing.
- When solid, their structure consists of the thermosetting resin interspersed with reinforcing fiber.

Thermoplastics

Advantages

- High impact resistance (10 X thermosets)
- Can be recycled and reused
- Easy repair
- High chemical resistance
- Aesthetically better surface finish
- Uncured thermoplastics do not need refrigerated storage
- Environmentally friendly production (less emissions)
- They can be blended with other thermoplastic materials

Disadvantages

- Heat and pressure requirement for fiber reinforcement
- High cost
- Heat and UV sensitivity
- Polymer degradation after constant remolding and remolding
- Structural failure on high stress or long-term load application
- Leaching of hazardous chemicals in the long run



Thermosets

Advantages

- High structural rigidity (hard but brittle)
- Heat and chemical resistance
- Economical raw materials
- Easy fiber reinforcement of liquid raw materials
- High durability
- Ability of thick and thin wall formation
- Better wetting and adhesive properties

Disadvantages

- Recycling inability
- Refrigerated storage required by liquid raw materials
- Repair difficulty
- Good surface finish is tough to accomplish
- Lengthy, complex process stages
- Structural failure on high-force impact
- Emission during production process
- Inability to blend thermosets by welding



Examples of Thermoplastic

Types	Uses
Polyethene (PE) 聚乙烯	Wide range of inexpensive uses including supermarket bags, plastic bottles.
Polypropene (PP) 聚丙烯	Food containers, appliances, car fenders (bumpers).
Nylon (Polyamide) 聚酰胺	Synthetic fiber, substituted for silk (in silk stockings), tufted carpet, machine gear wheels
Polyvinylchloride 聚氯乙烯 (PVC)	Plumbing pipes and guttering, shower curtains, window frames, flooring.
PET 聚酯	Carbonated drinks bottles, jars, plastic film, microwavable packaging.



Examples of Thermosets

Types	Uses
Vulcanized rubber (硫化橡胶)	Tyres, gloves and tubing
Bakelit (胶木)	Electrical insulators and plastic wear
Polyester (聚酯)	Glass-reinforced plastics / fibreglass (GRP)
Epoxy (环氧树脂)	Adhesive and in fibre reinforced plastic such as glass reinforced plastic and graphite-reinforced plastic.
Acrylonitrile butadiene styrene (ABS) (丙烯腈丁二烯苯乙烯塑胶)	Electronic equipment cases (e.g., computer monitors, printers, keyboards).



2. Classification by Chain Structures

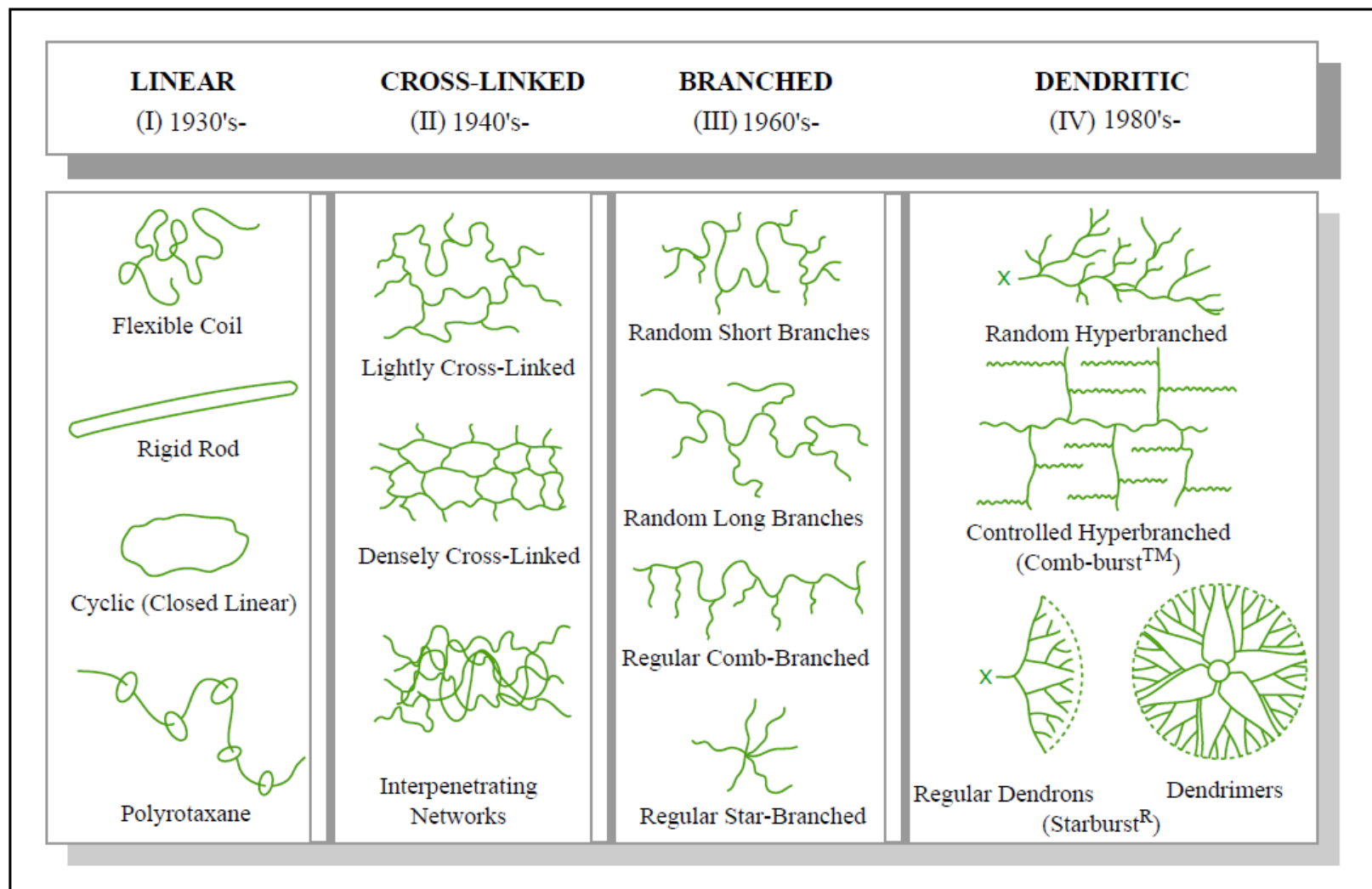


Figure by MIT OCW.



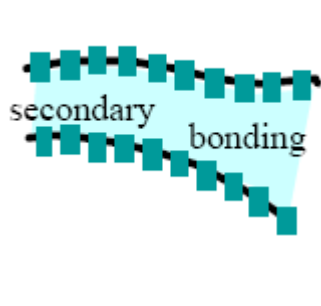
❑ **Linear chains** :a polymer consisting of a single continuous chain of repeat units

❑ **Branched chains** :a polymer that includes side chains of repeat units connecting onto the main chain of repeat units

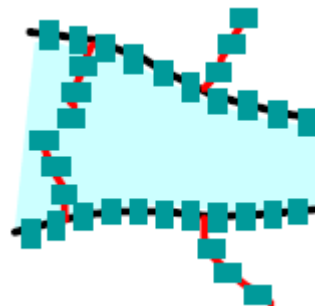
❑ **Hyper branched polymer** consist of a constitutional repeating unit including a branching groups

❑ **Cross linked polymer** :a polymer that includes interconnections between chains

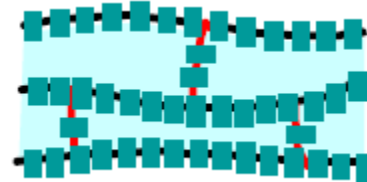
❑ **Net work polymer** :a cross linked polymer that includes numerous interconnections between chains



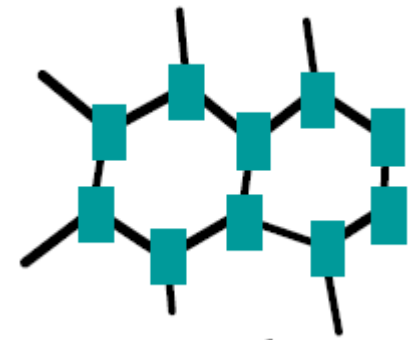
Linear



Branched



Cross-linked



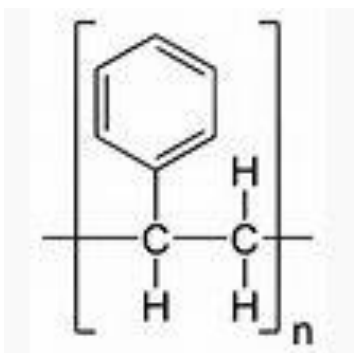
Network

Direction of increasing strength →

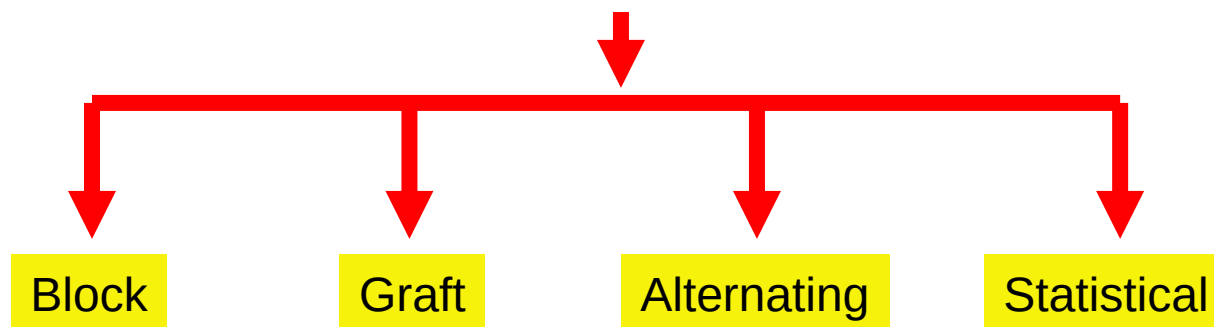


3. Classification by Monomer composition

Homopolymer



Copolymer



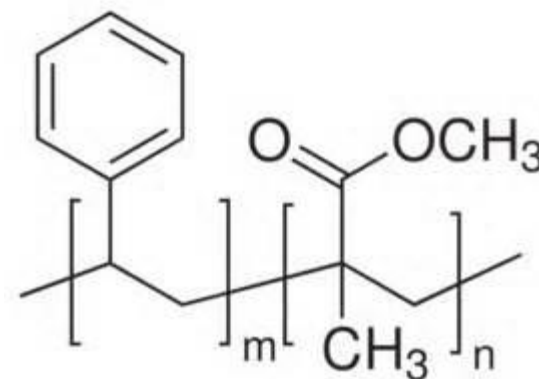
Homopolymer

Consist of only one type of constitutional repeating unit (A)

AAAAAAAAAAAAAAAA

copolymer

Consists of two or more constitutional repeating units (A.B)





□ Several classes of copolymer are possible

□ Statistical copolymer (Random)

ABAABABBBBAABAABB

two or more different repeating unit are distributed randomly

□ Alternating copolymer

ABABABABABABABAB

are made of alternating sequences of the different monomers

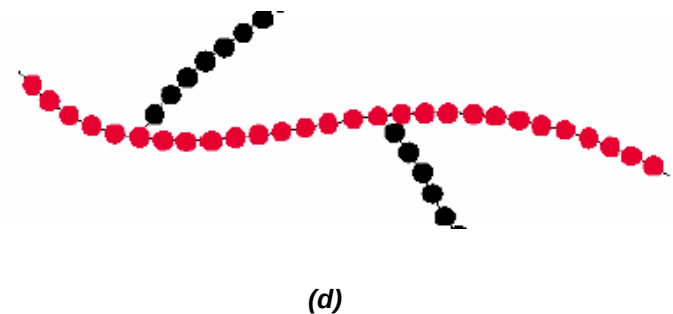
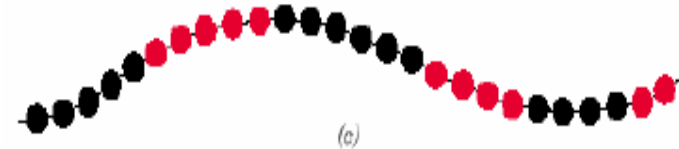
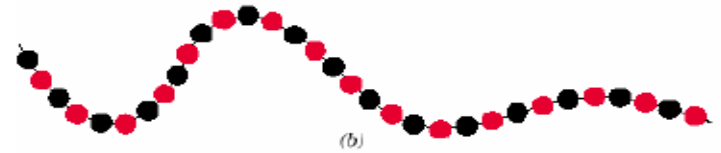
□ Block copolymer

AAAAAAAAABBBBBBBBBB

long sequences of a monomer are followed by long sequences of another monomer

□ Graft copolymer

Consist of a chain made from one type of monomers with branches of another type





3. Classification by Chain configuration and Conformation

Configuration/Conformation

- **Configuration**

- the arrangements of atoms on the polymer chain
- fixed relative positions of atoms

- **Conformation**

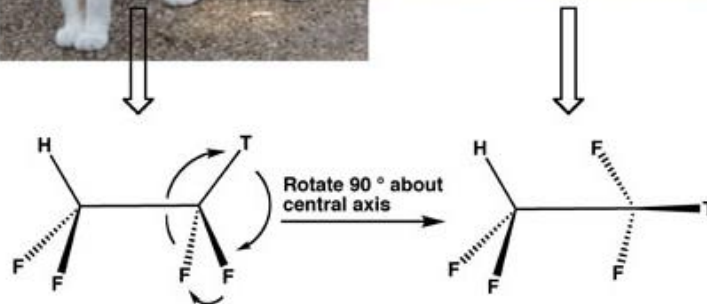
- the arrangement of the atoms in space
- regular linear chains can crystallise
- branched/bulky side-groups or irregular copolymers have longer repeat units and are thus less likely to crystallise but more likely to be amorphous



Different Configuration



Same cats - different conformations



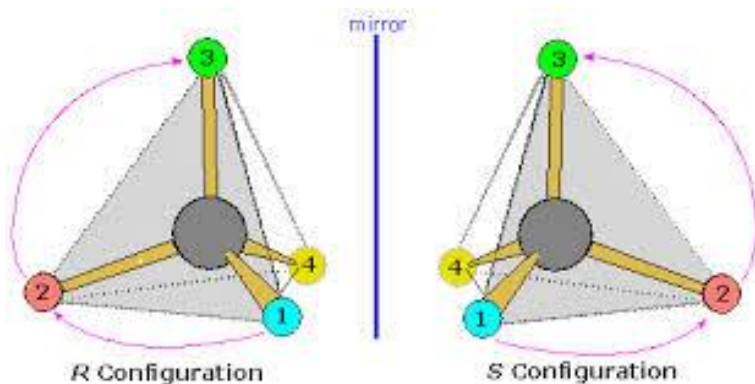
Different Conformation

Conformations can interconvert, but **configurations** can't be interconverted without breaking off body parts and moving them around.

➤ Configuration

Describes the spatial position of atoms within the molecule. The geometrical arrangement in polymers arising from the order of atoms determined by chemical bonds.

- The regularity and symmetry of the side-groups can affect strongly the properties of polymers.
- The configuration of a polymer cannot be altered unless chemical bonds are broken and reformed.
- asymmetric C-atom $\Leftrightarrow$ all substituents are different

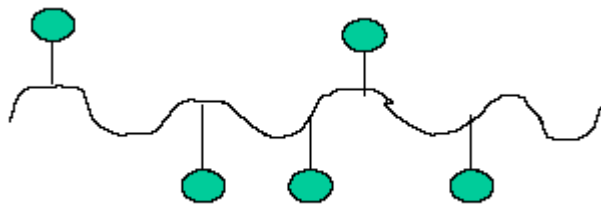




• Stereo-isomers---Tacticity

The orderliness of the succession of configurationally repeating units in the main chain of a polymer molecule (important for crystallization).

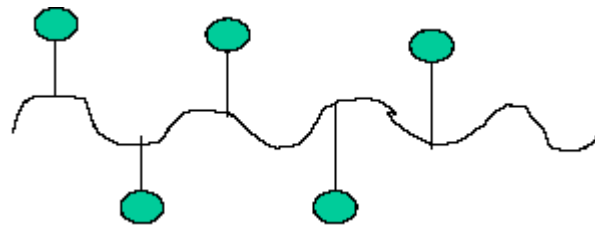
- if the radicals are linked in the same order, the configuration is called **isotactic**.
- in a stereoisomer in a **syndiotactic** configuration, the radical groups are at alternative sides in the chain
- in the **atactic** configuration, the radical groups are positioned at random



atactic

无规立构

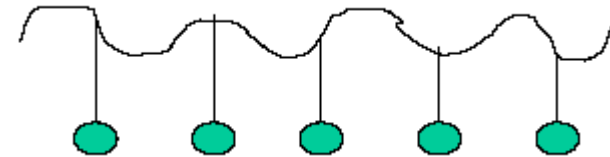
*Side groups on random
Sides of the backbone*



syndiotactic

间规立构

*Side groups on alternating
sides of the backbone*



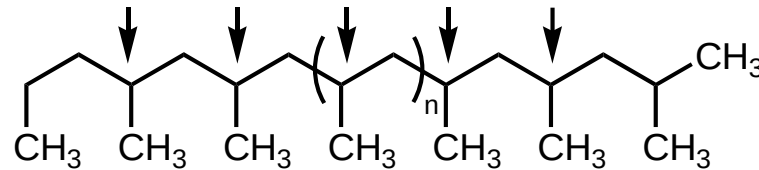
isotactic

全同立构

*Side groups on the same
side of the backbone*

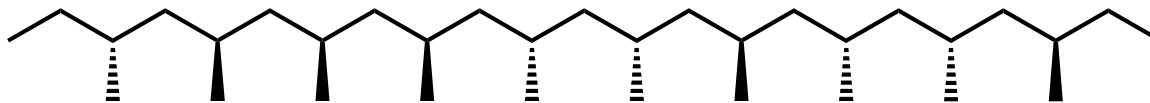


Polyolefins with side chains have stereocenters on every other carbon
聚烯烃 立体异构体



With so many stereocenters, the stereochemistry can be complex.
There are three main stereochemical classifications for polymers.

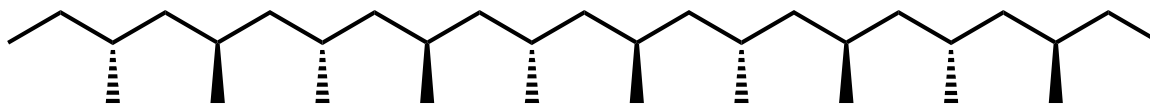
Atactic: random orientation



Isotactic: All stereocenters have same orientation



Syndiotactic: Alternating stereochemistry





- Tacticity affects physical properties
 - Atactic polymers will generally be amorphous, soft, flexible materials.
 - Isotactic and syndiotactic polymers will be more crystalline, thus harder and less flexible.
- Polypropylene (聚丙烯PP) is a good example
 - Atactic PP is a low melting, gooey(胶粘的)material.
 - Isoatactic PP is high melting (165°), crystalline, tough material that is industrially useful.
 - Syndiotactic PP has similar properties, but is very clear. It is harder to synthesize.



$\left(\text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} \right)_n$	Isotactic	Syndiotactic	Atactic
$T_m (^{\circ}\text{C})$	165	135	N/A
Density (g/mL)	0.92 – 0.94	0.89 – 0.91	0.85
Solubility (Hydrocarbons at Room Temp)	none	medium	high
Strength	high	medium	low

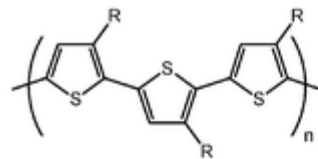


• Regio-isomers

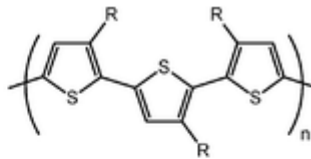
Asymmetry of 3-substituted thiophenes results in three possible couplings when two monomers are linked between the 2- and the 5-positions

- 2,5', or head–tail (HT), coupling
- 2,2', or head–head (HH), coupling
- 5,5', or tail–tail (TT), coupling

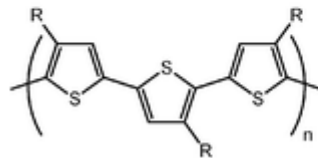
These three diads can be combined into four distinct triads:



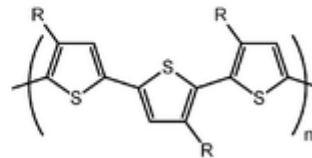
14a (HT-HT)



14b (HT-HH)



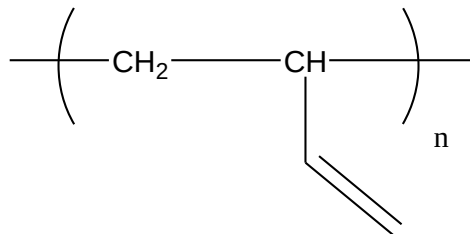
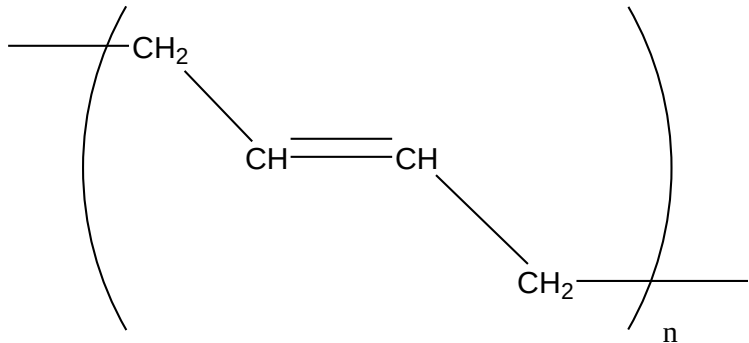
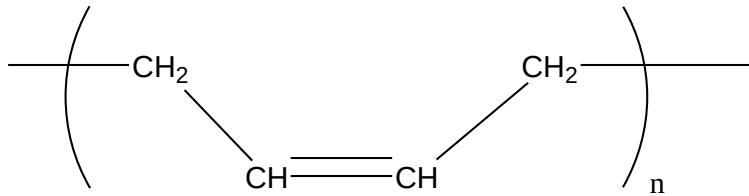
14c (TT-HT)



14d (TT-HH)



- Geometric isomers



- 1,3-butadiene

- cis-1,4-polybutadiene

- $T_g = -90\text{ }^{\circ}\text{C}$
- $T_m = \text{N/A}$ (amorphous)

- trans-1,4-polybutadiene

- $T_g = -63\text{ }^{\circ}\text{C}$
- $T_m = 100\text{ }^{\circ}\text{C}$

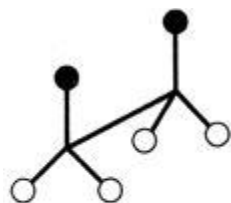
- 1,2-atactic-polybutadiene

- $T_g = 0\text{ }^{\circ}\text{C}$

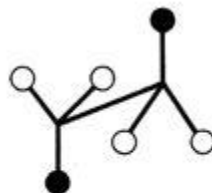


➤ Conformation

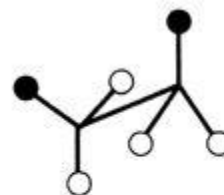
Defined by its sequence of bonds and torsion angles. The change in shape of a given molecule due to torsion about single (σ) bonds.



eclipsed conformation



anti conformation

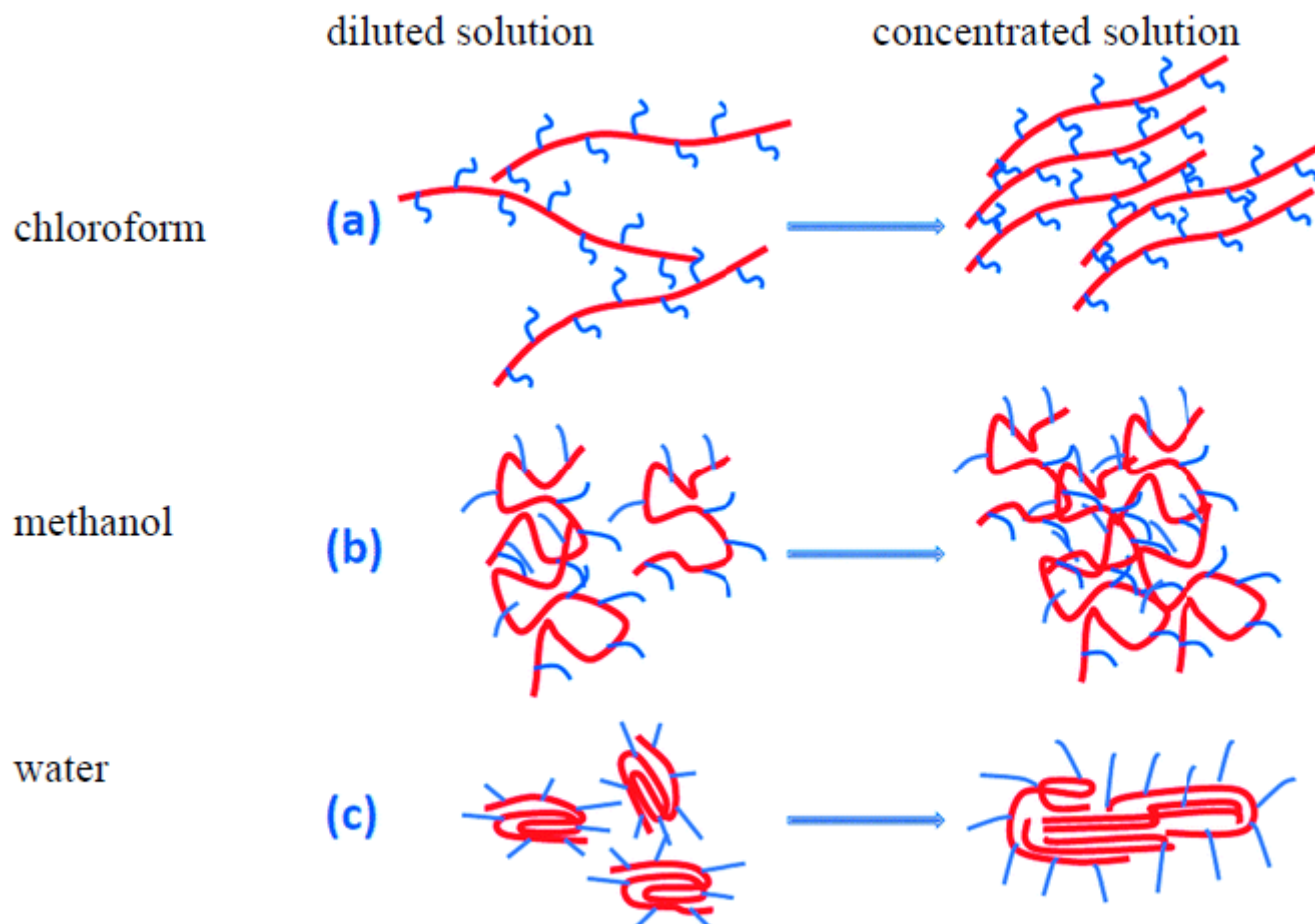


gauche conformation

● = methyl group



Example: solvent polarity effect on chain conformation of DEATG-PPV





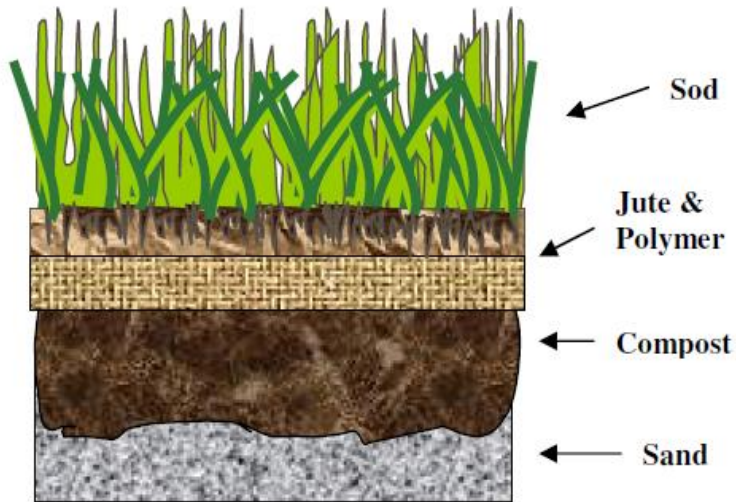
5.6 Characteristics of Polymers

General Characteristics :

1. Resistant to Chemicals (note not for all polymers).
2. Good Thermal stability.
3. Electrical Insulators (exception: conductive polymers).
4. Lightweight with varying degrees of strength.
5. Easily processable to produce thin fibers or intricate(复杂的)parts.



- Polymeric materials are used in and on soil to improve aeration, provide mulch, and promote plant growth and health.



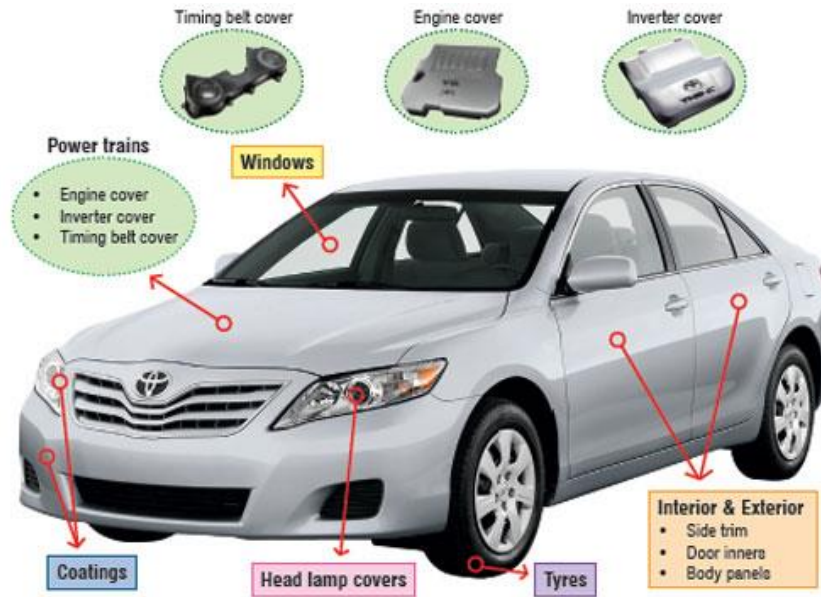


Medicine

- Many biomaterials, especially heart valve replacements and blood vessels, are made of polymers like Dacron, Teflon and polyurethane.

Consumer Science

- Plastic containers of all shapes and sizes are light weight and economically less expensive than the more traditional containers. Clothing, floor coverings, garbage disposal bags, and packaging are other polymer applications.



Industry

- Automobile parts, windshields for fighter planes, pipes, tanks, packing materials, insulation, wood substitutes, adhesives, matrix for composites, and elastomers are all polymer applications used in the industrial market.

Sports

- Playground equipment, various balls, golf clubs, swimming pools, and protective helmets are often produced from polymers.

5.7 Tradition Synthesis

5.7.1 Step Growth Polymerization

5.7.2 Chain Polymerization

- Free Radical Polymerization
- Ionic polymerization
- Coordination polymerization

5.8 Advanced Synthesis

- living polymerization
- NMP
- ATRP
- RAFT



5.7. 1 Step-Growth Polymerization

- I. Classification of polymerization reactions
- II. General types of step-growth polymerization
- III. Theoretical consideration

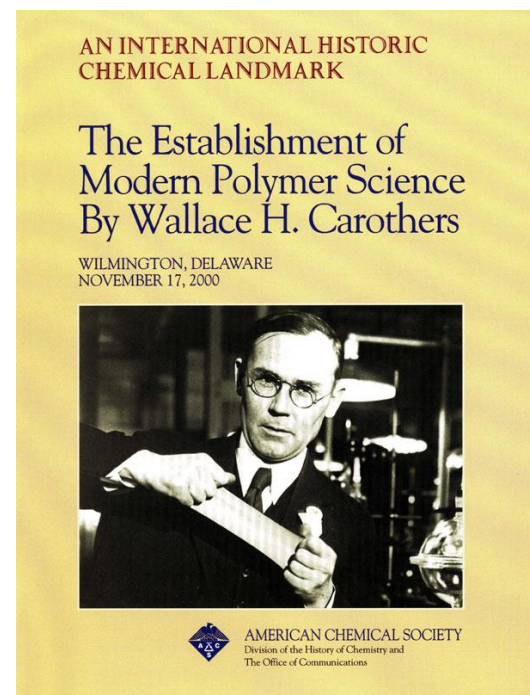


I. Classification of Polymerization Reactions

- Condensation Polymerization
- Addition Polymerization

---Introduced by Wallace Hume Carothers in 1929, according to the type of products

- a polymer only (addition)
- a polymer and a molecule with a low molecular weight (condensation)

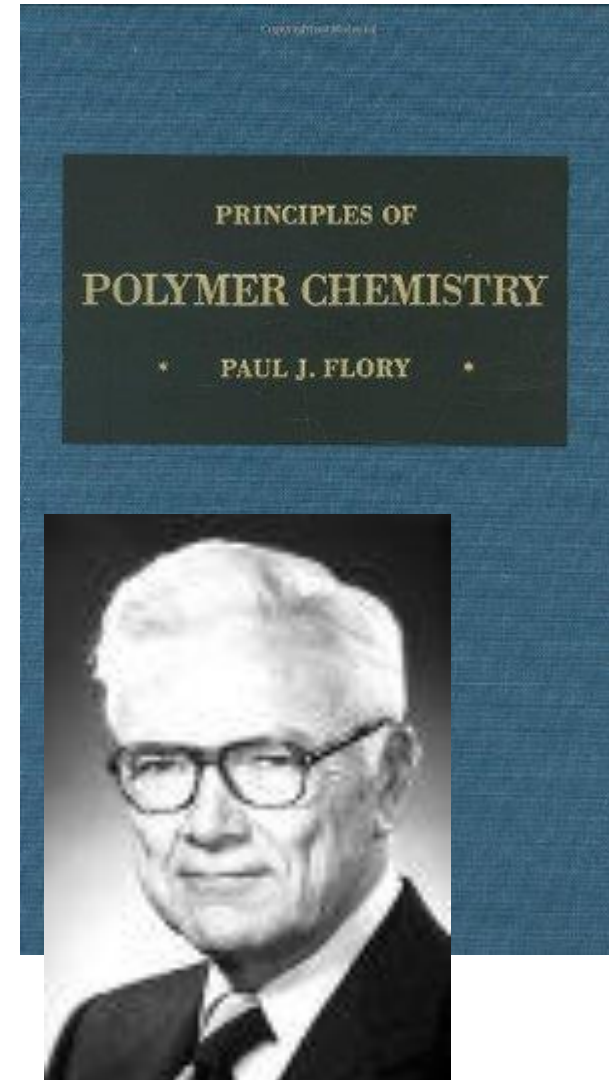


➤ Step-Growth Polymerization

➤ Chain Polymerization

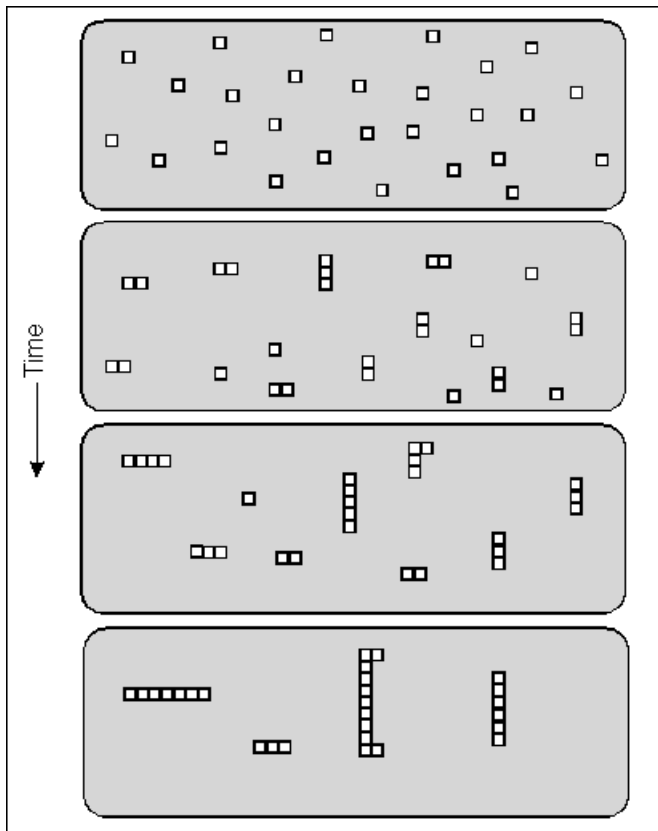
---Introduced by Paul John Flory in 1953,
according to the difference in reaction
mechanisms

- by functional groups (step-growth polymerization)
- by free-radical or ion (chain-growth polymerization)



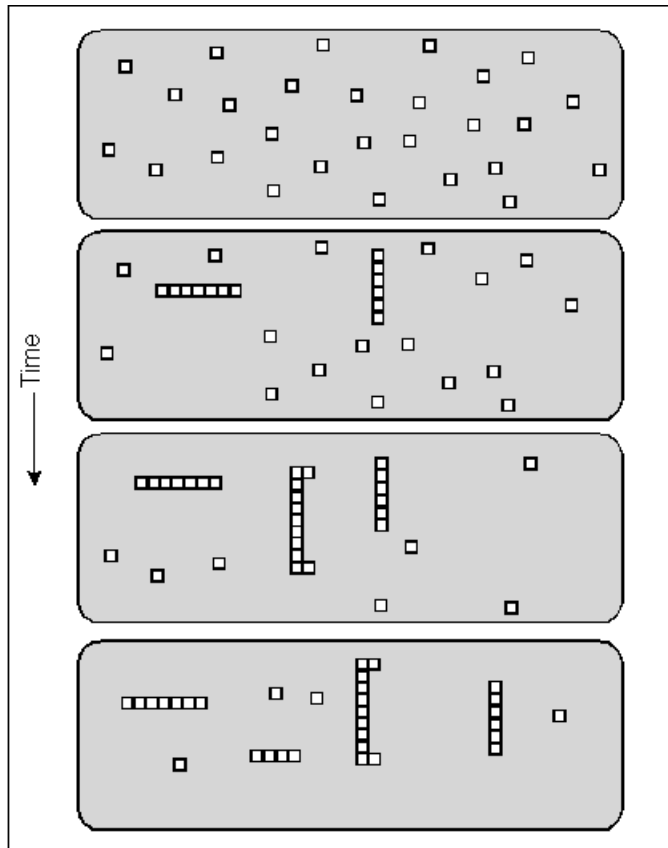


Step-growth Polymerization



- Any two molecular species can react.
- Monomer disappears early.
- Polymer MW rises throughout.
- Growth of chains is usually slow (minutes to days).
- Long reaction times increase MW, but yield of polymer hardly changes.
- All molecular species are present throughout.
- Usually (but not always) polymer repeat unit has fewer atoms than had the monomer.

Chain Polymerization



- Growth occurs only by addition of monomer to active chain end.
- Monomer is present throughout, but its concentration decreases.
- Polymer begins to form immediately.
- Chain growth is usually very rapid (second to microseconds).
- MW and yield depend on mechanism details.
- Only monomer and polymer are present during reaction.
- Usually (but not always) polymer repeat unit has the same atoms as had the monomer

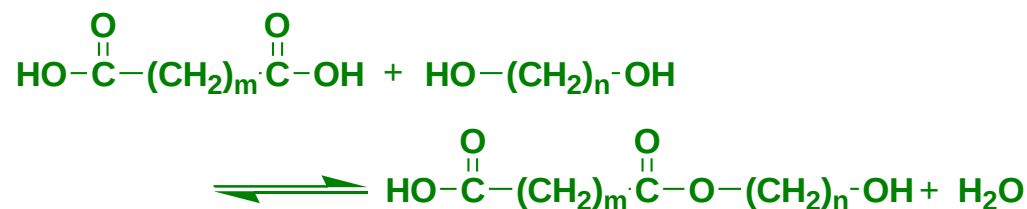


Chain Polymerization	Step-growth Polymerization
1. Unsaturated monomers	Monomers contain ≥ 2 functional groups
2. No products are eliminated	Elimination of H_2O , HCl , <i>etc.</i>
3. Only monomer and polymer are present during polymerization	Monomers and polymer are accompanied by dimers, trimers, and oligomeric species
4. Only monomers add to the growing polymer	All intermediate species are reactive, and contribute to the growing polymer
5. Mechanism involves reacting with double bond by active species like free radicals or ions	Involves simple elimination reaction between monomer functional groups
6. Rapidly yields a high MW polymer; crosslinking is achieved through use of monomers with two double bonds (<i>e.g.</i> , divinylbenzene)	Molecular weight is typically lower than chain polymerization. The presence of small amounts of multifunctional monomers results in extensive crosslinking (gels)
7. Examples: polyolefins, polydienes, vinyl polymers, acrylic polymers	Examples: polyesters, polyamides, polycarbonates, epoxies

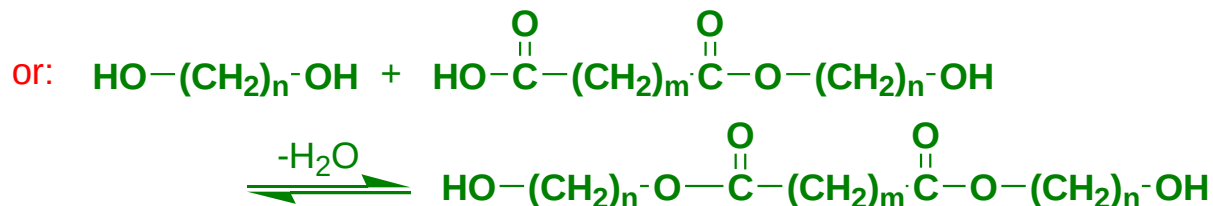
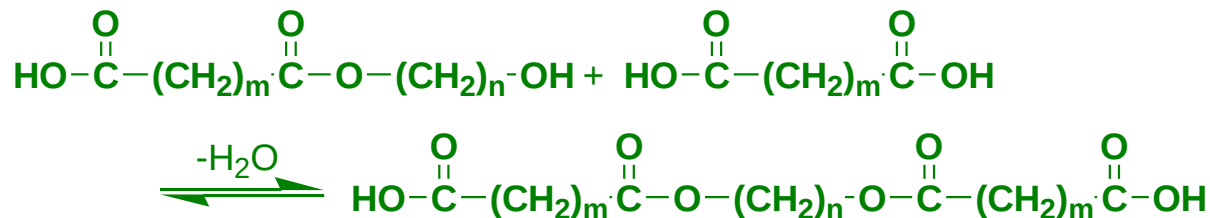


Making a Polymer by Step-growth Polymerization

- First step reaction between bifunctional acids and alcohols forms dimers.



- Dimers react with monomers to form trimers.



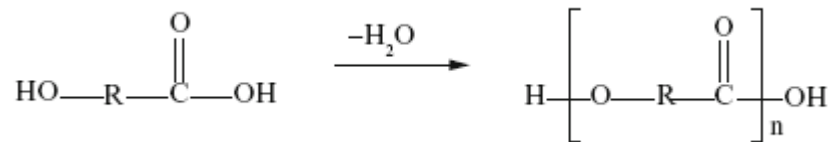
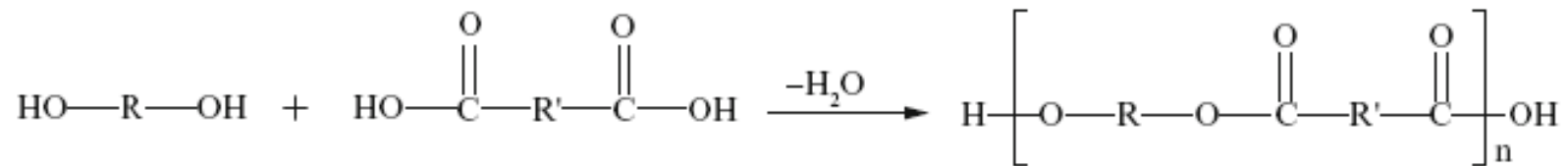
And so on:

Monomer is bifunctional

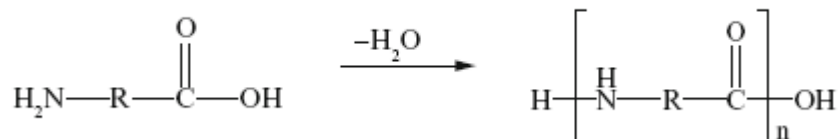
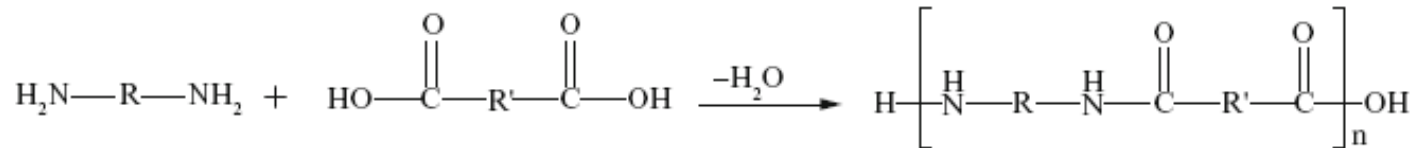


II. The general types of step-growth polymerization

(1). Polyester formation:

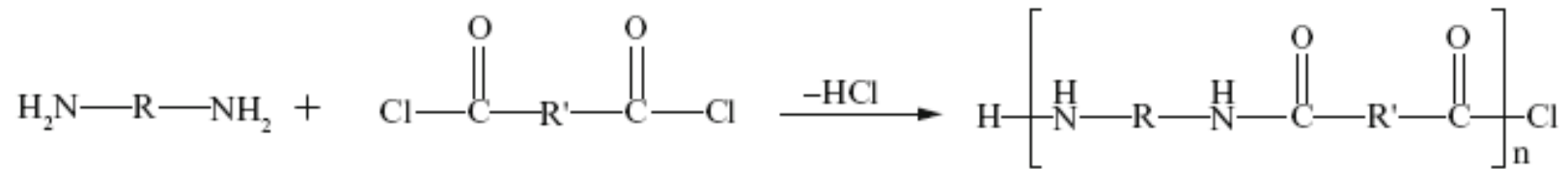


2. Polyamide formation:

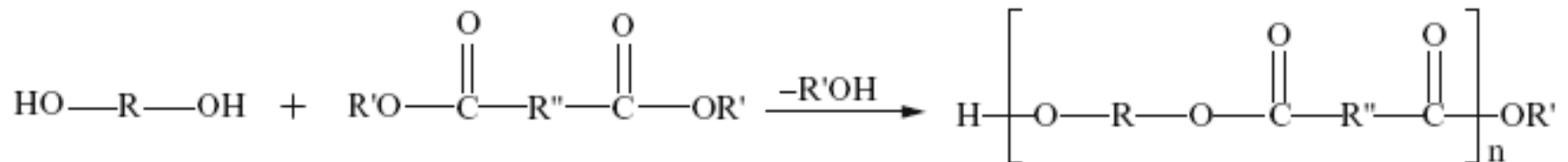




3. Polyamide formation through reaction of an acid chloride with a diamine:

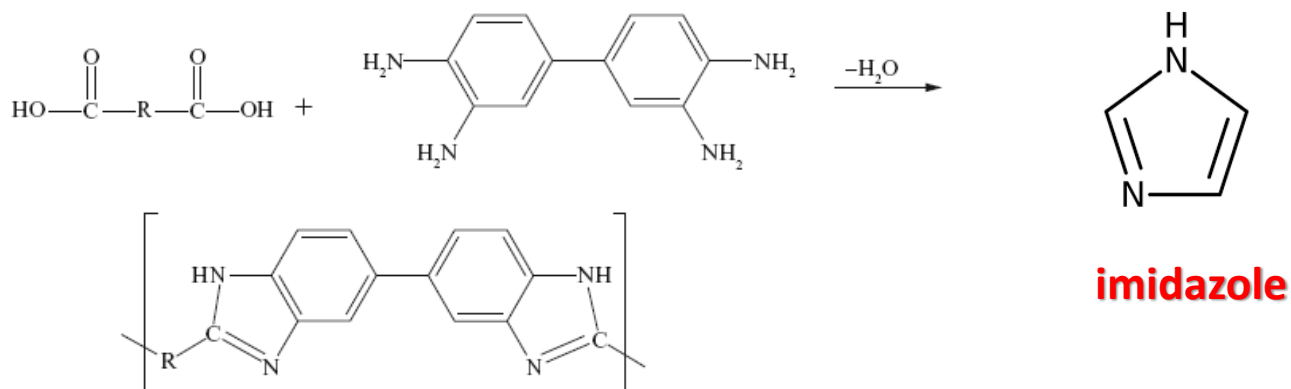


4. Transesterification involving a carboxylic acid ester and an alcohol:

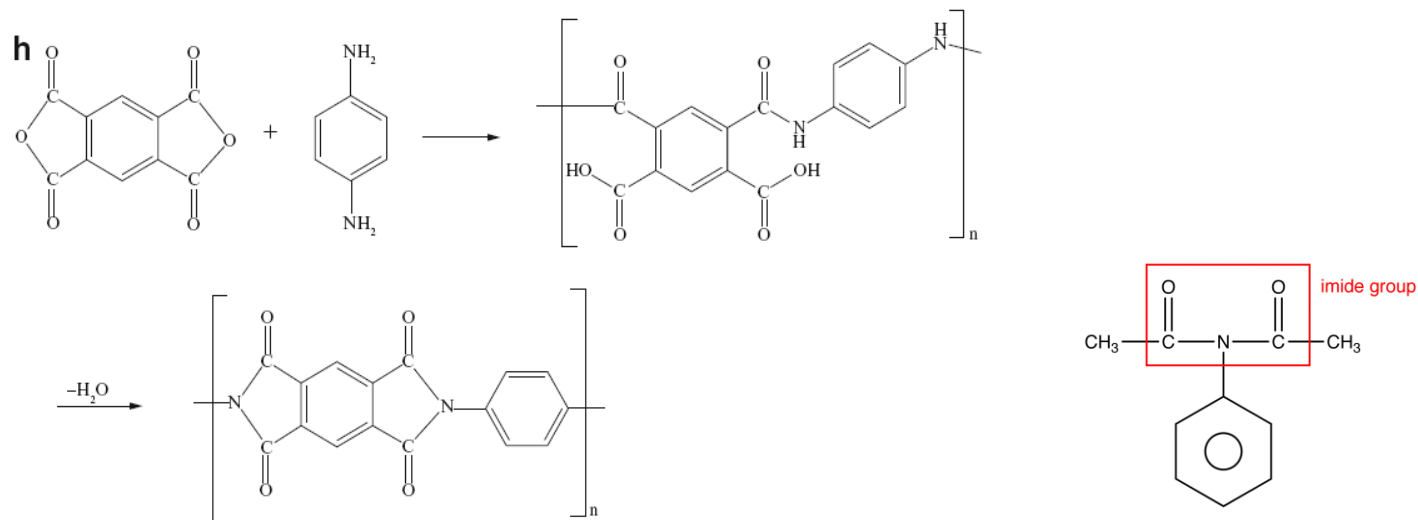




5. Polybenzimidazole (聚苯并咪唑) formation through reaction of a carboxylic acid and aromatic tetramines:

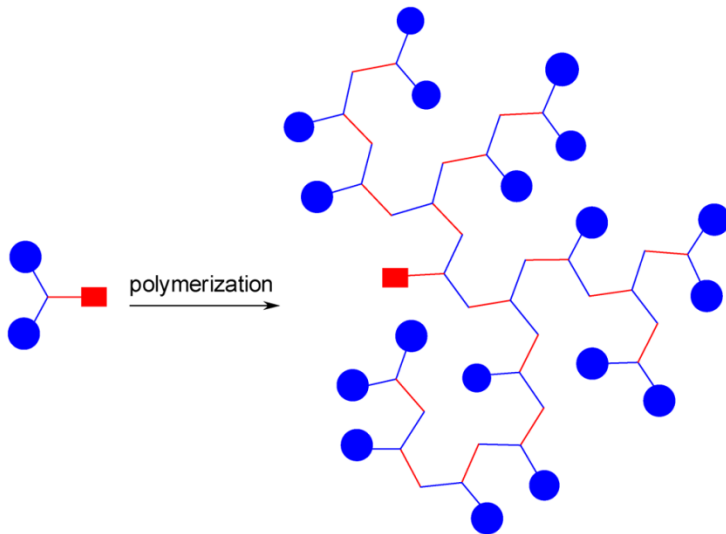


6. Polyimide (聚酰亚胺) formation from the reaction of dianhydrides and diamine:

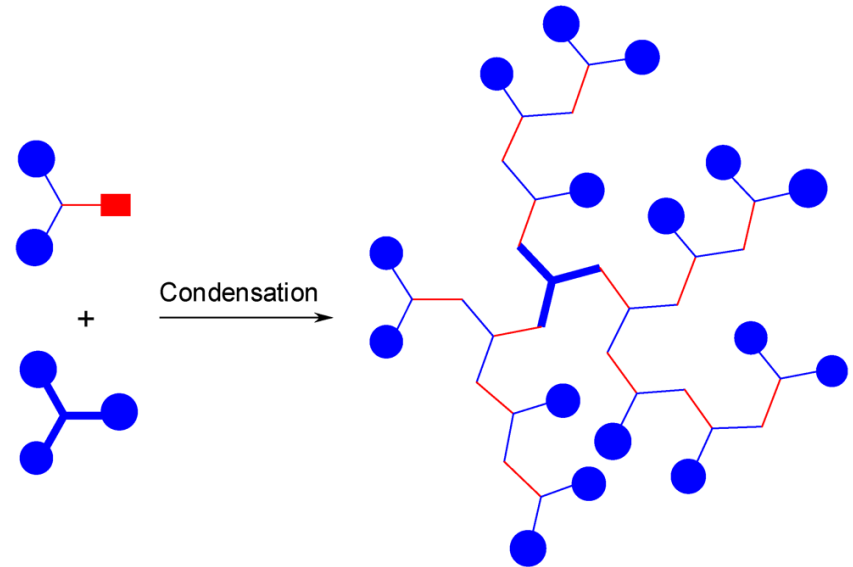




7. The formation of hyperbranched polymers via the AB_2 and $AB_2 + B_3$ approach



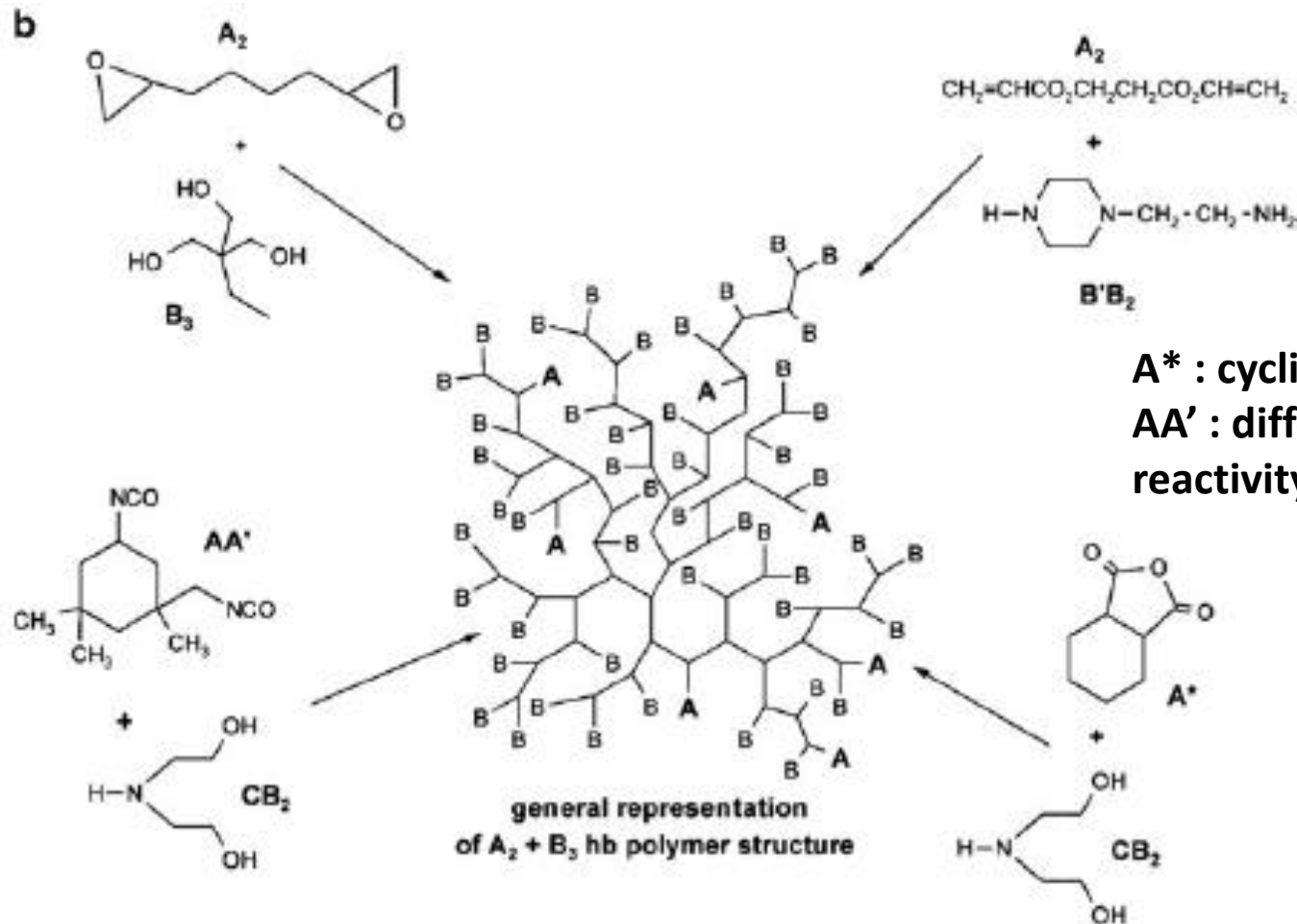
An A_2B monomer polycondensation.



Polycondensation of A_2B and A_3 monomers with excess A_2B monomer.



8. The synthesis of hyperbranched polymers by various $A_2 + B_3$ approaches



A^* : cyclic monomer;
 AA' : differences in the reactivity



Examples of step-growth polymerization:

Nylons-Polyamides

Extreme toughness and strength as strong as steel, as fine as spider's web, yet more elastic than any of the common natural fibers.



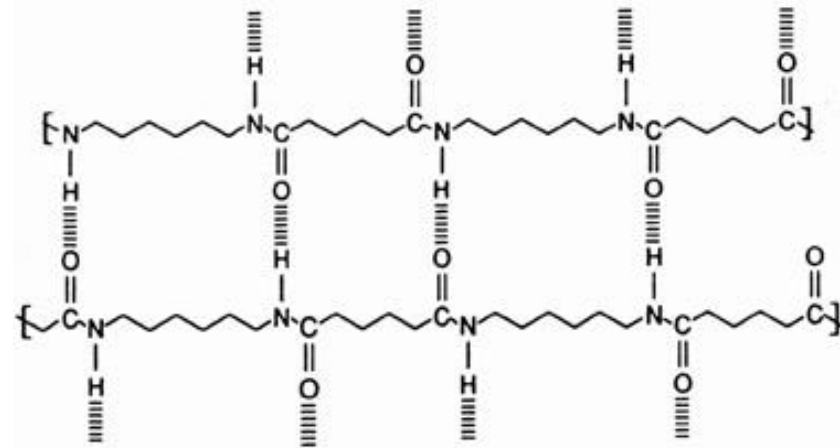
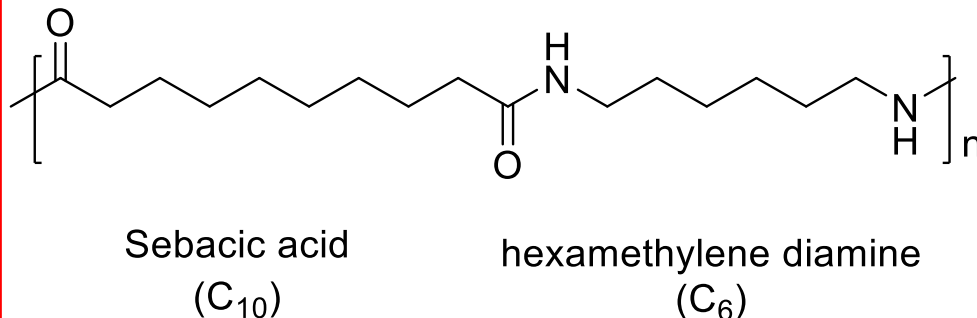


Synthesis of Nylon 6,10

Polyhexamethylenesebacamide

https://www.youtube.com/watch?feature=player_detailpage&v=yFEHKRdXb9Y

Nylon 6,10 (not nylon 10, 6)



Nylon polymer molecules linked by hydrogen bonds after cold drawing

Nomenclature of Nylon:

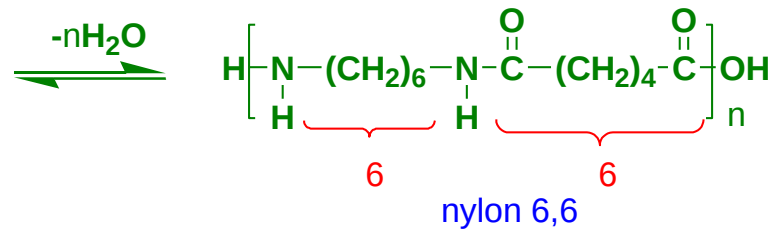
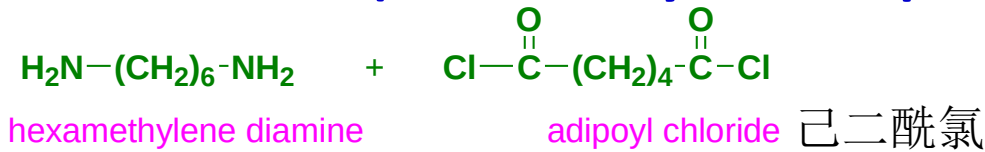
Typically, two numbers: Nylon m, n; made by polycondensation of diacid and diamine.

- First number (m):
number of carbons in diamine,
- Second number (n):
number of carbons in diacid.

Nylon-6,10 (PA610) is semicrystalline polyamide commonly used in monofilament form in applications such as bristles and brushes.



PA66 (Hexamethylene adipamide)

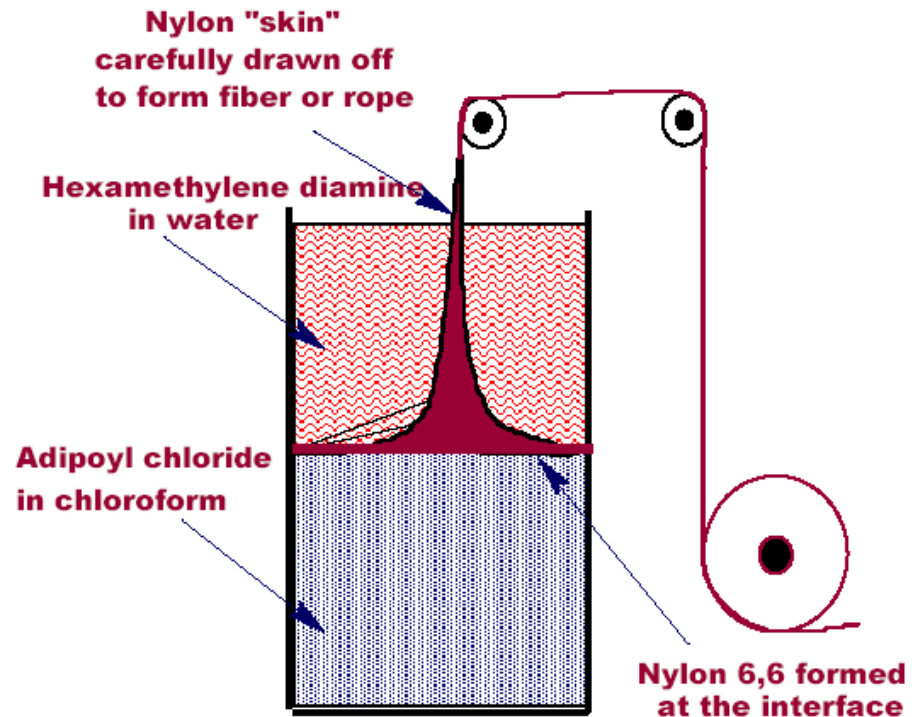


Preparation of PA66

To obtain a high M_w

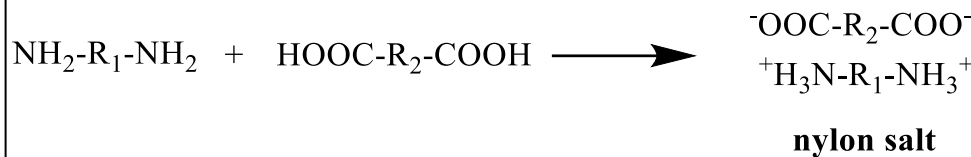
1. Formation of PA66 salt for stoichiometric equivalence
2. Polymerization of PA66 salt with removing small molecules

Nylon Rope Trick





Commercial Production of Nylons



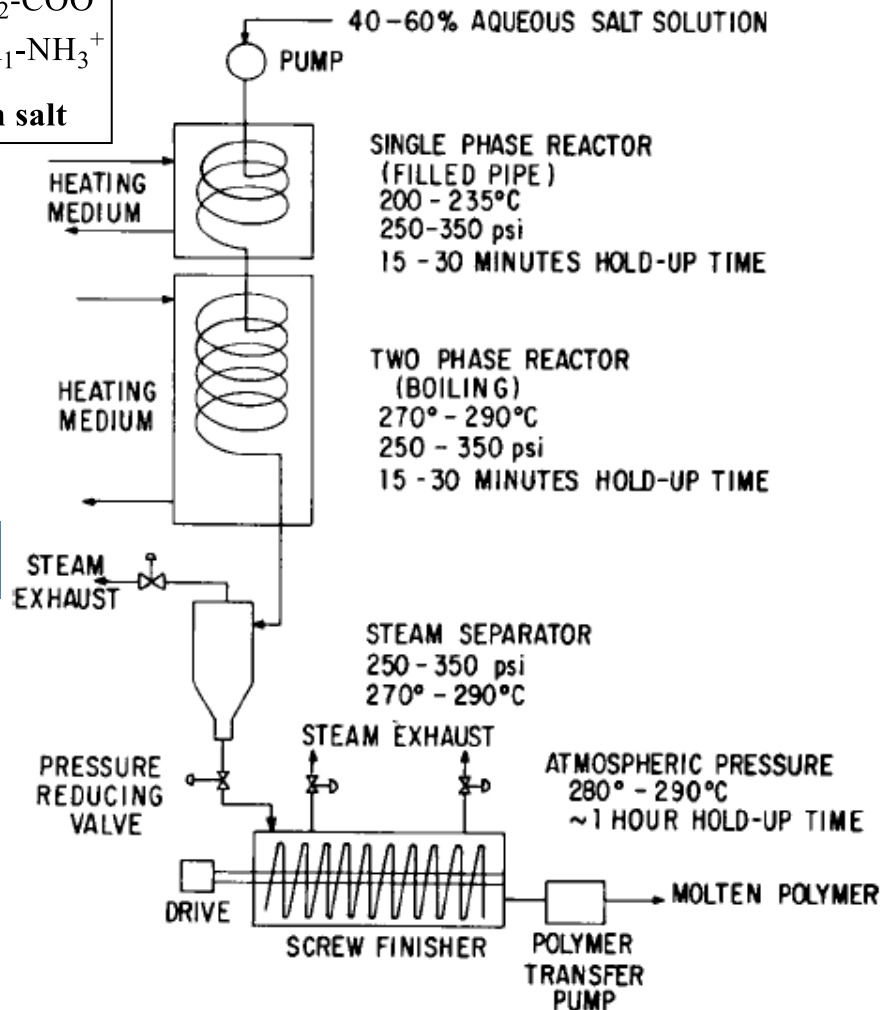
1. Commercial Processes start with nylon salt to ensure stoichiometric equivalence.
2. Further remove H_2O to make high M_w polymer

Water byproduct

http://www.nsf.gov/news/special_reports/chemistrynow/chem_nylon.jsp

On May 15, 1940, nylon stockings went on sale throughout the US, and in New York City alone four million pairs were sold in a matter of hours.

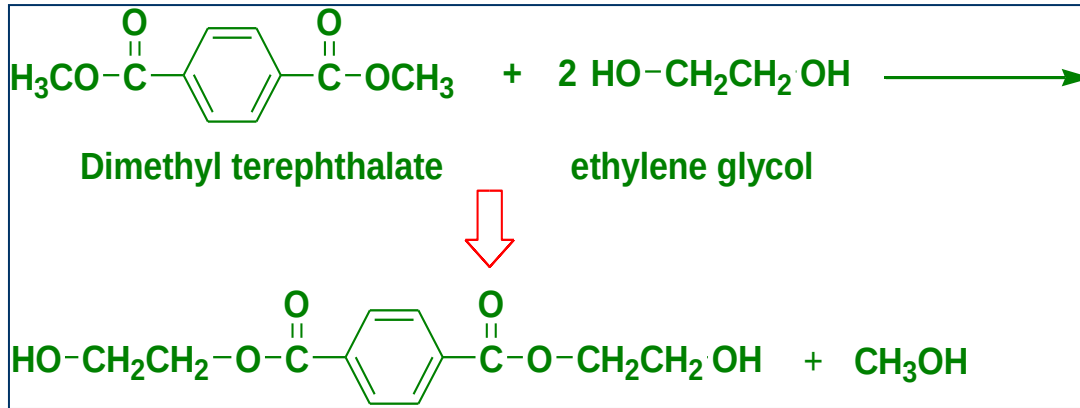
Odian, Principles of Polymerization, 4th ed.





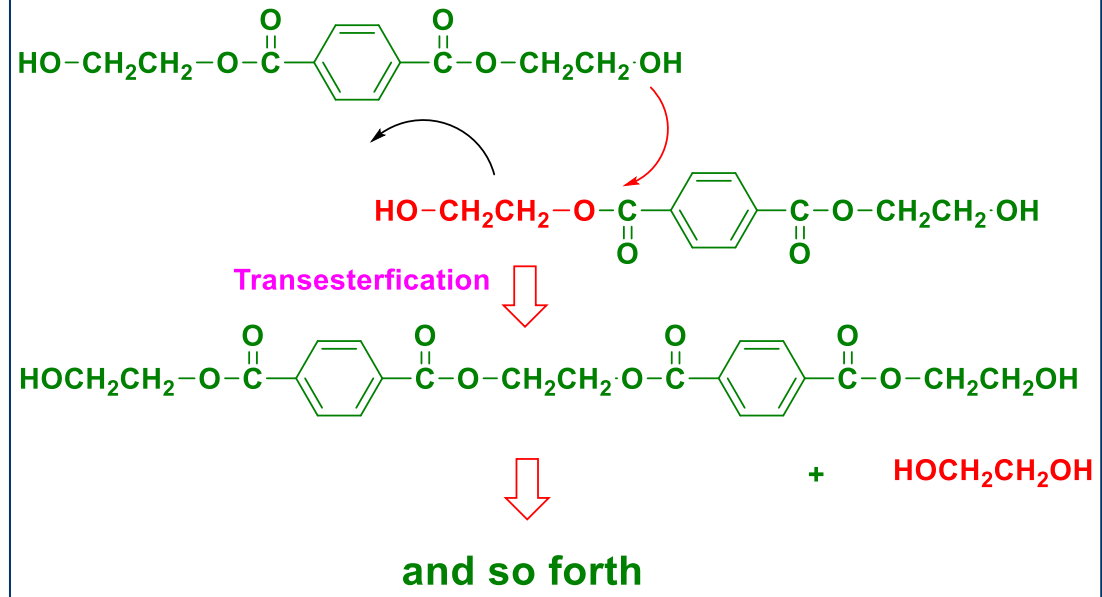
PET: poly ethylene terephthalate

聚对苯二甲酸乙二醇酯



Polyester- Transesterification

酯基转移





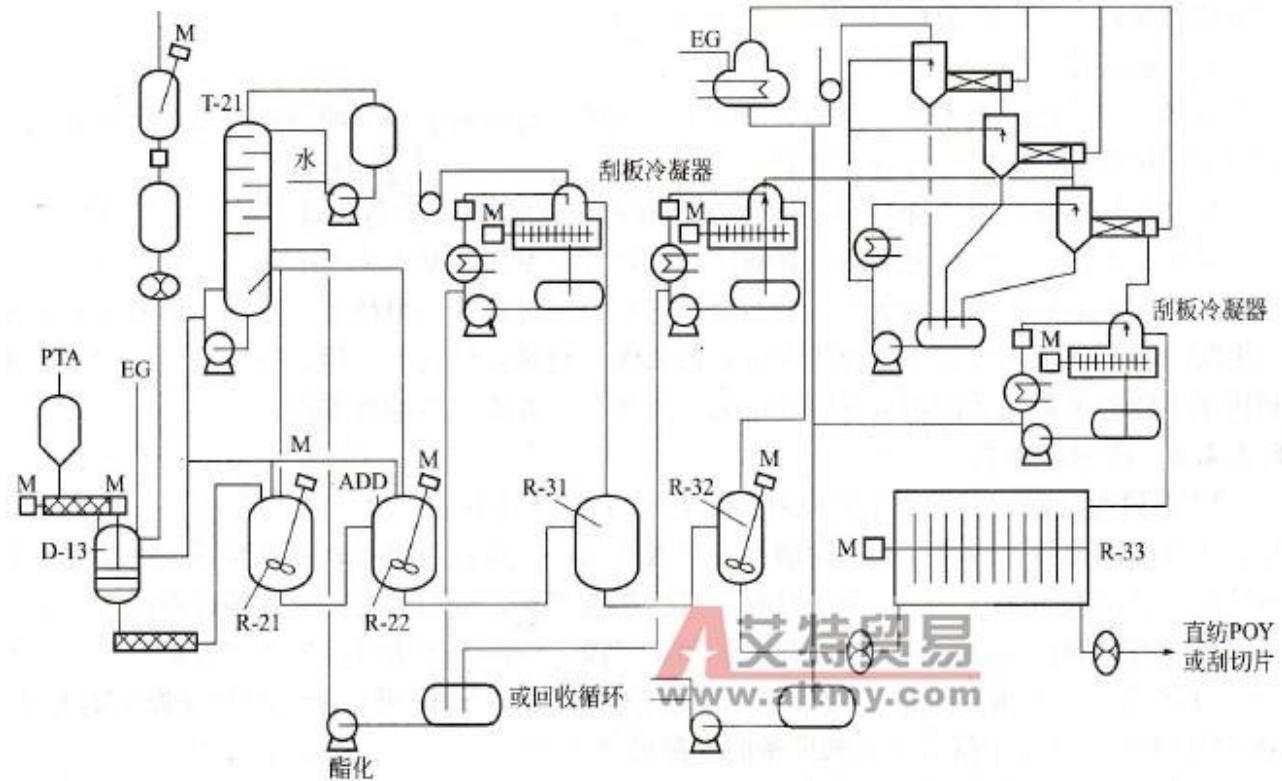
Production of PET

Further remove
ethylene glycol and
methanol to make high
 M_w polymer

对苯二甲酸(PTA)

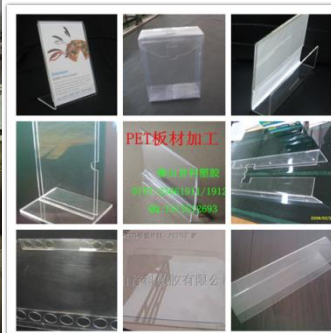
乙二醇(EG)

对苯二甲酸双-羟乙酯(DGT)单体



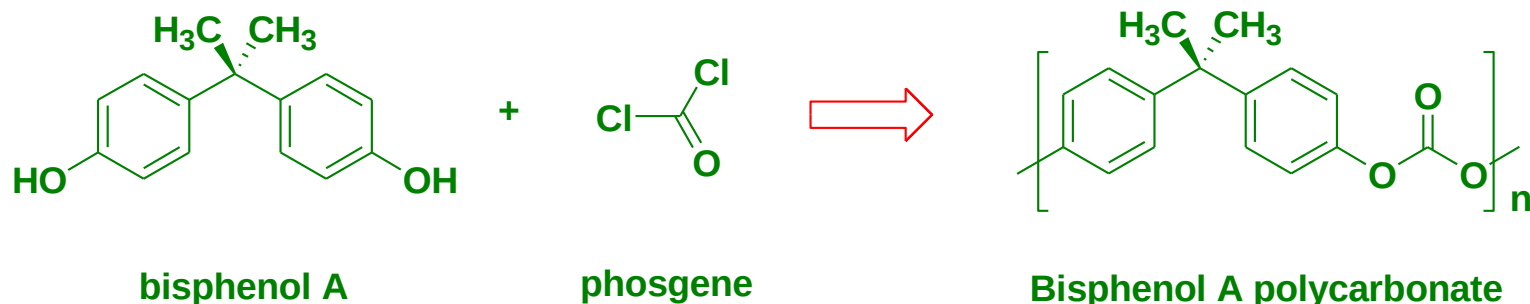
艾特贸易
www.aity.com

Processing of PET

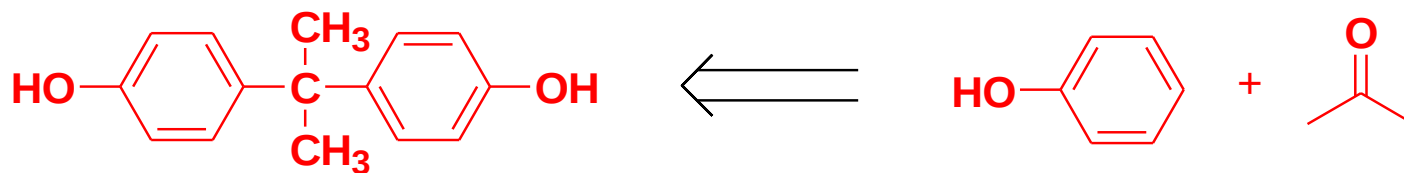


直接酯化法连续生产PET树脂工艺流程
D-13-浆料制备器；R-21，R-22-酯化反应器；
R-31，R-32-预缩聚反应器；
R-33-圆盘反应器；T-21-EG（乙二醇）回收塔

Polycarbonates (PC)



Bisphenol A synthesis:

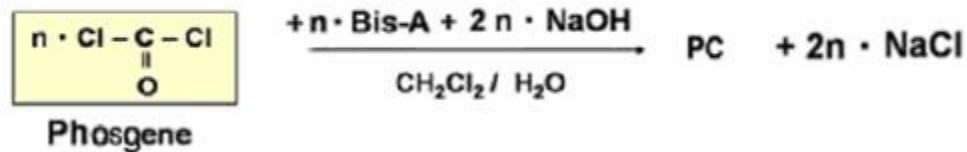


- Bisphenol A polycarbonate: optical clear, glassy polymer, tough materials.
- It is used in transparent street lighting, compact disk, football helmets.

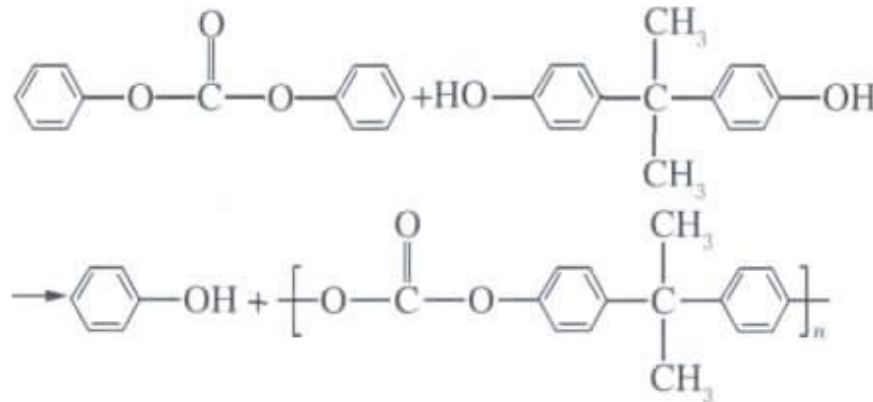
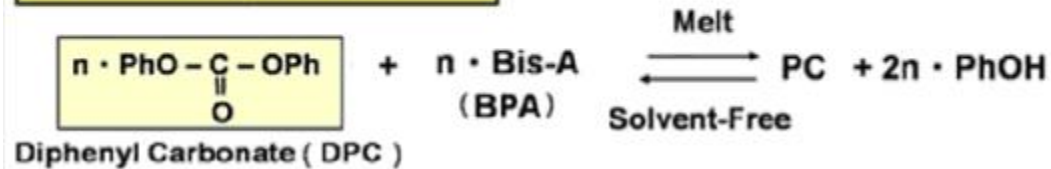


Overall Reaction Path to Polycarbonate

Phosgene Process



Non-Phosgene Process



Step Polymerization Reactions

- Carbonyl addition-elimination
 - Polyesters, polyamides, polycarbonates, polyimides...
- Aromatic addition-elimination
 - Polysulfones, polysulfides, polyetherketones
- Carbonyl addition-condensation
 - Phenol-formaldehyde and related polymers
 - Polymeric heterocycles
- Addition to multiple bonds or epoxides
 - Polyurethanes, epoxy polymers.
- Miscellaneous
 - Oxidative aromatic addition (polyphenylene oxide)
 - Acyclic diene metathesis (ADMET)
 - Aryl-aryl coupling
 - Reductive coupling (polysilanes)
 - Hydrolysis coupling (silicones)
 - Diels-Alder cycloaddition
 - Friedel-Crafts chemistry
 - SN2 reactions
 - and a lot of others...



III. Theoretical Consideration

- Two Classes of step-growth polymerization reactions:

Polycondensation:



Polyaddition:



- Equilibrium Constant, K :

$$K = \frac{[AB][C]}{[A][B]}$$

K is fixed for specific conditions;

- To increase the formation of AB , we must remove C from mixture (e.g. vacuum stripping).

$$[AB] = K \frac{[A][B]}{[C]}$$



For polyaddition:



no small molecules was removed

- Equilibrium Constant:

$$K = \frac{[AB]}{[A][B]} \quad [AB] = K[A][B]$$

- To increase the formation of AB, we need a larger K

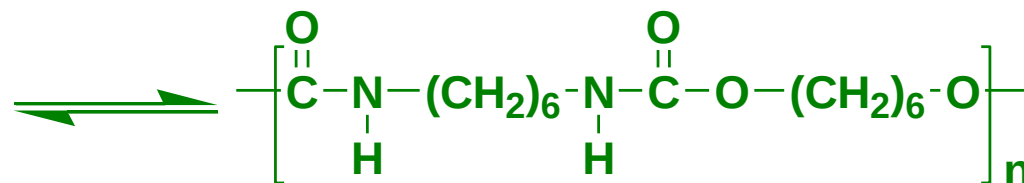
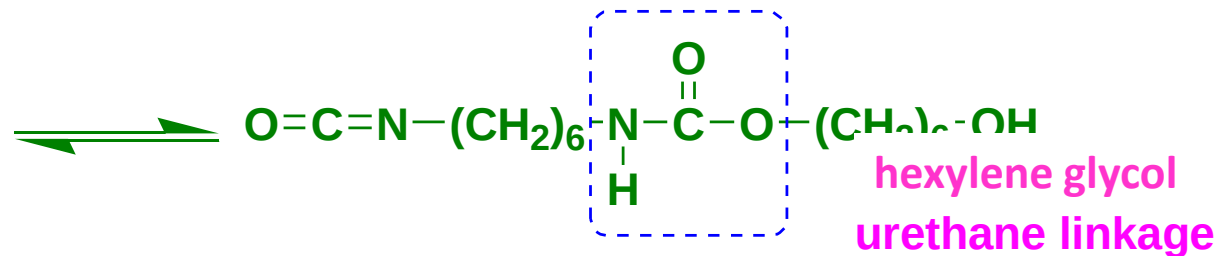


hexamethylene diisocyanate

+



ethylene glycol

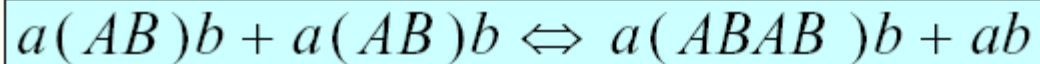
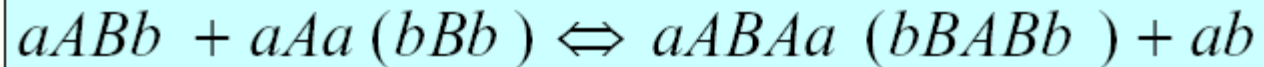


polyurethane

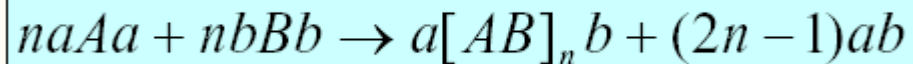


2-2 bifunctional system

Linear poycondensation

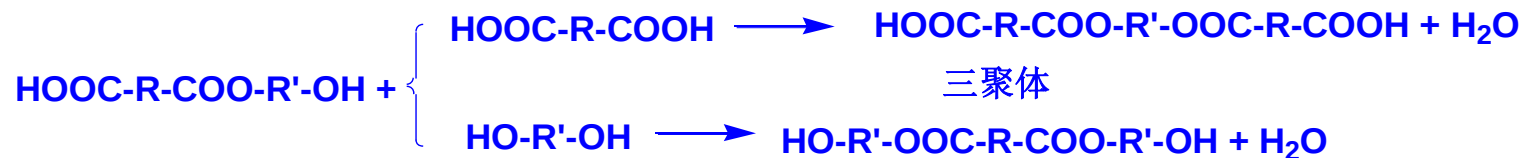
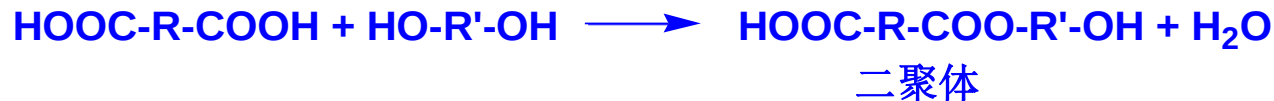


⋮



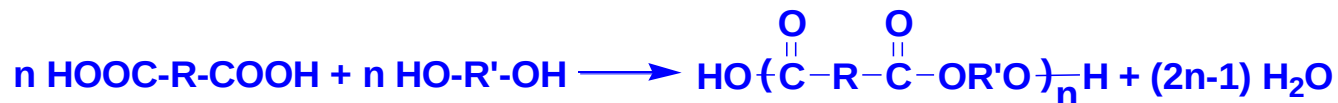


Poyester from diacids and diols



◦
◦
◦

◦
◦
◦





□ Molecular Weight of Polymer

The extent or fraction of reaction (p)

The fraction of the functional groups that has reacted at time t . p is also referred to as the extent or fraction of conversion. (The value of p is calculated from a determination of the amount of unreacted groups.)

$$[N_a] = [N_a]_0 - [N_a]_0 p = [N_a]_0 (1 - p)$$

$$p = \frac{[N_a]_0 - [N_a]}{[N_a]_0}$$

$[N_a]_0$ is the initial (at $t = 0$) concentration of the functional groups.

$[N_a]$ is the concentration of the functional groups at time t .



- For the case of stoichiometric amounts of diol and diacid, the number of unreacted carboxyl groups N_a is equal to the total number of molecules M present in the system at some time t .
- This is so because each molecule larger than a monomer will on the average have a hydroxyl at one end and a carboxyl at the other end, while each diacid monomer molecule contains two carboxyls and each diol monomer contains no carboxyls.

At time t :

N_a : *unreacted carboxyl groups*

M = *the total number of molecules present in the system*

$$N_a = M$$



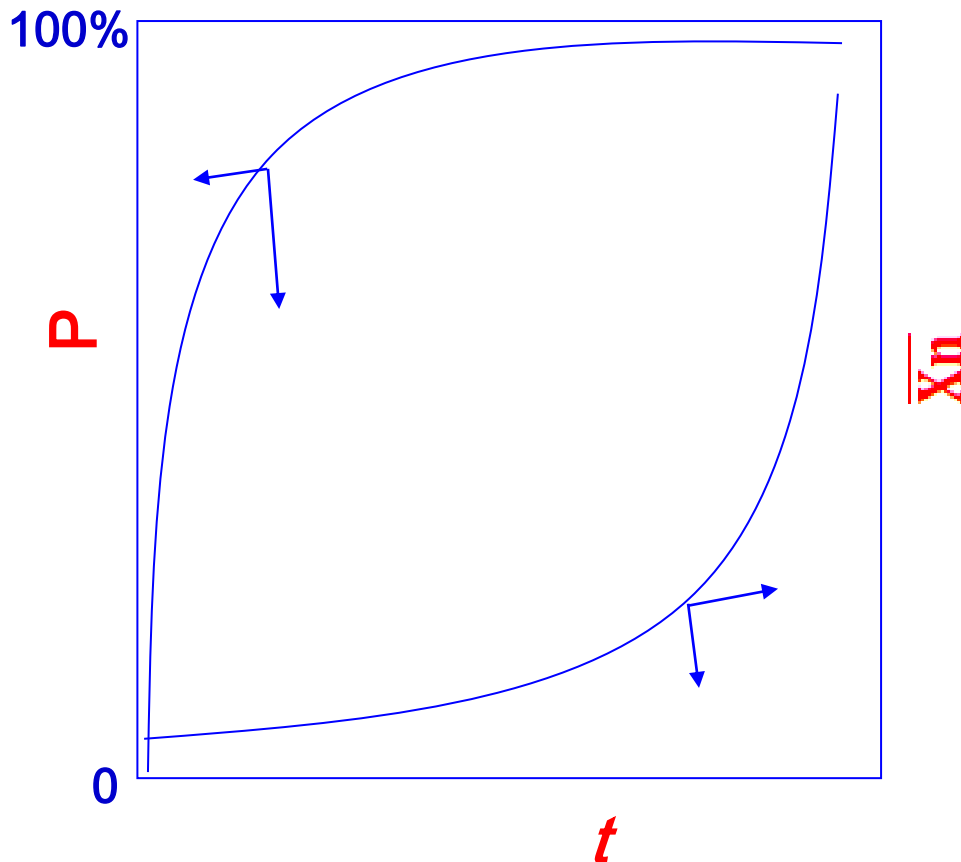
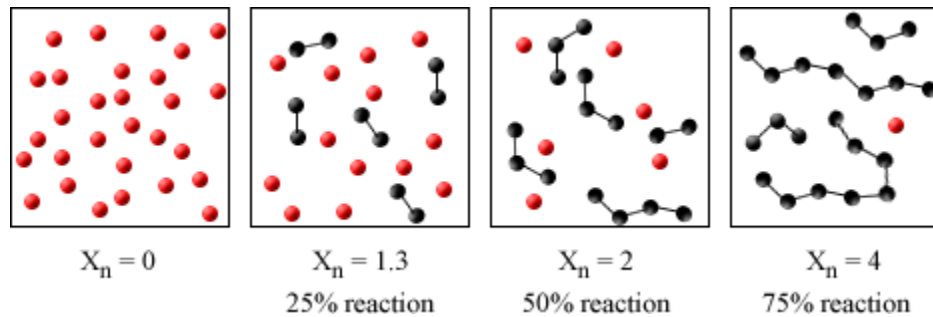
- The residue from each diol and each diacid (separately, not together) in the polymer chain is termed a structural unit (or a monomer unit).
- The repeating unit of the chain consists of two structural units, one each of the diol and diacid.
- The total number of structural units in any particular system equals the total number of bifunctional monomers initially present.

The number-average degree of polymerization $\overline{X}_n$ is defined as the average number of structural units per polymer chain.

$$\overline{X}_n = \frac{[N_a]_0}{M} = \frac{[N_a]_0}{[N_a]}$$

$$\overline{X}_n = \frac{1}{1-p}$$

Carothers' equation



$$P=0.9, \overline{X_n}=10$$

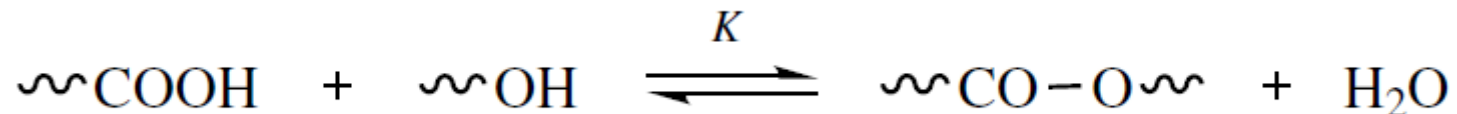
$$p=0.995, \overline{X_n}=200$$

As the reaction progresses, higher molecular weight species are formed but the degree of polymerisation increases very slowly. At 75% conversion the average $\overline{X_n}$ is still very low ($\overline{X_n} = 4$) and very high conversion rates are required in order to obtain high molecular weights.



□ Equilibrium Considerations---Closed System

- A polymerization in which the monomer(s) and polymer are in equilibrium is referred to as an equilibrium polymerization or reversible polymerization.
- By a closed system is meant one where none of the products of the forward reaction are removed. Nothing is done to push or drive the equilibrium point for the reaction system toward the polymer side.



$[M]_0$: the initial hydroxyl group or carboxyl group concentrations

p_e : the extent of reaction at equilibrium

$p_e[M]_0$: The concentration of ester groups [COO] or water [H₂O] at equilibrium

$$K = \frac{[\text{COO}][\text{H}_2\text{O}]}{[\text{COOH}][\text{OH}]} = \frac{(p_e[M]_0)^2}{([M]_0 - p_e[M]_0)^2}$$



$$K = \frac{[\text{COO}][\text{H}_2\text{O}]}{[\text{COOH}][\text{OH}]} = \frac{(p_e[\text{M}]_0)^2}{([\text{M}]_0 - p_e[\text{M}]_0)^2}$$

which simplifies to

Solving for p_e yields

$$K = \frac{p_e^2}{(1 - p_e)^2}$$



$$p_e = \frac{K^{1/2}}{1 + K^{1/2}}$$

$$\therefore \overline{X}_n = \frac{1}{1-p}$$



$$\overline{X}_n = 1 + K^{1/2}$$

In the process of polyester production, the equilibrium constant K is low ($K=1\sim 10$) ; therefore the $\overline{X}_n$ is very low if water is not removed in this system.



The equilibrium constant K

- Esters from Acids and alcohols $K = 1-10$
- Amides from Acids and amines $K = 10-1000$
- Amides or esters from acid chlorides $K > 10^4$



□ Equilibrium Considerations---Open, Driven System

$$K = \frac{[\text{COO}][\text{H}_2\text{O}]}{[\text{COOH}][\text{OH}]}$$

$$K = \frac{p[\text{H}_2\text{O}]}{[\text{M}]_0(1-p)^2}$$

$$\therefore \bar{X}_n = \frac{1}{1-p} \quad \Rightarrow \quad [\text{H}_2\text{O}] = \frac{K[\text{M}]_0}{\bar{X}_n(\bar{X}_n - 1)}$$

$(\bar{X}_n - 1)$ is close to $\bar{X}_n$ for large values of $\bar{X}_n$

$$\bar{X}_n = \sqrt{\frac{K[\text{M}]_0}{[\text{H}_2\text{O}]}}$$

To increase the $\bar{X}_n$ in the process, the byproduct (water) is required to be removed as much as possible from the polymerization system.



□ Stoichiometric Control---Bifunctional systems

For linear polymerization

Desired molecular weight can be obtained by

- Adjusting the stoichiometric ratio of the two monomers so that one of the reactants is present in slight excess



- Addition of a small amount of a monofunctional monomer





- For the polymerization of the bifunctional monomers A-A and B-B, where B-B is present in excess.

r : stoichiometric ratio

$[N_a]_0, [N_b]_0$: the numbers of functional group a and b

$$r = \frac{[N_a]_0}{[N_b]_0} < 1$$

The average number of structural unit is:

$$([N_a]_0 + [N_b]_0)/2 = [N_a]_0(1 + 1/r)/2$$

The extent of reaction p is defined as the fraction of the limiting groups (A groups) that have reacted at a particular time. The fraction of B groups that have reacted is given by rp .

The total numbers of unreacted A and B groups are $[N_a]_0(1 - p)$ and $[N_b]_0(1 - rp)$, respectively.

The total number of polymer chain ends is given by the sum of the total number of unreacted A and B groups.

$$\overline{X_n} = \frac{[N_a]_0(1 + 1/r)/2}{\{[N_a]_0(1 - p) + [N_b]_0(1 - rp)\}/2} = \frac{1 + r}{1 + r - 2rp}$$

If $r=1$

The two bifunctional monomers are present in stoichiometric amounts



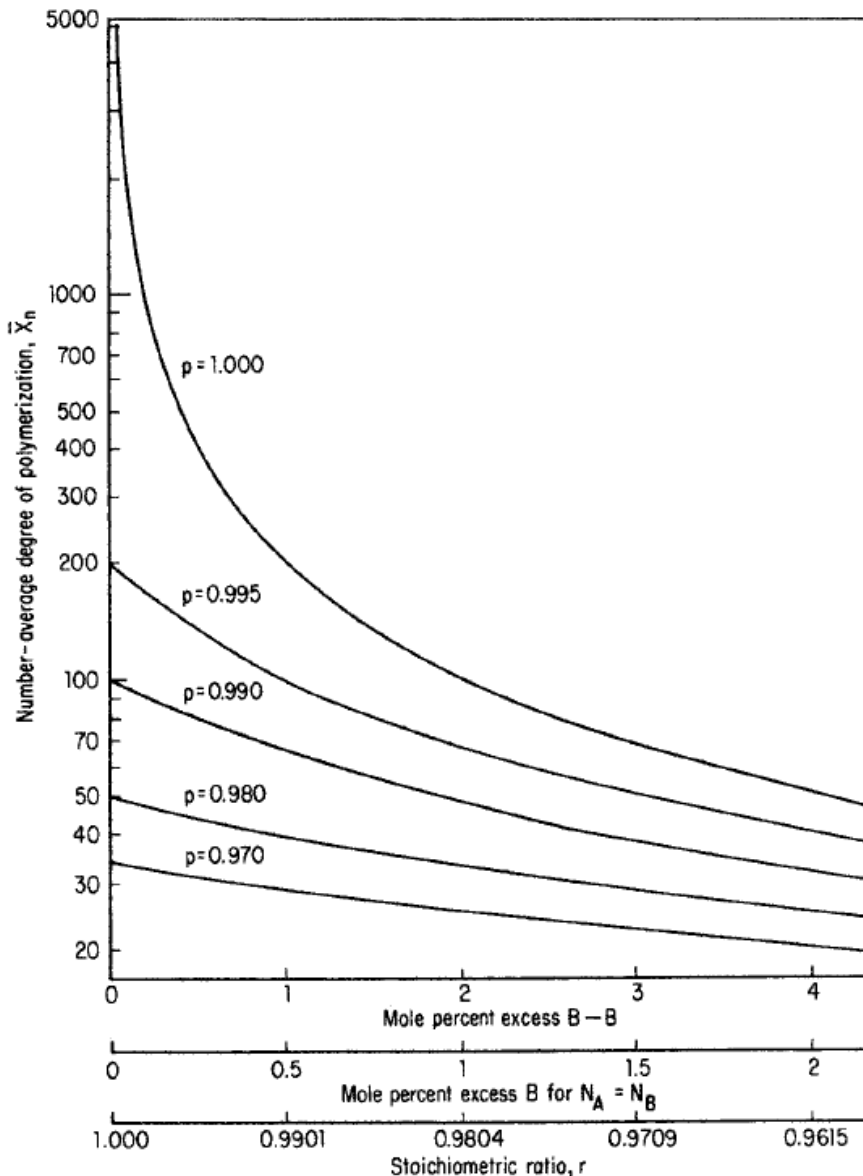
$$\overline{X_n} = \frac{1}{1 - p}$$

If $p=1$

For 100% conversion



$$\overline{X_n} = \frac{1 + r}{1 - r}$$



$$\text{If } r = \frac{1000}{1001} \quad p = 100\%$$

$$\overline{X}_n = 2001$$

$$\text{If } r = \frac{100}{101} \quad p = 100\%$$

$$\overline{X}_n = 201$$



$r \downarrow$

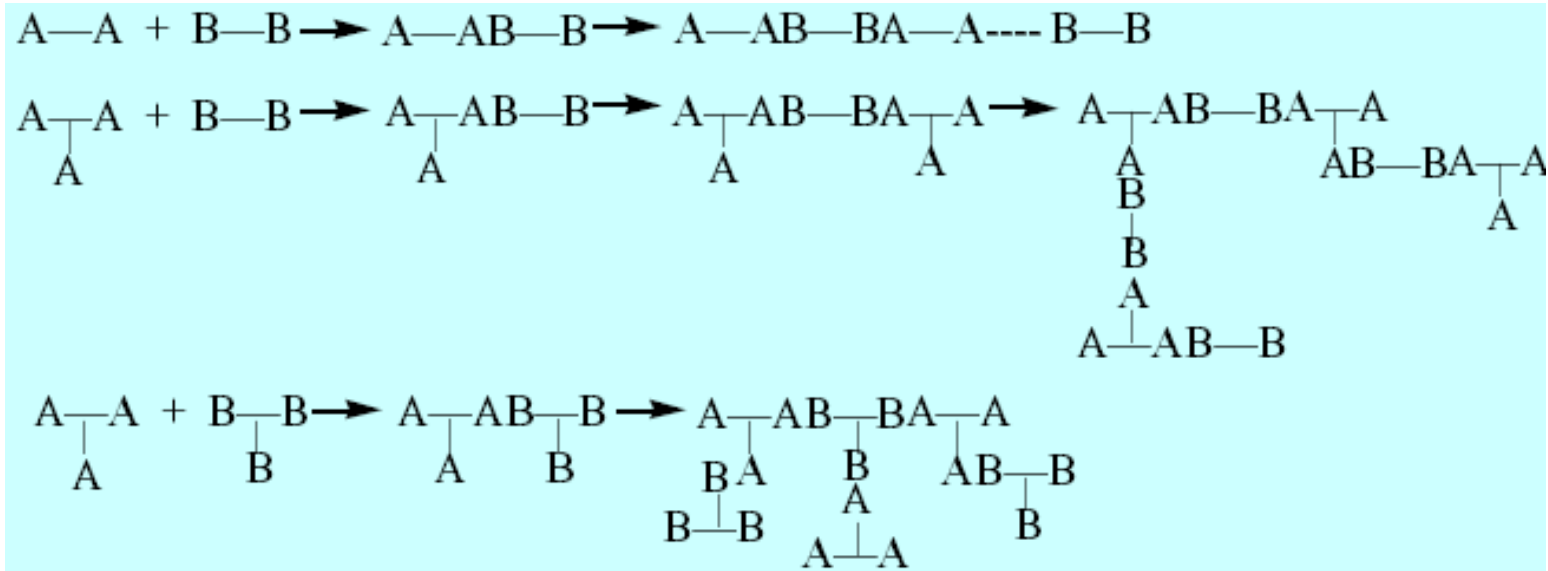
$\overline{X}_n$ decrease significantly

It is clear that step polymerizations will almost always be carried out to at least 98% reaction, since a degree of polymerization of at least approximately 50–100 is usually required for a useful polymer. Higher conversions and the appropriate stoichiometric ratio are required to obtain higher degrees of polymerization.



Carothers theory of gelation---Functionalities > 2

The resulting polymers will first be branched instead of linear, and then form network structures by crosslinking.





- Carothers equation: $\overline{X}_n \rightarrow \infty$
- Gel point
Crosslinking is distinguished by the occurrence of gelation in the polymerization. At this point, the system loses fluidity.
- Carothers derived a relationship between the extent of reaction at the gel point (p) and the average functionality (f_{avg}) of the polymerization system for the case where the two functional groups A and B are present in equivalent amounts

$$f_{avg} = \frac{\sum N_i f_i}{\sum N_i}$$

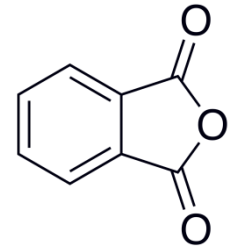
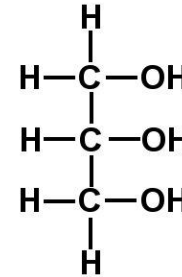
N_i : the number of molecules of monomer i with functionality f_i ,

$\sum N_i$: the summations are over all the monomers present in the system.



For example, a system consisting of 2 mol of glycerol (a triol) and 3 mol of phthalic acid (a diacid):

$$f_{avg} = \frac{\sum N_i f_i}{\sum N_i} = \frac{2 \times 3 + 3 \times 2}{2 + 3} = 2.4$$



In a system containing equivalent numbers of A and B groups:

N_0 : the number of monomer molecules present initially;

$N_0 f_{avg}$: the total number of functional groups;

N : the number of molecules after reaction has occurred;

$2(N_0 - N)$: the number of functional groups that have reacted.

$$p = \frac{2(N_0 - N)}{N_0 f_{avg}}$$



$$\bar{X}_n = \frac{N_0}{N}$$



$$\bar{X}_n = \frac{2}{2 - pf_{\text{avg}}}$$

$$p = \frac{2}{f_{\text{avg}}} - \frac{2}{\bar{X}_n f_{\text{avg}}}$$

Carothers equation

At the gel point where the number-average degree of polymerization becomes infinite. The critical extent of reaction p_c at the gel point is given by:

$$p_c = \frac{2}{f_{\text{avg}}}$$

This equation can be used to calculate the extent of reaction required to reach the onset of gelation in a mixture of reacting monomers from its average functionality.



For example, a system consisting of 2 mol of glycerol (a triol) and 3 mol of phthalic acid (a diacid):

$$f_{avg} = \frac{\sum N_i f_i}{\sum N_i} = \frac{2 \times 3 + 3 \times 2}{2 + 3} = 2.4$$

$$p_c = \frac{2}{f_{avg}} = \frac{2}{2.4} = 0.833$$



Applications of Carothers equations

$$\bar{X}_n = \frac{2}{2 - pf_{\text{avg}}}$$

To calculate the degree of polymerization of reaction system.

If $\bar{X}_n < 0$ in the calculation: system past the gel point at that conversion.

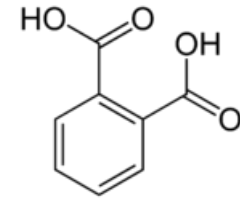
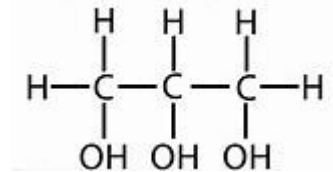
$$p_c = \frac{2}{f_{\text{avg}}}$$

To calculate the critical extent of reaction at the gel point.

p is the extent of reaction of the deficient functional group.



For example, a mixture of 1 mol of glycerol and 5 mol of phthalic acid.



$$f_{avg} = \frac{\sum N_i f_i}{\sum N_i} = \frac{1 \times 3 + 5 \times 2}{1 + 5} = 2.17$$

$$p_c = \frac{2}{f_{avg}} = \frac{2}{2.17} = 0.922$$

- Both of these conclusions are grossly in error.
- It is apparent that the gross stoichiometric imbalance between the A and B functional groups in this system ($r = \frac{3}{10}$) precludes the formation of any but extremely low-molecular-weight species.
- The polymerization will stop with the large portion of the acid functional groups being unreacted.



□ Stoichiometric Control---Functionalities > 2

Extension to Nonstoichiometric Reactant Mixtures

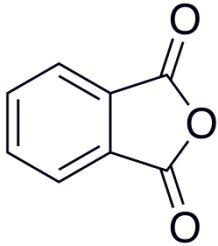
Ternary mixture: N_A moles of A_{f_A} , N_C moles of A_{f_C} ($f_C > 2$), N_B moles of B_{f_B} . A and C have same functional groups (such as A); Number of A functional group is smaller than that of B functional group. (i.e. B groups are in excess)

$$f_{\text{avg}} = \frac{2(N_A f_A + N_C f_C)}{N_A + N_C + N_B} \quad (\text{Only calculate the reacting groups})$$

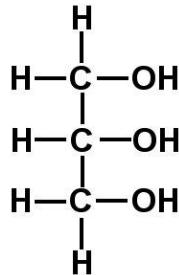
The extent of polymerization depends on the deficient reactant. The excess of other reactant results in lowering of functionality.

Calculate the extent of reaction at which gelation occurs for the following mixtures:

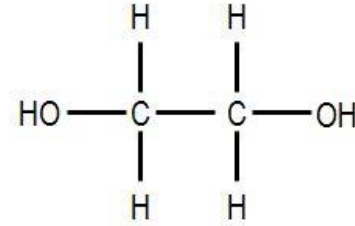
- Phthalic anhydride and glycerol in stoichiometric amounts
- Phthalic anhydride and glycerol in the molar ratio 1.500:0.980
- Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.990:0.02
- Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.500:0.700



$$f_B = 2$$



$$f_C = 3$$



$$f_A = 2$$

	$N_B f_B$	$N_C f_C$	$N_A f_A$
a	$3 \times 2 = 6$	$2 \times 3 = 6$	
b	$1.5 \times 2 = 3$	$0.98 \times 3 = 2.94$	
c	$1.5 \times 2 = 3$	$0.99 \times 3 = 2.97$	$0.02 \times 2 = 0.04$
		3.01	
d	$1.5 \times 2 = 3$	$0.500 \times 3 = 1.5$	$0.700 \times 2 = 1.4$
		2.9	

	f_{avg}	$p_c = \frac{2}{f_{\text{avg}}}$
a	$\frac{6 + 6}{3 + 2} = 2.4$	0.83
b	$\frac{2.94 \times 2}{1.5 + 0.98} = 2.37$	0.84
c	$\frac{3 \times 2}{1.5 + 0.99 + 0.02} = 2.39$	0.84
d	$\frac{2.9 \times 2}{1.5 + 0.5 + 0.7} = 2.15$	0.93



2. Step-growth polymerization and Carothers' equation

p : the fraction of functional group conversion

官能团的反应率

$$p = \frac{[N_a]_0 - [N_a]}{[N_a]_0}$$

$[N_a]_0$ is the initial (at $t = 0$) concentration of the functional group a .

$[N_a]$ is the concentration of the functional group a at time t .

$\overline{X}_n$: 聚合物的聚合度 The number-average degree of polymerization

$\overline{X}_n$: 每条高分子链所含结构单元数的平均值

Average number of structural units contained in a polymer chain

$$\overline{X}_n = \frac{\text{结构单元总数}}{\text{大分子个数}} = \frac{\text{the total number of structural units}}{\text{the number of polymer chains}}$$

2-2 体系的聚合物 $a[A-B]_nb$ 由两种结构单元 (A、B) 组成重复单元 (A-B)

结构单元数是重复单元数的两倍, 因此 $\overline{X}_n = 2n$

$$t = 0 \quad aAa : bBb = 1:1$$

$[N_a]_0$ is the initial (at $t = 0$) concentration of the functional group a .

能够组成结构单元的个数为 $[N_a]_0$



$[N_a]$ is the concentration of the functional group a at time t .

大分子的个数为 N_a

$$\overline{X_n} = \frac{\text{结构单元总数}}{\text{大分子个数}} = \frac{[N_a]_0}{[N_a]} \quad \because p = \frac{[N_a]_0 - [N_a]}{[N_a]_0} \quad \therefore \overline{X_n} = \frac{1}{1-p}$$

r: 官能团的化学计量比 $r = \frac{[N_a]_0}{[N_b]_0} < 1$

$[N_a]_0, [N_b]_0$: the numbers of functional group a and b

$$\overline{X_n} = \frac{\text{结构单元总数}}{\text{大分子个数}} = \frac{[N_a]_0}{[N_a]} \rightarrow \text{变为求平均值}$$

The total number of structural units

$$\text{结构单元总数} = ([N_a]_0 + [N_b]_0)/2 = ([N_a]_0 + [N_a]_0/r)/2 = \{[N_a]_0(1 + 1/r)\}/2$$

The total number of polymer chains

$$\text{大分子个数} = (N_a + N_b)/2 = \{[N_a]_0(1 - p) + [N_b]_0(1 - rp)\}/2$$

$$\overline{X_n} = \frac{[N_a]_0(1 + 1/r)/2}{\{[N_a]_0(1 - p) + [N_b]_0(1 - rp)\}/2} = \frac{1 + r}{1 + r - 2rp}$$



$$\overline{X}_n = \frac{1+r}{1+r-2rp}$$

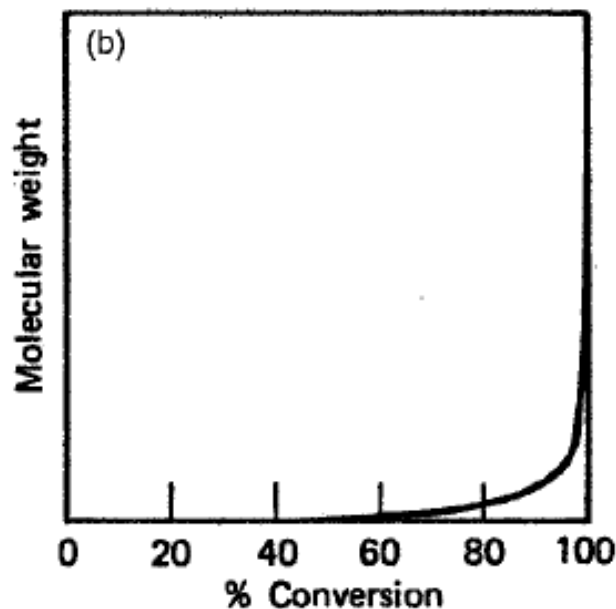
Carothers' equation

If $r=1$

The two bifunctional monomers are present in stoichiometric amounts

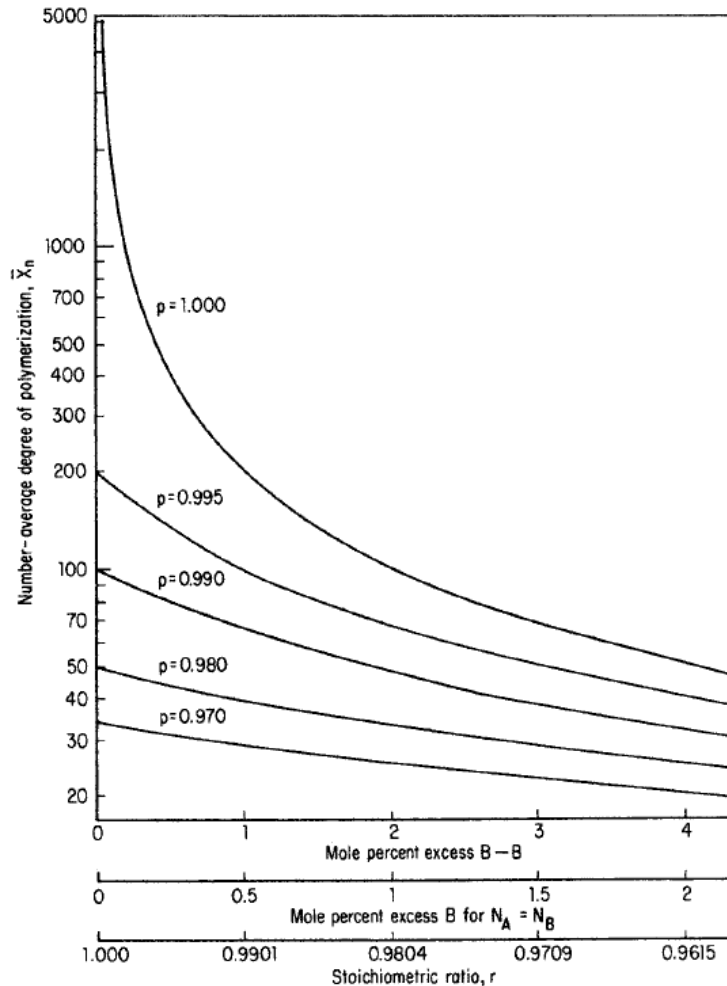


$$\overline{X}_n = \frac{1}{1-p}$$





If $r < 1$



$$\text{If } r = \frac{1000}{1001} \quad p = 100\%$$

$$\bar{X}_n = 2001$$

$$\text{If } r = \frac{100}{101} \quad p = 100\%$$

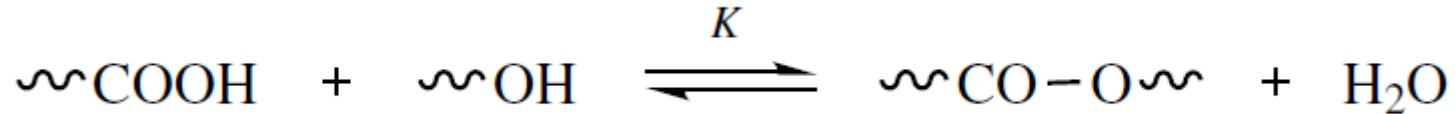
$$\bar{X}_n = 201$$



$\bar{X}_n$ decrease significantly

It is clear that step polymerizations will almost always be carried out to at least 98% reaction, since a degree of polymerization of at least approximately 50–100 is usually required for a useful polymer. Higher conversions and the appropriate stoichiometric ratio are required to obtain higher degrees of polymerization.

To increase the M_w in the process of polyester production, the byproduct (water) is required to be removed as much as possible from the polymerization system. Why? Can you explain it in your own words or show an example with simulation data.



$$K = \frac{[\text{COO}][\text{H}_2\text{O}]}{[\text{COOH}][\text{OH}]} = \frac{p[\text{H}_2\text{O}]}{[M]_0(1-p)^2}$$

$[M]_0$: the initial hydroxyl group or carboxyl group concentrations

p_e : the extent of reaction at equilibrium

$p_e[M]_0$: The concentration of ester groups $[\text{COO}]$ at equilibrium

$$\therefore \overline{X}_n = \frac{1}{1-p} \quad \Rightarrow \quad [\text{H}_2\text{O}] = \frac{K[M]_0}{\overline{X}_n(\overline{X}_n - 1)}$$

$(\overline{X}_n - 1)$ is close to $\overline{X}_n$ for large values of $\overline{X}_n$

$$\overline{X}_n = \sqrt{\frac{K[M]_0}{[\text{H}_2\text{O}]}}$$

To increase the $\overline{X}_n$ in the process, the byproduct (water) is required to be removed as much as possible from the polymerization system.

In order to get a polymer with a high molecular weight through step-growth polymerization

- An extremely high conversion of functional groups
- Accurate stoichiometry
- Remove the small by-products



$$\overline{X}_n \rightarrow \infty$$

In a system containing equivalent numbers of A and B groups:

$$f_{\text{avg}} = \frac{\sum N_i f_i}{\sum N_i}$$

N_i : the number of molecules of monomer i with functionality f_i ,

$\sum N_i$: the summations are over all the monomers present in the system.

$$p = \frac{2(N_0 - N)}{N_0 f_{\text{avg}}} \quad \overline{X}_n = \frac{N_0}{N} \quad \Rightarrow \quad \overline{X}_n = \frac{2}{2 - p f_{\text{avg}}} \quad \Rightarrow \quad p = \frac{2}{f_{\text{avg}}} - \frac{2}{\overline{X}_n f_{\text{avg}}}$$

N_0 : the number of monomer molecules present initially;

$N_0 f_{\text{avg}}$: the total number of functional groups;

N : the number of molecules after reaction has occurred;

$2(N_0 - N)$: the number of functional groups that have reacted.

$$\overline{X}_n \rightarrow \infty$$

$$p_c = \frac{2}{f_{\text{avg}}}$$

This equation can be used to calculate the extent of reaction required to reach the onset of gelation in a mixture of reacting monomers from its average functionality.

Functionalities > 2 & Nonstoichiometric Reactant Mixtures

Ternary mixture: N_A moles of A_{f_A} , N_C moles of A_{f_C} ($f_C > 2$), N_B moles of B_{f_B} . A and C have same functional groups (such as A); Number of A functional group is smaller than that of B functional group. (i.e. B groups are in excess)

$$f_{\text{avg}} = \frac{2(N_A f_A + N_C f_C)}{N_A + N_C + N_B} \quad (\text{Only calculate the reacting groups})$$

The extent of polymerization depends on the deficient reactant. The excess of other reactant results in lowering of functionality.

5.7 Tradition Synthesis

5.7.1 Step Growth Polymerization

5.7.2 Chain Polymerization

- Free Radical Polymerization
- Ionic polymerization
- Coordination polymerization

5.8 Advanced Synthesis

- living polymerization
- NMP
- ATRP
- RAFT



5.7.2 Chain Polymerization

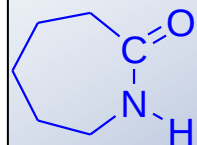
1. Chain-growth polymerization: One monomer addition at a time, with an active site.
2. During reaction, high- M_w polymer and monomer coexist—unlike step-growth polymer, don't need to drive reaction to 99.9% conversion to obtain high M_w polymer.

Monomers for chain polymerization

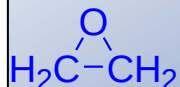
$\text{CH}_2=\text{CHX}$ Various olefins and vinyl monomers

$\text{CH}_2=\text{CX}-\text{CH}=\text{CH}_2$ Various Dienes

$\text{HC}\equiv\text{CR}$ Acetylene



Caprolactam



Ethylene oxide

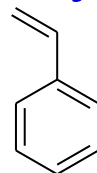
Types of chain polymerizations:

1. Free-radical polymerization.
2. Cationic polymerization.
3. Anionic polymerization.
4. Coordination polymerization.
5. Ring-open polymerization.



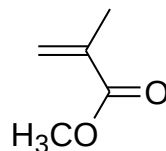
Monomers for free-radical polymerization

Styrene-conjugated unsaturation;
radical stabilized by π -delocation



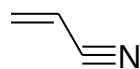
makes polystyrene, PS,

Methyl methacrylate (甲基丙烯酸甲酯)-conjugated; works generally for acrylates and methacrylates



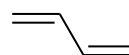
makes poly(methyl methacrylate), PMMA; Lucite(透明树脂), plexiglas (树脂玻璃); acrylates and methacrylates are also basis of latex paints (乳胶漆)

Acrylonitrile (丙烯腈)-conjugated;
typically copolymerized with styrene



makes poly(acrylonitrile), PAN: orlon (腈纶); acrylic fibers

1,3-butadiene-conjugated



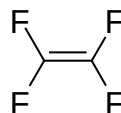
makes polybutadiens, PB: synthetic rubber

Vinyl chloride-not conjugated but polar; Cl provides radical stability



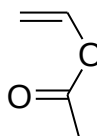
makes poly(vinyl chloride), PVC: Plastic pipes; tygon tubing

Tetrafluoroethylene-somewhat polar



makes poly(tetrafluoro ethylene), PTFE: Teflon.

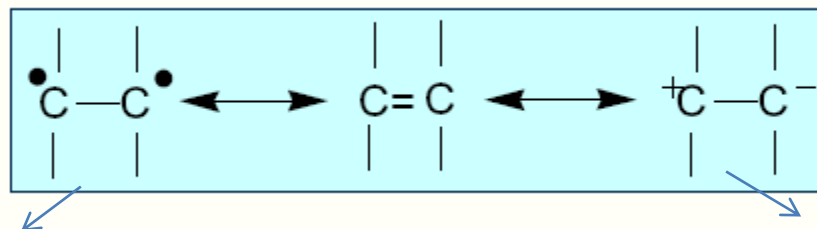
Vinyl Acetate(醋酸乙烯)-conjugated by ester groups.



makes polyvinyl alcohol.

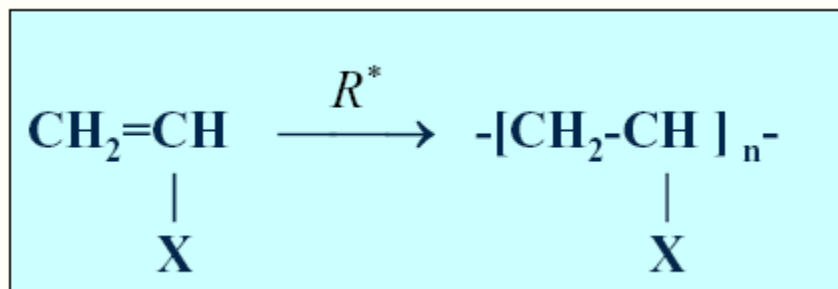


Effect of substituted groups on the polymerization mechanism

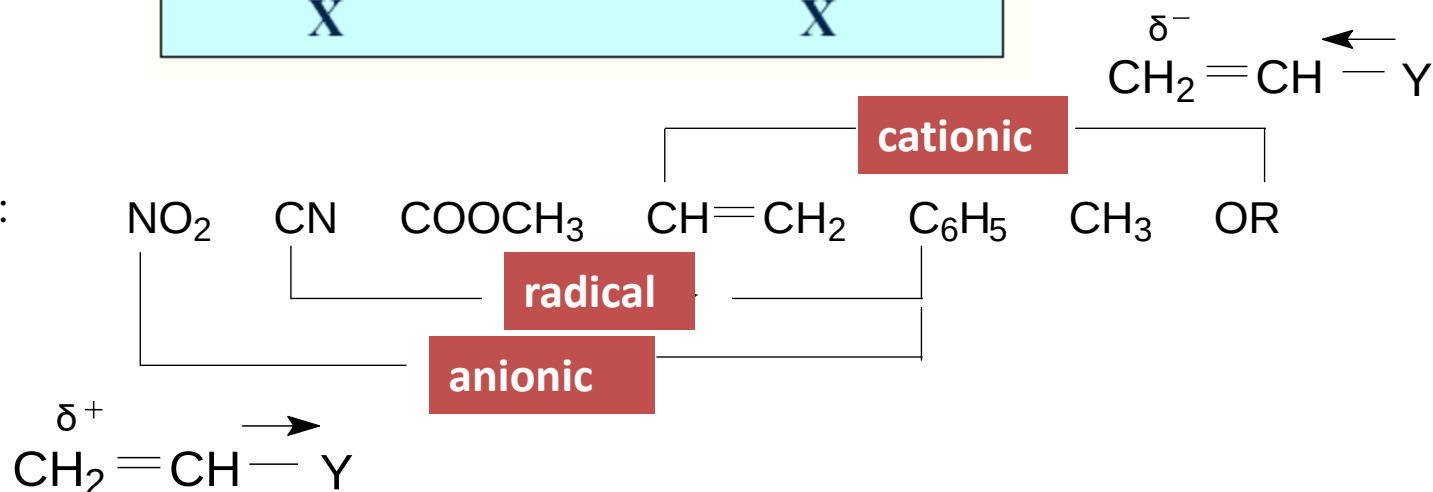


Free-radical polymerization

Anionic or cationic polymerization



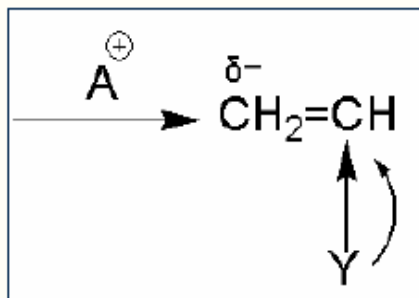
取代基 X:



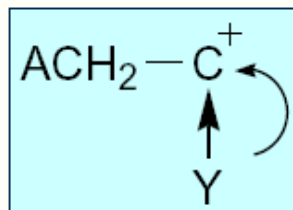


Monomers not amenable to free-radical polymerization

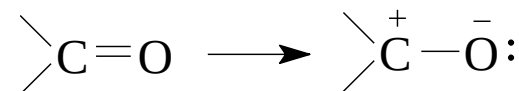
a Electron-donating Substituent



Cationic polymerization

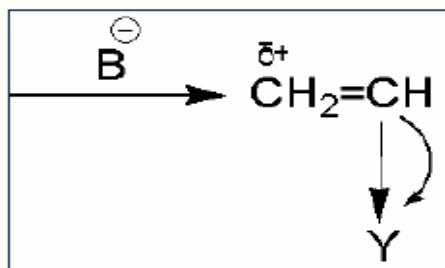


c

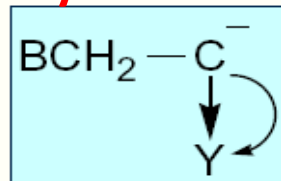


Cationic or anionic polymerization

b Electron-withdrawing Substituent



Anionic polymerization

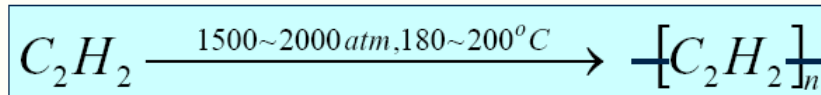


d

Propylene ($\text{CH}_2=\text{CH}-\text{CH}_3$)-nor
do higher α -olefins, or
distributed olefins (e.g.
isobutylene ($\text{CH}_2=\text{C}(\text{CH}_3)_2$)
Polymerizable by coordination
polymerization or cationic
polymerization

Monomers amenable to free-radical polymerization only at high P:

e





Chain Polymerizations of olefins – types of the active site

Monomer	Free-Radical	Anionic	Cationic	coordination
$\text{CH}_2=\text{CH}_2$	High P			+
$\text{CH}_2=\text{CHCH}_3$				+
$\text{CH}_2=\text{CHCH}_2\text{CH}_3$				+
$\text{CH}_2=\text{C}(\text{CH}_3)_2$			+	+
$\text{CH}_2=\text{CHCH}=\text{CH}_2$	+	+	+	+
$\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}=\text{CH}_2$	+	+	+	+
$\text{CH}_2=\text{CClCH}=\text{CH}_2$	+			
$\text{CH}_2=\text{CHC}_6\text{H}_5$	+	+	+	+
$\text{CH}_2=\text{C}(\text{CH}_3)\text{C}_6\text{H}_5$	+	+	+	+
$\text{CH}_2=\text{CHCl}$	+			
$\text{CH}_2=\text{CCl}_2$	+	+		



Chain Polymerizations of olefins – types of the active site

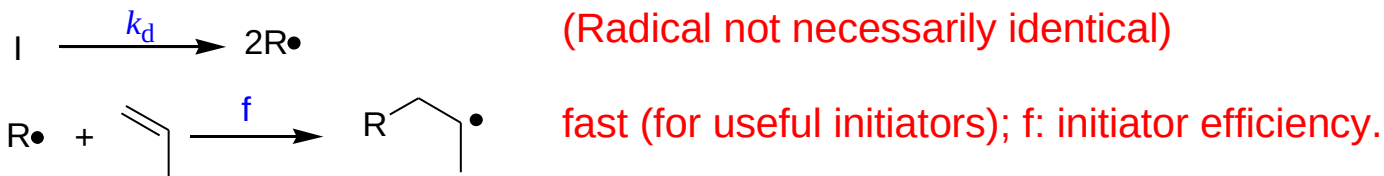
Continued table

Monomer	Free-Radical	Anionic	Cationic	coordination
$\text{CH}_2=\text{CHF}$	+			
$\text{CF}_2=\text{CF}_2$	+			
$\text{CF}_2=\text{CFCF}_3$	+			
$\text{CH}_2=\text{CF}_2$	+			
$\text{CH}_2=\text{CH}-\text{OR}$			+	
$\text{CH}_2=\text{CHOCOCH}_3$	+			
$\text{CH}_2=\text{CHCOOCH}_3$	+	+		+
$\text{CH}=\text{C}(\text{CH}_3)\text{COOCH}_3$	+	+		+
$\text{CH}_2=\text{CHCN}$	+	+		
$\text{CH}_2=\text{C}(\text{CN})_2$		+		
$\text{CH}_2=\text{CHNO}_2$		+		

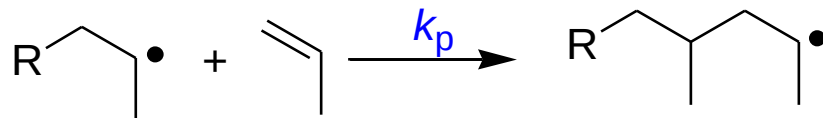


Steps in free-radical polymerization

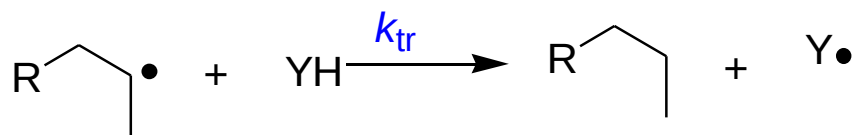
1. Initiation: homolytic cleavage(均裂) to generate radical (through heat or UV), then first addition of monomer:



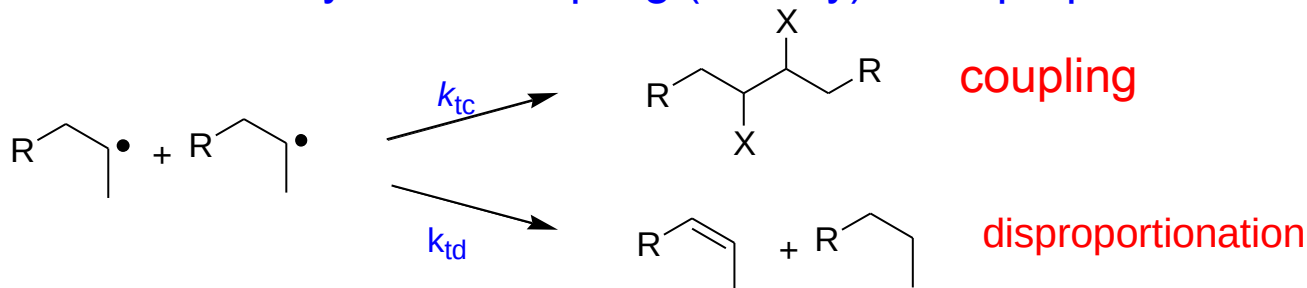
2. Propagation: repeat many times to form high- M_w polymer; “chain growth”:



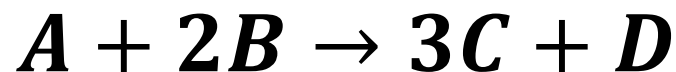
3. Transfer: often happens, whether desired or not:.



4. Termination-by either coupling (usually) or disproportionation (sometimes):



Reminding



The volume/surface area of the reaction system is constant.
At some instant:

$$v = \frac{d[D]}{dt} = \frac{1}{3} \frac{d[C]}{dt} = -\frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt}$$

v : the rate of reaction

Rate laws and rate constants

A rate law shows the relationship between the reaction rate and the concentrations of reactants.

For a general reaction occurring at a fixed temperature:



The rate law is:

$$v = k[A]^m [B]^n \dots$$

k: rate constant

- Specific for a given reaction at a given temperature: Dependent on T
- Does not change as the reaction proceeds.
- Be found by experiment.

m, n: reaction orders

First-order reactions



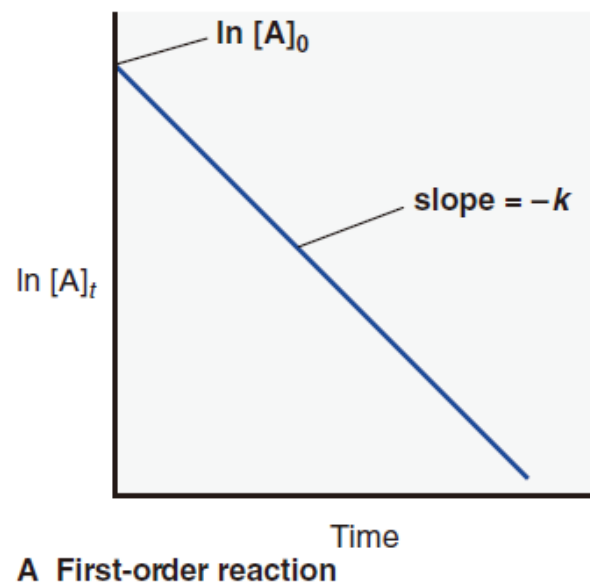
$$\frac{d[A]}{dt} = -k[A] \quad \Longrightarrow \quad \frac{d[A]}{[A]} = -k dt$$

$$\int_{[A]_0}^{[A]} \frac{d[A]}{[A]} = -k \int_0^t dt$$

$[A]_0$ is the initial concentration of A (at $t = 0$)

$$\ln \left(\frac{[A]}{[A]_0} \right) = -kt \quad [A] = [A]_0 e^{-kt}$$

$$\ln[A] = \ln[A]_0 - kt$$



Half-life

- Half-life is defined as the time required for one-half of a reactant to react.
- Because $[A]$ at $t_{1/2}$ is one-half of the original $[A]$, $[A]_t = 0.5 [A]_0$.

For the first-order reaction:

$$kt_{1/2} = -\ln\left(\frac{\frac{1}{2}[A]_0}{[A]_0}\right) = -\ln\frac{1}{2} = \ln 2$$

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

For a first-order reaction, the half-life of a reactant is independent of its initial concentration.

Arrhenius equation

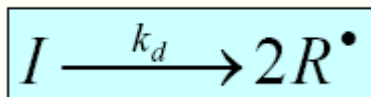
$$k = Ae^{-E_a/RT}$$

E_a : activation energy

A: frequency factor



Initiation

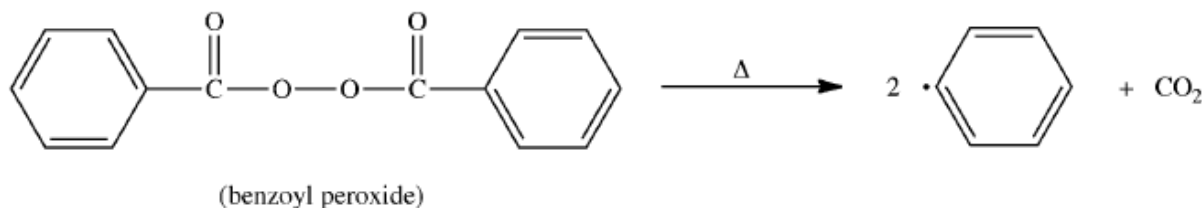


$$R_d \equiv -\frac{d[I]}{dt} = k_d[I]$$

I : initiator

$R^\bullet$: primary radical

R_d, K_d : the rate and its rate constant of initiator decomposition



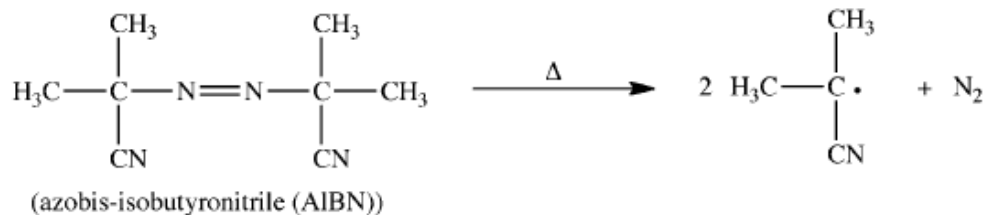
Arrhenius equation:

$$k_d = A_d e^{-E_d/RT}$$

$$E_d = 10^5 \sim 150 \text{ kJ/mol}$$

$$k_d = 10^{-4} \sim 10^{-6} \text{ s}^{-1}$$

$$A_d = 10^{13} \sim 10^{14}$$



Generation of free radicals from a variety of initiators

Initiator half-life

$t_{1/2}$ defined as the time for the concentration of initiator to decrease to one half its original value.

$$R_d \equiv -\frac{d[I]}{dt} = k_d[I]$$



$$-k_d t = \ln \frac{[I]}{[I]_0} = -\ln 2$$



$$t_{1/2} = \frac{\ln 2}{k_d} = \frac{0.693}{k_d}$$

❖ $t_{1/2} < 1\text{hr}$:

High-reactivity initiator, e.g. DCPD(1hr)

❖ $1\text{hr} < t_{1/2} < 6\text{hr}$:

Middle-reactivity initiator, e.g. BPPD (2.1 hr)

❖ $t_{1/2} > 6\text{hr}$:

Low-reactivity initiator, e.g. AIBN (16 hr)

- Table from Odian, “principles of polymerization,” 4th ed.

TABLE 3-2 Half-Lives of Initiators^{a,b}

Initiator	Half-Life at					
	50°C	70°C	85°C	100°C	130°C	175°C
Azobisisobutyronitrile	74 h	4.8 h	—	7.2 min	—	—
Benzoyl peroxide	—	7.3 h	1.4 h	20 min	—	—
Acetyl peroxide	158 h	8.1 h	1.1 h	—	—	—
<i>t</i> -Butyl peracetate	—	—	88 h	13 h	18 min	—
Cumyl peroxide	—	—	—	—	1.7 h	—
<i>t</i> -Butyl peroxide	—	—	—	218 h	6.4 h	—
<i>t</i> -Butyl hydroperoxide	—	—	—	338 h	—	4.81 h

^a Data from Brandrup and Immergut [1989], Brandrup et al. [1999], and Huyser [1970].

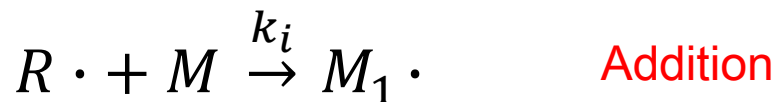
^b Half-life ($t_{1/2}$) values are for benzene or toluene solutions of the initiators.

The second part of initiation:

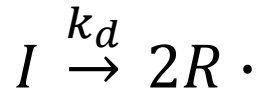
Combination of the primary free radical and the first monomer



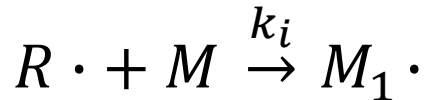
✓ The total Initiation



Assume decomposition is the rate limiting step. i.e. $k_i \gg k_d$



Decomposition



Addition

Assume decomposition is the rate limiting step. i.e. $k_d \gg k_i$

Then we should only have to consider k_d

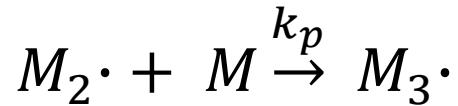
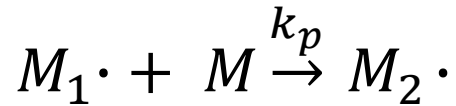
$$-\frac{d[I]}{dt} = \frac{1}{2} \frac{d[M_1 \cdot]}{dt} = k_d[I]$$

Only a fraction f of radicals initiate chain growth

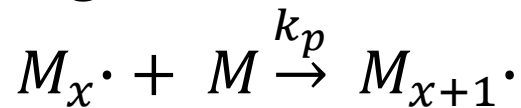
$$r_i = \frac{d[M_1 \cdot]}{dt} = 2fk_d[I]$$



- Propagation



In general



Assumption: reactivity is independent of chain length

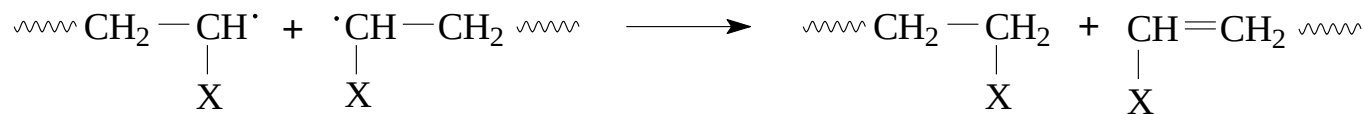
$$r_p = -\frac{d[M]}{dt} = k_p[M][M\cdot]$$



• Termination



Coupling



Disproportionation

The overall rate at which free-radicals are consumed is given by

$$-\frac{1}{2} \frac{d[M\cdot]}{dt} = k_t [M\cdot][M\cdot]$$

$$r_t = -\frac{d[M\cdot]}{dt} = 2k_t [M\cdot][M\cdot]$$

Both reaction are second order

where, $k_t = k_{tc} + k_{td}$ is the overall rate constant for termination



$$r_i = \frac{d[M_1 \cdot]}{dt} = 2fk_d[I]$$

Initiation

$$r_p = -\frac{d[M]}{dt} = k_p[M][M \cdot]$$

Propagation

$$r_t = -\frac{d[M \cdot]}{dt} = 2k_t[M \cdot][M \cdot]$$

Termination

Problem: We do not know $[M \cdot]$

Solution: Assume a steady state concentration of transient species

$$[M \cdot] = \text{constant}$$

This means that radicals are consumed at the same rate as they are generated.

$$r_i = r_t$$



$$r_i = r_t \quad 2fk_d[I] = 2k_t[M \cdot]^2$$

$$[M \cdot] = \left[\frac{fk_d[I]}{k_t} \right]^{1/2}$$

Rate of propagation = Rate of polymerization

$$r_p = R_p \quad r_p = -\frac{d[M]}{dt} = k_p[M][M \cdot]$$

$$R_p = k_p \left[\frac{fk_d}{k_t} \right]^{1/2} [I]^{1/2} [M]$$

If we want to increase R_p , increase $[M]$ or $[I]$



Rate constants of elementary reactions

$$k_d = 10^{-4} \sim 10^{-6},$$

$$E_d = 105 \sim 150 \text{ kJ/mol},$$

$$k_p = 10^2 \sim 10^4,$$

$$E_p = 16 \sim 33 \text{ kJ/mol},$$

$$k_t = 10^6 \sim 10^8$$

$$E_t = 8 \sim 21 \text{ kJ/mol}$$

Rate of Initiation:

$$R_i = 10^{-8} \sim 10^{-10} \text{ mol. L}^{-1} \cdot \text{s}^{-1}$$

Rate of Propagation:

$$R_p = k_p [M][M\cdot] = 10^{-4} \sim 10^{-6} \text{ mol/L. s}$$

Rate of Termination:

$$R_t = k_t [M\cdot]^2 = 10^{-8} \sim 10^{-10} \text{ mol/L. s}$$



Total rate constants of radical polymerization

Arrhenius equation: $k = Ae^{-E/RT}$

$$R_p = k_p \left[\frac{fk_d}{k_t} \right]^{1/2} [f[I]]^{1/2} [M] = k [f[I]]^{1/2} [M]$$

$$k = k_p \left[\frac{k_d}{k_t} \right]^{1/2}$$

$$k = A_p \left(\frac{A_d}{A_t} \right)^{1/2} e^{-[(E_p - E_t/2) + E_d/2]/RT}$$

Total activity energy of radical polymerization:

$$E = (E_p - E_t/2) + E_d/2$$

In general,

$E_d \approx 125$ kJ/mol,

$E_p \approx 29$ kJ/mol,

$E_t \approx 17$ kJ/mol,

So total activity energy $E \approx 83$ kJ/mol.



**When $E > 0$, the total rate constant k increases with increasing T .
In general, k increases 1~1.5 times /per 10°C**

Case 1 : For a polymer, $T_1 = 90^\circ\text{C}$, $T_2 = 100^\circ\text{C}$, $E = 80 \text{ kJ/mol}$, $R = 8.314 \text{ J/mol} \cdot \text{K}$, please calculate k_2/k_1

Answer :

$$\frac{k_2}{k_1} = \frac{e^{-E/RT_2}}{e^{-E/RT_1}} = e^{\frac{E}{R}(\frac{1}{T_1} - \frac{1}{T_2})} = e^{\frac{80000}{8.314}(\frac{1}{363} - \frac{1}{373})} = e^{0.71} = 2$$

E_d dominates in the activity energy of radical polymerization

$$E = (E_p - E_t/2) + E_d/2$$

Comparing with increasing T , selecting a initiator with a lower E_d has more significant effect on the polymerization rate.

e.g. Oxidation –reduction system can keep a higher polymerization rate at a lower temperature.

Case 2 : The first system : $K_2S_2O_8$ $E_d = 140$ kJ/mol;
second : $K_2S_2O_8 - Fe^{2+}$ $E_d = 50$ kJ/mol,
 $E_p = 29$ kJ/mol, $E_t = 17$ kJ/mol, $T = 60^\circ C$,
 $R = 8.314$ kJ/mol.K , please calculate k_2/k_1 .

**Case 2: The first system system: $\text{K}_2\text{S}_2\text{O}_8$ $E_d = 140 \text{ kJ/mol}$;
second: $\text{K}_2\text{S}_2\text{O}_8\text{—F}_e^{2+}$ $E_d = 50 \text{ kJ/mol}$,
 $E_p = 29 \text{ kJ/mol}$, $E_t = 17 \text{ kJ/mol}$, $T = 60^\circ\text{C}$,
 $R = 8.314 \text{ kJ/mol}\cdot\text{K}$, please calculate k_2/k_1 .**

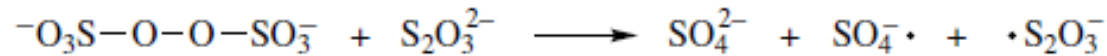
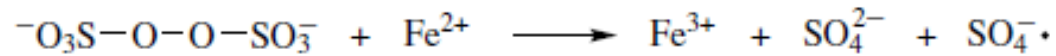
Answer:

$$E_1 = (E_p - \frac{E_t}{2}) + \frac{E_d}{2} = 29 - \frac{17}{2} + \frac{140}{2} = 90.5$$

$$E_2 = (E_p - \frac{E_t}{2}) + \frac{E_d}{2} = 29 - \frac{17}{2} + \frac{50}{2} = 45.5$$

$$\frac{k_2}{k_1} = e^{-(E_2 - E_1)/RT} = e^{(90.5 - 45.5) \times 1000 / (8.314 \times 333)} = e^{16.25} = 1.14 \times 10^7$$

Selecting a initiator with a lower E_d has more significant effect on the polymerization rate .



- Many oxidation–reduction reactions produce radicals that can be used to initiate polymerization. This type of initiation is referred to as redox initiation, redox catalysis, or redox activation.
- A prime advantage of redox initiation is that radical production occurs at reasonable rates over a very wide range of temperatures, depending on the particular redox system, including initiation at moderate temperatures of 0–50 °C and even lower. This allows a greater freedom of choice of the polymerization temperature than is possible with the thermal homolysis of initiators. Some redox polymerizations can be initiated photolytically as well as thermally.

$$[M \cdot] = \text{constant} \quad \Rightarrow \quad R_i = R_t$$

This means that radicals are consumed at the same rate as they are generated.

$$R_p = k_p \left[\frac{f k_d}{k_t} \right]^{1/2} [I]^{1/2} [M]$$

R_p : rate of polymerization

k_d : rate constant of initiation

k_p : rate constant of propagation

k_t : rate constant of termination

f : the fraction of radicals initiate chain growth

$[I]$: the concentration of initiator

$[M]$: the concentration of monomer



Define chain length

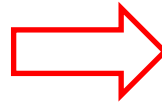
The kinetic chain length ν of a radical chain polymerization is defined as the average number of monomer molecules consumed (polymerized) per each radical, which initiates a polymer chain.

This quantity will obviously be given by the ratio of the polymerization rate to the initiation rate or to the termination rate.

$$\nu = \frac{R_p}{R_i} = \frac{R_p}{R_t}$$

$$R_p = -\frac{d[M]}{dt} = k_p[M][M\cdot]$$

$$R_t = -\frac{d[M\cdot]}{dt} = 2k_t[M\cdot][M\cdot]$$



$$\nu = \frac{k_p[M]}{2k_t[M\cdot]}$$

OR

$$R_p = R_p = -\frac{d[M]}{dt} = k_p[M][M\cdot] \quad \Rightarrow$$

$$\nu = \frac{k_p^2[M]^2}{2k_t R_p}$$



Consider a time period:

1. 100 chains are started;
2. 1,000,000 monomers are reacted in this time period

The kinetic chain length of these chains

$$v = \frac{1,000,000}{100} = 10,000$$

Coupling termination

$$\overline{X}_n = \frac{1,000,000}{100/2} = 20,000 = 2v$$

Disproportion termination

$$\overline{X}_n = \frac{1,000,000}{100} = 10,000 = v$$

The number-average degree of polymerization $\overline{X}_n$, defined as the average number of monomer molecules contained in a polymer molecule, is related to the kinetic chain length.

If the propagating radicals terminate by coupling, a dead polymer molecule is composed of two kinetic chain lengths.

$$\overline{X}_n = 2v$$

For termination by disproportionation the kinetic chain length is synonymous with the number-average degree of polymerization.

$$\overline{X}_n = v$$

b : the average number of initiator fragments per polymer molecule

$$1 < b < 2$$



- The fractions of propagating chains, a and $(1 - a)$, respectively, which undergo termination by coupling and disproportionation can be related to b , the average number of initiator fragments per polymer molecule.
- For a reaction system composed of n propagating chains, coupling yields an initiator fragments and $an/2$ polymer molecules, while disproportionation yields $(1 - a)n$ initiator fragments and $(1 - a)n$ polymer molecules.
- The average number of initiator fragments per polymer molecule, defined as the total initiator fragments divided by the total number of polymer molecules, is given as

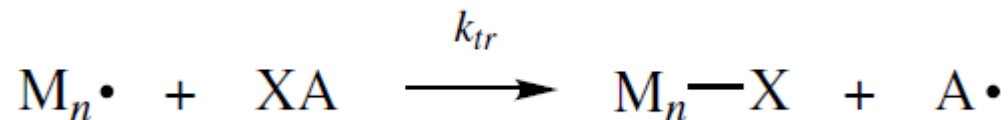
$$b = \frac{an + (1 - a)n}{an/2 + (1 - a)n} = \frac{2}{2 - a}$$

The general relationship between the degree of polymerization is:

$$\bar{X}_n = bv = \frac{2v}{(2 - a)}$$

Effect of Chain Transfer

- In many polymerization systems the polymer molecular weight is observed to be lower than predicted on the basis of the experimentally observed extents of termination by coupling and disproportionation.
- This effect is due to the premature termination of a growing polymer by the transfer of a hydrogen or other atom or species to it from some compound present in the system—the monomer, initiator, or solvent, as the case may be.
- These radical displacement reactions are termed chain-transfer reactions.



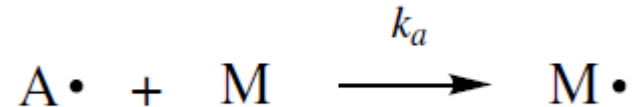
$$R_{tr} = k_{tr}[M\cdot][XA]$$

k_{tr} is the chain-transfer rate constant



Reinitiation

Chain transfer results in the production of a new radical A, which then reinitiates polymerization.



k_a is the chain-transfer rate constant

- Chain transfer is a chain-breaking reaction; it results in a decrease in the size of the propagating polymer chain.
- The effect of chain transfer on the polymerization rate is dependent on whether the rate of reinitiation is comparable to that of the original propagating radical.



TABLE 3-3 Effect of Chain Transfer on R_p and $\bar{X}_n$

Case	Relative Rate Constants for Transfer, Propagation, and Reinitiation		Type of Effect	Effect on R_p	Effect on $\bar{X}_n$
	$k_p \gg k_{tr}$	$k_a \simeq k_p$			
1	$k_p \gg k_{tr}$	$k_a \simeq k_p$	Normal chain transfer	None	Decrease
2	$k_p \ll k_{tr}$	$k_a \simeq k_p$	Telomerization	None	Large decrease
3	$k_p \gg k_{tr}$	$k_a < k_p$	Retardation	Decrease	Decrease
4	$k_p \ll k_{tr}$	$k_a < k_p$	Degradative chain transfer	Large decrease	Large decrease

Chain transfer is important in that it may alter the molecular weight of the polymer product in an undesirable manner. On the other hand, controlled chain transfer may be employed to advantage in the control of molecular weight at a specified level.



$$\bar{X}_n = bv = \frac{2v}{(2-a)}$$



$$\bar{X}_n = \frac{R_p}{(2-a)(R_t/2)}$$

$$v = \frac{R_p}{R_i} = \frac{R_p}{R_t}$$

The degree of polymerization must now be redefined as the polymerization rate divided by the sum of the rates of all chain-breaking reactions (i.e., the normal termination mode plus all chain-transfer reactions).

$k_{tr,M}$: chain-transfer rate constant for that transfer to monomers;

$k_{tr,I}$: chain-transfer rate constant for that transfer to initiators;

$k_{tr,S}$: chain-transfer rate constant for that transfer to compound S, which referred to as a chain transfer agent;



The number average degree of polymerization follows:

$$\bar{X}_n = \frac{R_p}{(2 - a)(R_t/2) + k_{tr,M}[M\cdot][M] + k_{tr,S}[M\cdot][S] + k_{tr,I}[M\cdot][I]}$$

$$R_p = -\frac{d[M]}{dt} = k_p[M][M\cdot]$$

A chain – transfer constant C for a substance is defined as the ratio of the rate constant k_{tr} for the chain transfer of a propagating radical with that substance to the rate constant k_p for propagation of the radical.

The chain-transfer constants for monomer, chain-transfer agent, and initiator are then given by

$$C_M = \frac{k_{tr,M}}{k_p} \quad C_S = \frac{k_{tr,S}}{k_p} \quad C_I = \frac{k_{tr,I}}{k_p}$$



$$\frac{1}{\bar{X}_n} = \frac{(2 - a)R_i}{2R_p} + C_M + C_S \frac{[S]}{[M]} + C_I \frac{[I]}{[M]}$$

Mayo Equation

Monomer transfer constants C_M 

XVII



TABLE 3-4 Monomer Chain-Transfer Constants^a

Monomer	$C_M \times 10^4$
Acrylamide	0.6, 0.12 ^b
Acrylonitrile	0.26–0.3
Ethylene	0.4–4.2
Methyl acrylate	0.036–0.325
Methyl methacrylate	0.07–0.10
Styrene	0.60–0.92 ^c
Vinyl acetate	1.75–2.8
Vinyl chloride	10.8–12.8

^a All C_M are for 60°C except where otherwise noted.

^b Value at 25°C.

^c Data from Braks and Huang [1978]; all other data from Brandrup et al. [1999].

- The monomer chain-transfer constants are generally small for most monomers—in the range 10^{-5} to 10^{-4} .
- Chain transfer to monomer places the upper limit to the polymer molecular weight that can be obtained, assuming the absence of all other transfer reactions.

For example: The C_M value of vinyl chloride is sufficiently high that the maximum number-average molecular weight that can be achieved is 50,000–100,000. This limit is still reasonable from the practical viewpoint—PVC of this molecular weight is a very useful product.



Initiator transfer constants C_I

Different initiators have varying transfer constants and many peroxides have significant chain-transfer constants.

TABLE 3-5 Initiator Chain-Transfer Constants^{a,b}

Initiator	C_I for Polymerization of		
	Styrene	Methyl Methacrylate	Acrylamide
2,2-Azobisisobutyronitrile ^c	0.091–0.14	0	—
<i>t</i> -Butyl peroxide	0.0003–0.00086	—	—
Lauroyl peroxide (70°C)	0.024	—	—
Benzoyl peroxide	0.048–0.10	0–0.02	—
<i>t</i> -Butyl hydroperoxide	0.035	—	—
Cumyl hydroperoxide	0.063	0.33	—
Persulfate (40°C)	—	—	0.0028

^a Data from Brandrup et al. [1999].

^b All C_I values are for 60°C except where otherwise noted.

^c Ayrey and Haynes [1974]; Braks and Huang [1978].

The decrease of polymer size due to chain transfer to initiator is much less than indicated from the C_I values because it is the quantity $C_I[I]/[M]$, which affects $\overline{X}_n$. The initiator concentrations are quite low (10^{-4} – 10^{-2} M) in polymerization, and the ratio $[I]/[M]$ is typically in the range 10^{-3} – 10^{-5} .



Transfer to chain-transfer agent C_S

TABLE 3-6 Transfer Constants for Chain-Transfer Agents^{a,b}

Transfer Agent	$C_S \times 10^4$ for Polymerization of	
	Styrene	Vinyl Acetate
Benzene	0.023	1.2
Cyclohexane	0.031	7.0
Heptane	0.42	17.0 (50°C)
Toluene	0.125	21.6
Ethylbenzene	0.67	55.2
Isopropylbenzene	0.82	89.9
<i>t</i> -Butylbenzene	0.06	3.6
<i>n</i> -Butyl chloride	0.04	10
<i>n</i> -Butyl bromide	0.06	50
2-Chlorobutane	1.2	—
Acetone	4.1	11.7
Acetic acid	2.0 (40°C)	1.1
<i>n</i> -Butyl alcohol	1.6	20
Ethyl ether	5.6	45.3
Chloroform	3.4	150
<i>n</i> -Butyl iodide	1.85	800
Butylamine	7.0	—
Triethylamine	7.1	370
Di- <i>n</i> -butyl sulfide	22	260
Di- <i>n</i> -butyl disulfide	24	10,000
Carbon tetrachloride	110	10,700
Carbon tetrabromide	22,000	28,700 (70°C)
1-Butanethiol	210,000	480,000

^aData from Brandrup et al. [1999] and Eastmond [1976a,b,c].

^bAll values are for 60°C unless otherwise noted.

- Aliphatic hydrocarbons such as cyclohexane with strong $C-H$ bonds show low transfer constants.
- Benzene has an even lower C_S value because of the stronger $C-H$ bonds.
- The presence of the weaker benzylic hydrogens in toluene, ethylbenzene, and isopropylbenzene leads to higher C_S values relative to benzene.



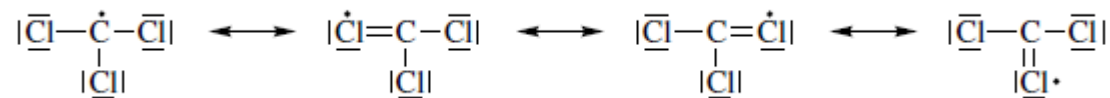
The benzylic CH is abstracted easily because the resultant radical is resonance-stabilized.

- The transfer constant for t-butylbenzene is low, since there are no benzylic $C-H$ bonds present.
- Acids, carbonyl compounds, ethers, amines, and alcohols have transfer constants higher than those of aliphatic hydrocarbons, corresponding to $C-H$ breakage and stabilization of the radical by an adjacent O, N, or carbonyl group.
- The weak $S-S$ bond leads to high transfer constants for disulfides.





- The high C_S values for carbon tetrachloride and carbon tetrabromide are due to the weak carbon–halogen bonds. These bonds are especially weak because of the excellent stabilization of the trihalocarbon radicals formed by resonance involving the halogen free pairs of electrons:



Two evidences for C_S

(I)

- The absolute value of the transfer constant for any one compound may change very significantly depending on the monomer being polymerized.

The order of radical reactivity is:

Vinyl chloride(氯乙烯) > vinyl acetate(醋酸乙烯酯) > acrylonitrile(丙烯腈) > methyl acrylate(丙烯酸甲酯) > methyl methacrylate(甲基丙烯酸甲酯) > styrene(苯乙烯) > 1,3-butadiene(丁二烯).



(II)

- The order of reactivity of a series of transfer agents usually remains the same irrespective of the monomer when the transfer agents are relatively neutral in polarity.
- It is apparent that the electron-rich (electron-donor) transfer agent triethylamine 三乙胺 has enhanced reactivity with the electron-poor (electron-acceptor) monomers, especially acrylonitrile. The electron-poor (electron-acceptor) transfer agent carbon tetrachloride 四氯化碳 has enhanced reactivity with the electron-rich (electron-donor) monomers vinyl acetate 醋酸乙烯酯 and styrene 苯乙烯 and markedly lowered reactivity with the electron-poor monomers.

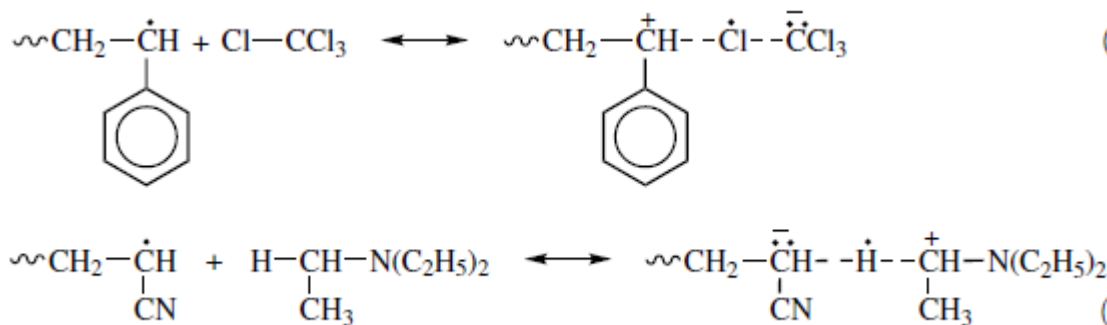


TABLE 3-7 Polar Effects in Chain Transfer^{a,b,c}

Monomer	Chain-Transfer Agent			
	CCl ₄		(C ₂ H ₅) ₃ N	
	$C_S \times 10^4$	k_{tr}	$C_S \times 10^4$	k_{tr}
Vinyl acetate	10,700	2400	370	85
Acrylonitrile	0.85	0.17	3800	760
Methyl acrylate	1.25 ^c	0.26 ^c	400	84
Methyl methacrylate	2.4	0.12	1900	98
Styrene	110	1.8	7.1	0.12

^a C_S values from Brandrup et al. [1999], Eastmond [1976a,b,c] and are for 60°C unless otherwise noted.

^b k_{tr} values were calculated from Eq. 3-107 using k_p values from Table 3-11.

^c C_S value is at 80°C. The k_{tr} was calculated using the k_p value for 60°C.



Practical Utility of Mayo Equation

$$\frac{1}{\bar{X}_n} = \frac{(2 - a)R_i}{2R_p} + C_M + C_S \frac{[S]}{[M]} + C_I \frac{[I]}{[M]}$$

Mayo Equation

The use of the Mayo equation allows one to quantitatively determine the effects of normal termination and transfer to monomer, initiator, and solvent on the molecular weight of the product and then adjust the reaction parameters to obtain the desired molecular weight.



Case 1. A reaction system that yields a molecular weight lower than needed.

- The initiation rate would be lowered if the low molecular weight is due to an excessively high initiation rate.
- If the initiator or solvent are too active in transfer, a less active initiator or solvent would be chosen.
- If the monomer itself is too active in transfer, it sets the upper limit of the molecular weight that can be achieved as long as all other factors are optimized.



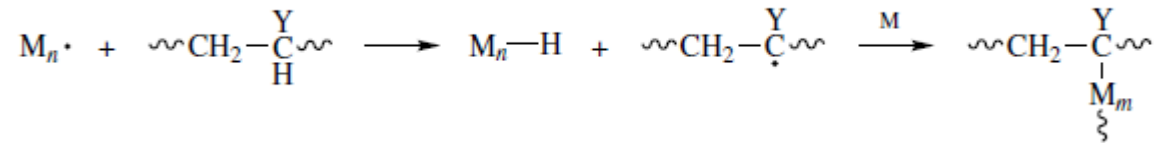
Case 2. A reaction system that yields too high a polymer molecular weight.

- The simplest method of lowering molecular weight is to add a chain-transfer agent. When used in this manner, transfer agents are called regulators or modifiers.
- Transfer agents with transfer constants of one or greater are especially useful, since they can be used in low concentrations.
- For example, low-molecular-weight acrylic ester polymers have been used as plasticizers.



Chain transfer to polymer

Transfer to polymer results in the formation of a radical site on a polymer chain. The polymerization of monomer at this site leads to the production of a branched polymer.

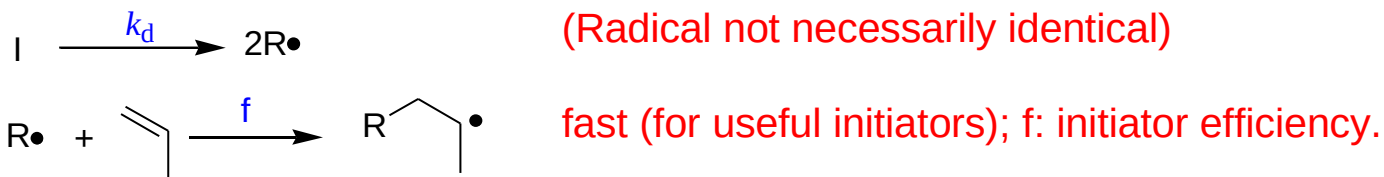


- Transfer to polymer does not necessarily lead to a decrease in the overall degree of polymerization.
- As branching drastically decreases the crystallinity of a polymer, the effect of chain transfer to polymer plays a very significant role in determining the physical properties and the ultimate applications of a polymer.

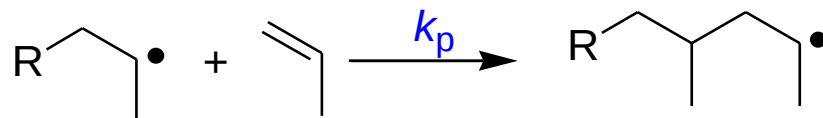


Four steps in free-radical polymerization

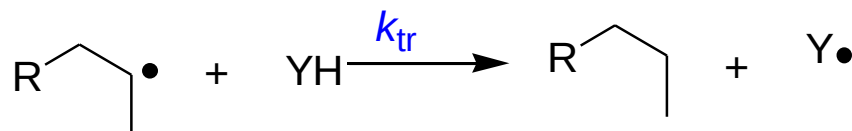
1. Initiation: homolytic cleavage(均裂) to generate radical (through heat or UV), then first addition of monomer:



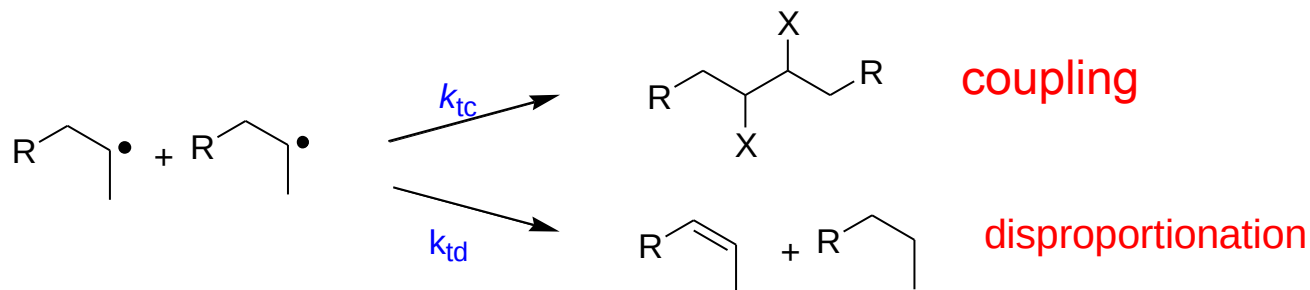
2. Propagation: repeat many times to form high- M_w polymer; “chain growth”:



3. Transfer: often happens, whether desired or not:.



4. Termination-by either coupling (usually) or disproportionation (sometimes):



- 聚合速率和哪些因素有关

$$[M \cdot] = \text{constant} \quad \Rightarrow \quad R_i = R_t$$

This means that radicals are consumed at the same rate as they are generated.

$$R_p = k_p \left[\frac{k_d}{k_t} \right]^{1/2} [fI]^{1/2} [M] = k [fI]^{1/2} [M]$$

R_p : rate of polymerization

k_d : rate constant of initiation

k_p : rate constant of propagation

k_t : rate constant of termination

f : the fraction of radicals initiate chain growth

$[I]$: the concentration of initiator

$[M]$: the concentration of monomer

$$k = k_p \left[\frac{k_d}{k_t} \right]^{1/2}$$

Case 1 : For a polymer, $T_1 = 90^\circ\text{C}$, $T_2 = 100^\circ\text{C}$, $E = 80 \text{ kJ/mol}$, $R = 8.314 \text{ J/mol} \cdot \text{K}$, please calculate k_2/k_1

Arrhenius equation: $k = Ae^{-E/RT}$

$$R_p = k_p \left[\frac{fk_d}{k_t} \right]^{1/2} [I]^{1/2} [M] = k [fI]^{1/2} [M]$$

$$k = Ae^{-E/RT}$$

$$\frac{k_2}{k_1} = \frac{e^{-E/RT_2}}{e^{-E/RT_1}} = e^{\frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)} = e^{\frac{80000}{8.314} \left(\frac{1}{363} - \frac{1}{373} \right)} = e^{0.71} = 2$$

Case 2 : The first system : $\text{K}_2\text{S}_2\text{O}_8$ $E_d = 140 \text{ kJ/mol}$;
 second : $\text{K}_2\text{S}_2\text{O}_8\text{—F}_e^{2+}$ $E_d = 50 \text{ kJ/mol}$,
 $E_p = 29 \text{ kJ/mol}$, $E_t = 17 \text{ kJ/mol}$, $T = 60^\circ\text{C}$,
 $R = 8.314 \text{ kJ/mol.K}$, please calculate k_2/k_1 .

Arrhenius equation: $k = Ae^{-E/RT}$

$$R_p = k_p \left[\frac{fk_d}{k_t} \right]^{1/2} [I]^{1/2} [M] = k [fI]^{1/2} [M]$$

$$k = k_p \left[\frac{k_d}{k_t} \right]^{1/2}$$

$$k = A_p \left(\frac{A_d}{A_t} \right)^{1/2} e^{-[(E_p - E_t/2) + E_d/2]/RT}$$

$$E_1 = (E_p - \frac{E_t}{2}) + \frac{E_d}{2} = 29 - \frac{17}{2} + \frac{140}{2} = 90.5$$

$$E_2 = (E_p - \frac{E_t}{2}) + \frac{E_d}{2} = 29 - \frac{17}{2} + \frac{50}{2} = 45.5$$

$$\frac{k_2}{k_1} = e^{-(E_2 - E_1)/RT} = e^{(90.5 - 45.5) \times 1000 / 8.314 \times 333} = e^{16.25} = 1.14 \times 10^7$$



- 影响分子量的因素

The kinetic chain length ν of a radical chain polymerization is defined as the average number of monomer molecules consumed (polymerized) per each radical, which initiates a polymer chain.

$$\nu = \frac{R_p}{R_i} = \frac{R_p}{R_t}$$

Coupling termination

$$\overline{X_n} = 2\nu$$

Disproportion termination

$$\overline{X_n} = \nu$$



The fractions of propagating chains, a and $(1 - a)$, respectively, which undergo termination by coupling and disproportionation

$$\overline{X}_n = \frac{R_p}{(2 - a)(R_i/2)}$$

只考虑了termination step



既termination又考虑chain transfer对聚合度的影响

$$\frac{1}{\overline{X}_n} = \frac{(2 - a)R_i}{2R_p} + C_M + C_S \frac{[S]}{[M]} + C_I \frac{[I]}{[M]}$$

Mayo Equation

chain - transfer constant C

$$C_M = \frac{k_{tr,M}}{k_p} \quad C_S = \frac{k_{tr,S}}{k_p} \quad C_I = \frac{k_{tr,I}}{k_p}$$

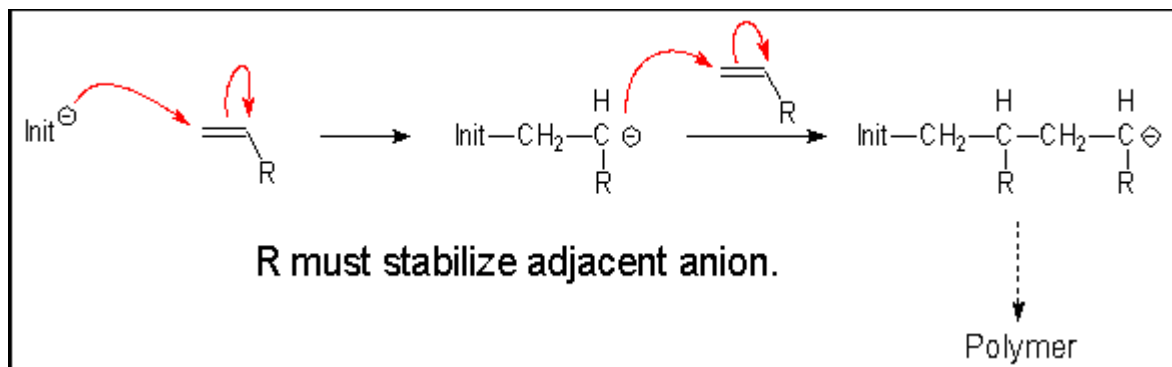


Ionic polymerization vs free-radical polymerization

- Ionic polymerization is more complex than free-radical polymerization.
- Whereas free radical polymerization is non-specific, the type of ionic polymerization procedure and catalysts depend on the nature of the substituent (R) on the vinyl (ethenyl) monomer.
- Anionic initiation, requires the R group to be electron withdrawing in order to promote the formation of a stable carbanion.
- Cationic initiation is therefore usually limited to the polymerization of monomers where the R group is electron-donating. This helps stabilize the delocalization of the positive charge through the π orbitals of the double bond.

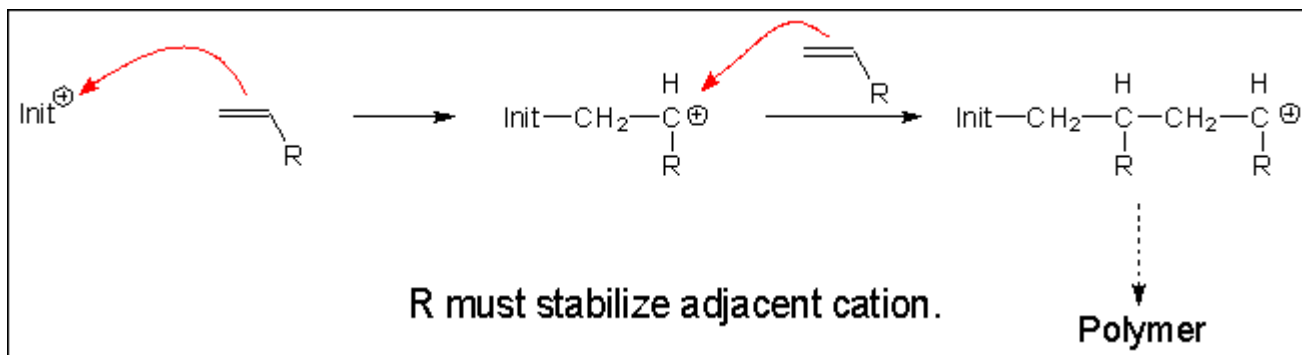


Anionic Polymerization -the growing chain end bears a *negative* charge



The mechanism of anionic polymerization is a kind of repetitive conjugate addition reaction .
(the "Michael reaction" in organic chemistry)

Cationic Polymerization -the growing chain end bears a *positive* charge



The mechanism of cationic polymerization is a kind of repetitive alkylation reaction.



5.7 Tradition Synthesis

5.7.1 Step Growth Polymerization

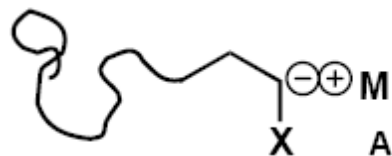
5.7.2 Chain Polymerization

- Free Radical Polymerization
- Ionic polymerization
 - Anionic polymerization
 - Cationic polymerization
 - Ring-opening polymerization
- Coordination polymerization



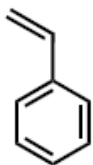
□ Anionic polymerization

Active site: an carbanion

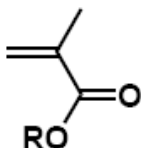
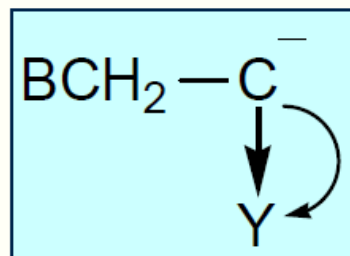
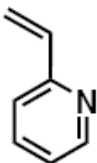
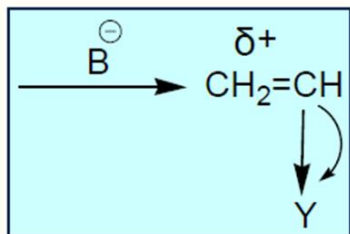
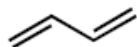


Alkali metal counterion; commonly Li

Some monomers :



- Monomers with electron-withdrawing substituents
- Electron-withdrawing substituents facilitate the attack of an anionic species by decreasing the electronic density on the double bond.
- They stabilize the propagating anionic species by resonance



BUT:

- monomer selection VERY limited because carbanions are very reactive:
- Must rigorously exclude water, O₂ from polymerization;
- Use only hydrocarbons, ethers, 3° amines (NR₃) as solvents.



Anionic polymerization of monomers with suitable initiators

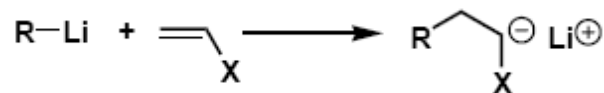
引发剂	单	体	Q	e	σ
SrR ₂ ; CaR ₂	a	α -甲基苯乙烯			-0.161
Na; NaR		苯乙烯	1	0	0.009
Li; LiR		丁二烯	1.28	0	
RMgX	b	丙烯酸甲酯	1.33	1.41	0.385
t-ROLi		甲基丙烯酸甲酯	1.92	1.20	
ROK	c	丙烯腈	2.70	1.91	0.663
ROLi		甲基丙烯腈	3.33	1.74	
强碱		甲基乙烯酮	3.45	1.51	0.502
吡啶	d	硝基乙烯			0.778
NR ₃		亚甲基丙二酸二乙酯			
弱碱		α -氰基丙烯酸乙酯			1.150
ROR		α -氰基-2,4-己二烯酸乙酯			
H ₂ O		偏二氰基乙烯	100		1.256

Match between initiators and monomers

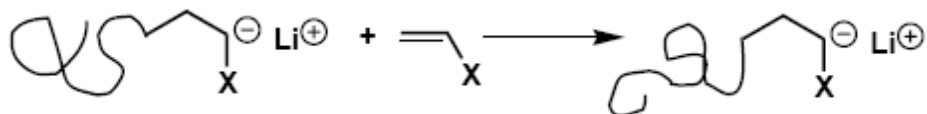


General features of anionic polymerization

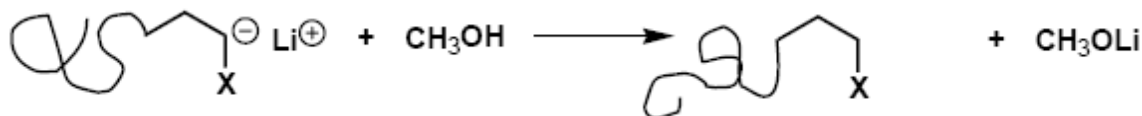
• **Initiation:** most common initiators are butyl-lithium isomers (*n*-, *sec*-, *tert*-) – these are soluble in hydrocarbon solvents (alkanes, aromatics) but are unstable in air (easily handled in a glovebox)



• **Propagation:** just as with radical polymerization – simple chain growth polymerization

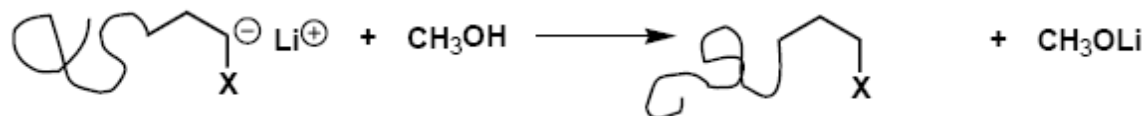


• **Termination:** NONE; if exclude groups which will react with alkyl-lithium; “living polymerization”





- If initiation is fast relative to propagation (typically the case with butyllithium as initiator), get a very narrow molecular weight distribution – “horse race polymerization” where all chains out of the “gate” at the same time, all run at the same speed.
- In practice, can routinely get $M_w/M_n < 1.05$
- When polymerization completed, can easily terminate by adding alcohol, e.g., MeOH:





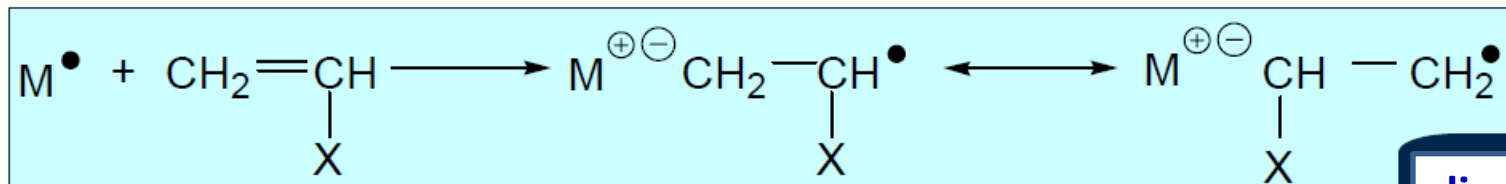
Anionic polymerization---initiation

The active initiators:

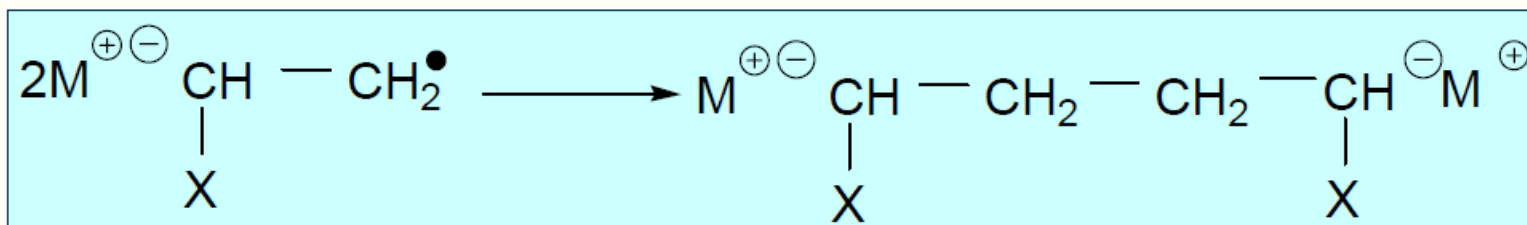
- Alkali metals: electron-transfer initiators
- Organometallic compounds: anionic initiators



- Electron-transfer initiation



dicarbanion



The reaction involves:

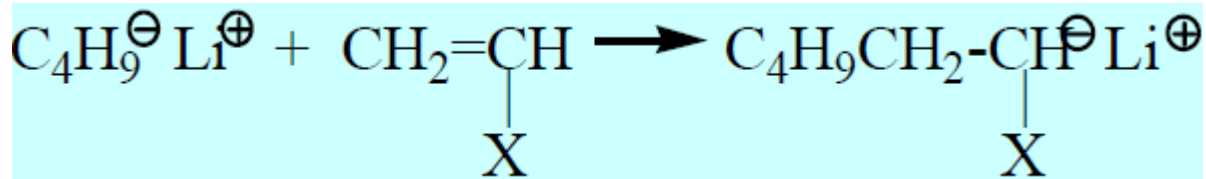
- ✓ The radical-anions are formed by transferring of an electron from the alkali metal to monomers;
- ✓ The radical-anions dimerize to form the dicarbanion.

- Anionic initiators

(Nucleophilic initiators, *principles of polymerization*, by George Odian, ISBN 047127400-3)

Metal alkyl compounds

Alkyl lithium compounds are probably the most useful of these initiators





Feature of anionic polymerization :

- Quick initiation
- Slow propagation
- No termination

However, it can be terminated by Impurities and transfer agents.



Anionic polymerization---propagation



- Propagation occurs with complete consumption of monomer to form living polymers;
- The propagating anionic centers remain intact because transfer of proton or other positive species from the solvent does not occur;
- Polymerization without termination is apparent in several different ways:
 - If a reaction is high purified, the color of carbanions does not change at 100% conversion.
 - Propagation proceeds by adding more monomers after 100% conversion, and M_w is increased.



Anionic polymerization---termination

Termination by

- Impurities: moisture, oxygen, carbon dioxide, etc
- Deliberately added transfer agents: water, alcohol, etc



Most anion polymerizations are carried out in an inert atmospheres with rigorously cleaned reagents and glassware since trace impurities leads to termination

Kinetics of anionic polymerization

Polymerization rate:

$$R_p = k_p [M^-][M] = k_p [C][M]$$

Where $[M^-]$ is the total concentration of all types of living anionic propagating centers (free ions and ion pairs)= $[C]$ is the concentration of initiators



$$\overline{X}_n = \frac{p[M]_0}{[M^-]_0} = \frac{p[M]_0}{[C]}$$

If $p = 100\%$

$$\overline{X}_n = \frac{[M]_0}{[C]}$$

At **100%** conversion, the number-average degree of polymerization for a living anion polymerization is **the ratio of the concentration of reacted monomers to the concentration of living ends.**

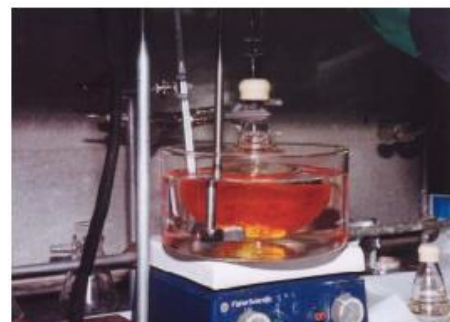


Usual procedures in many labs and production facilities:

1. Premix monomer and solvent, as “clean” as possible.
2. Add initiator dropwise until “color” of living ends persists, thus killing any trace impurities

3. Now add calculated charge of initiator:

$$[I]_0 = \frac{[M]_0}{X_n}$$



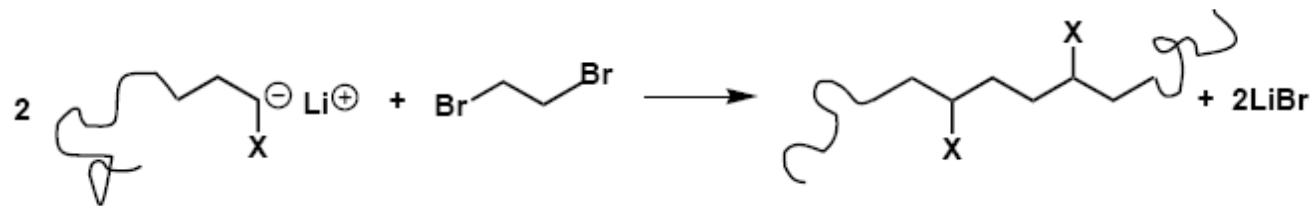
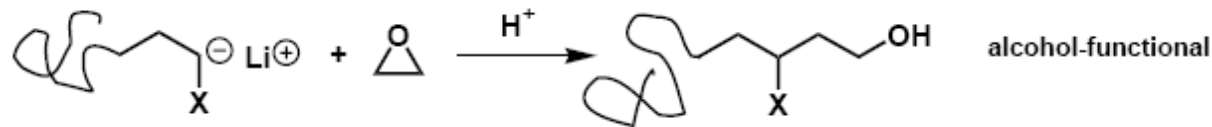
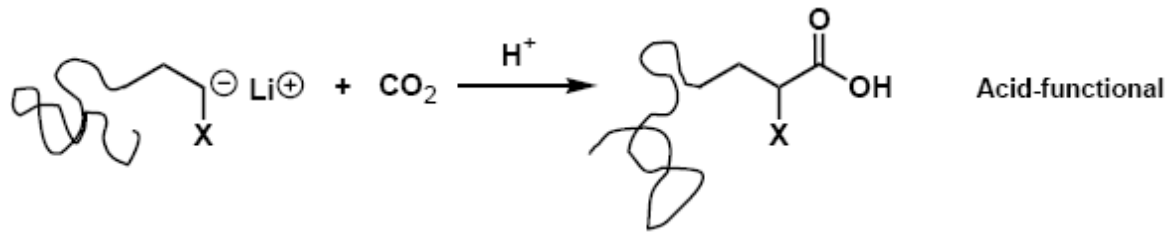
Styryl-lithium anion is bright orange in color

4. Allow to polymerize to essentially completion ($p \rightarrow 1$). For alkyl-lithium in hydrocarbon solvents, rates are modest, so often run polymerizations at slightly elevated temperatures ($\sim 60^\circ\text{C}$). In ethers (THF), rates very fast, so must run polymerization cold (typically dry ice temperatures, -78°C).



5. Terminate polymerization. Besides “straight” termination by alcohol, can terminate with more interesting agents:

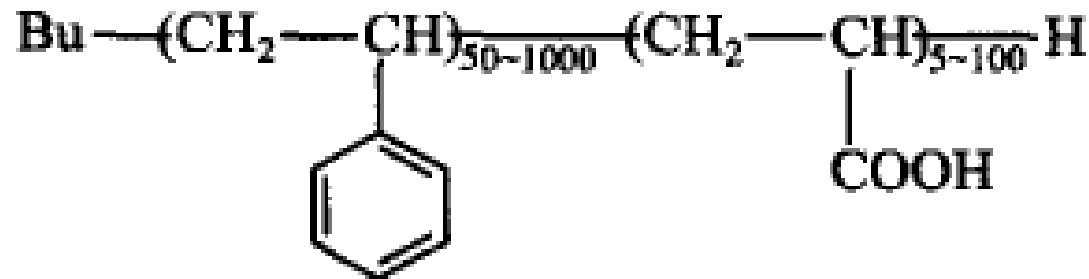
Functional termination:





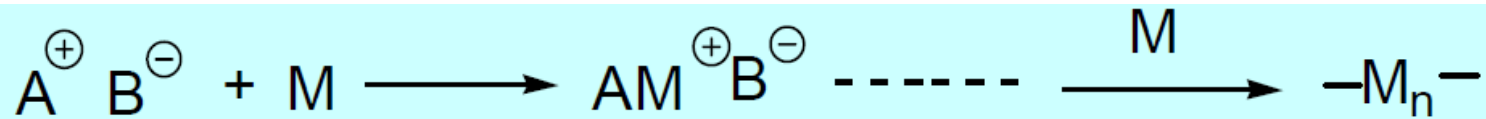
Application of anion polymerization

- Preparation of telechelic polymers远螯聚合物 or functional polymers
- Preparation of block copolymers





□ Cationic polymerization

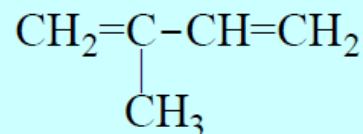
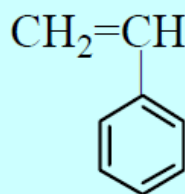
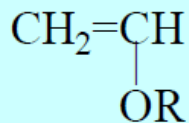
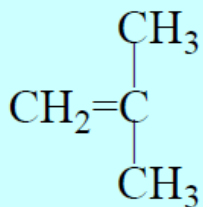


$A^{\oplus}$: Active center of cation

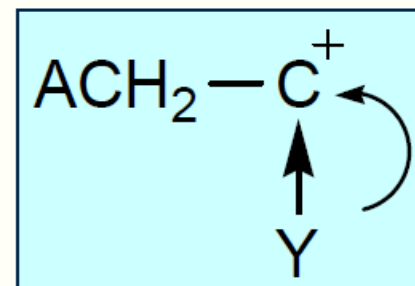
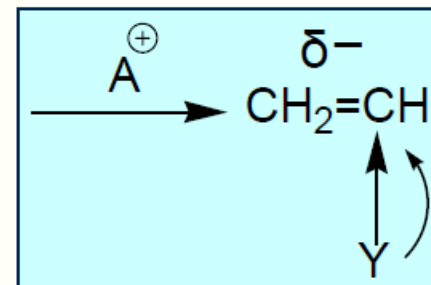
$B^{\ominus}$: Counterion

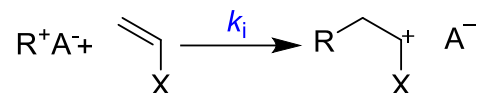
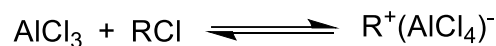
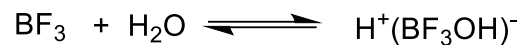
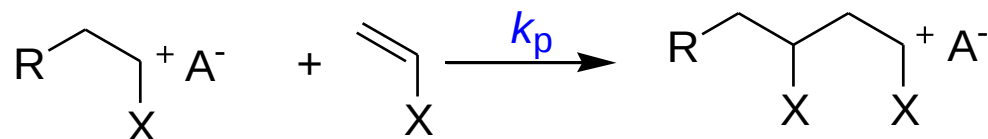
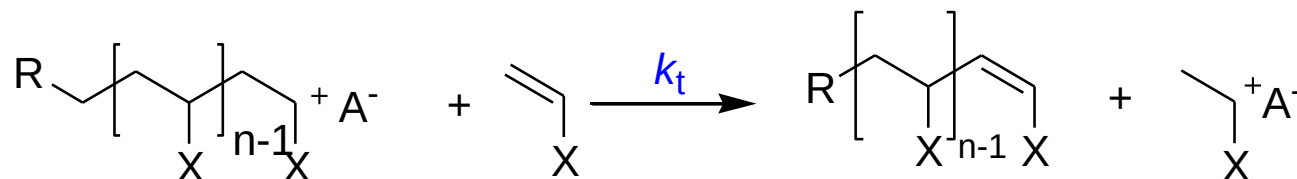
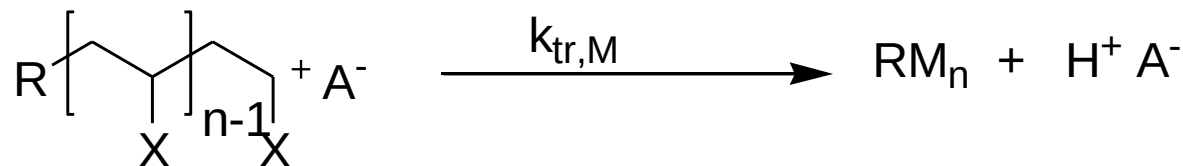


Monomers with electron-donating substituents



- Electron-donating substituents facilitate the attack of an cationic species by increasing the electronic density on the double bond.
- They stabilize the propagating cationic species by resonance



**1. Initiation:****2. Propagation: repeat many times to form high- M_w polymer; “chain growth”:****3. Transfer to monomer****4. Termination-by lose one proton from the end group.**

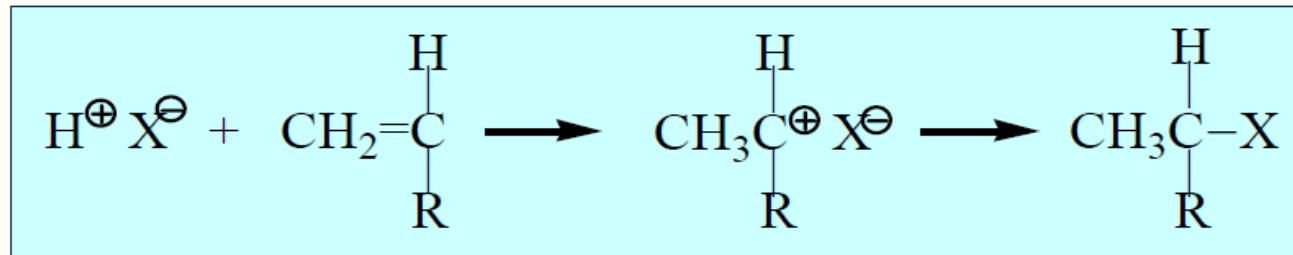


Cationic polymerization-initiation

Initiators:

- Protonic acids

such as H_2SO_4 , H_3PO_4 , HClO_4



- Lewis acids

such as AlCl_3 、 BF_3 、 SnCl_4 、 ZnCl_2

Co-initiator:

- Proton donor (RH) : such as H_2O 、 ROH 、 HX 、 RCOOH ;
- carbocation(RX) : such as RX 、 RCOX 、 $(\text{RCO})_2\text{O}$



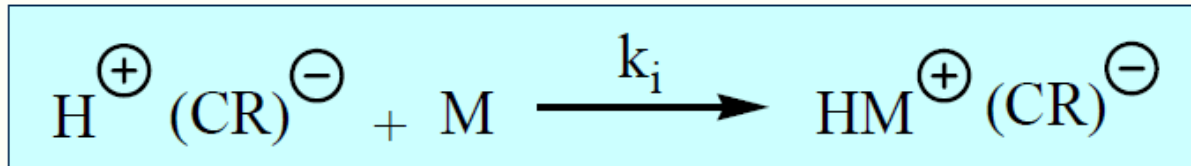
An initiating system : initiator – co-initiator complex



C: initiator (Lewis)

RH: co-initiator

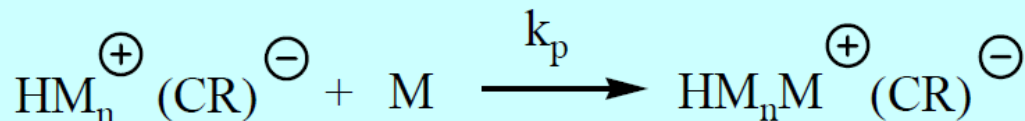
$H^{\oplus} (CR)^{\ominus}$ **initiator – co-initiator complex**



The complex donates a proton to monomer (M)



Cationic polymerization-propagation

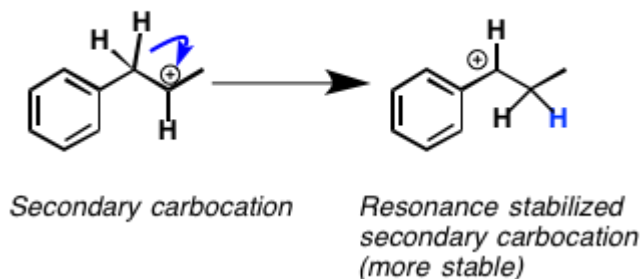


This addition proceeds by insertion of monomer between the carbocation and its negative counterion.

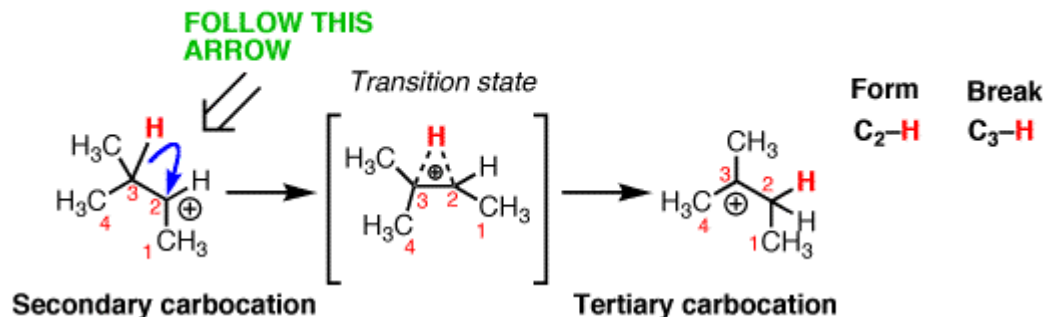
Isomerization polymerization

The propagation reaction can be complicated in some cases by the occurrence of intramolecular rearrangement due to 1,2-hydride ion or 1,2-methide shifts.

Another favorable hydride shift



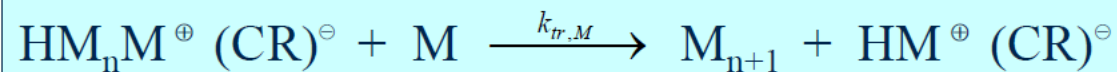
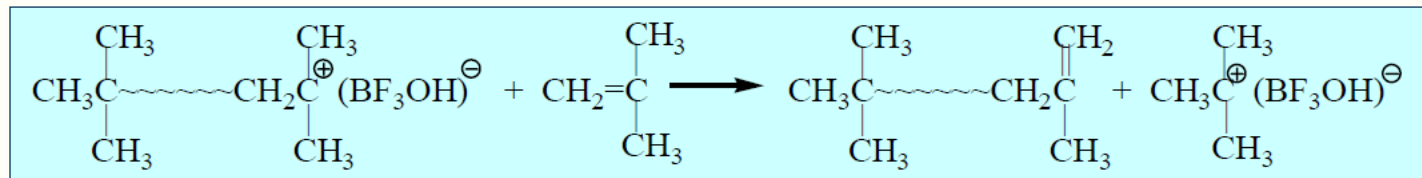
One pathway rearrangements can occur through is a *hydride shift*.





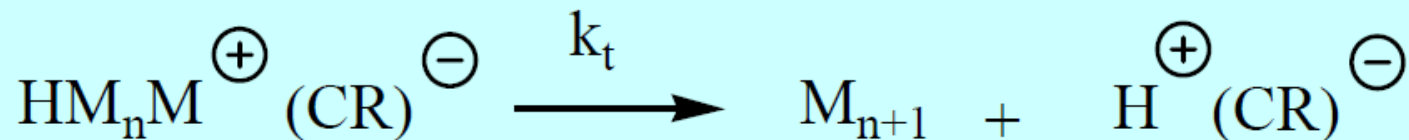
Cationic polymerization-chain transfer and termination

Chain transfer to monomers

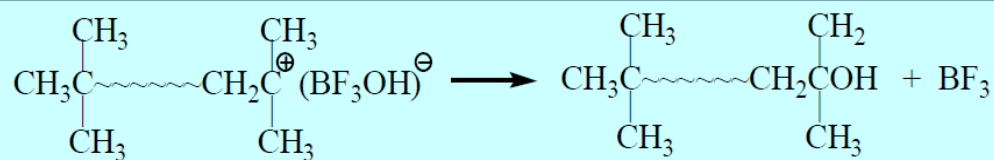
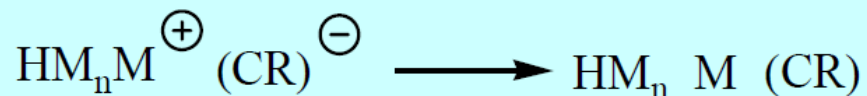




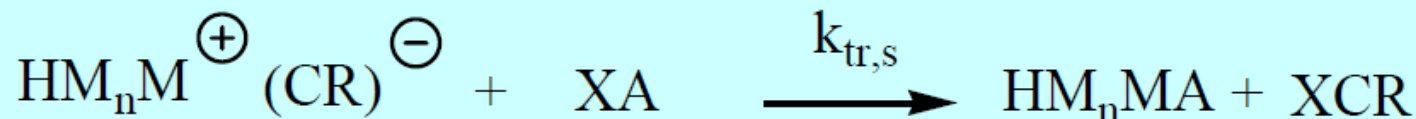
- Spontaneous termination



- Combination with counterion or an anion fragment from the counterion



- Adding chain-transfer agents, such as water, alcohol, acids, etc





Feature of cationic polymerization :

- Quick initiation
- Quick propagation
- Easy chain transfer
- Hard termination



Cationic vinyl polymerization is plagued by numerous side reactions, which lead to chain transfer mostly.

- Difficult to achieve high M_w (*initiator can give rise to many separate chains because of chain transfer).

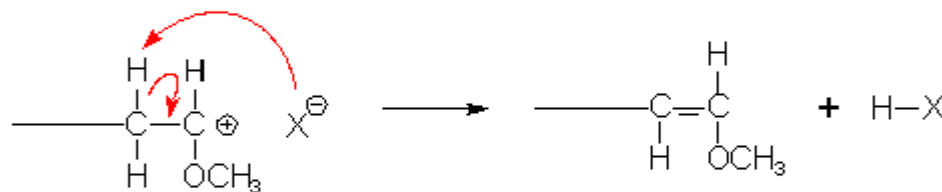
- These side reactions can be minimized.

But! not eliminated by running the reaction at low temperature.

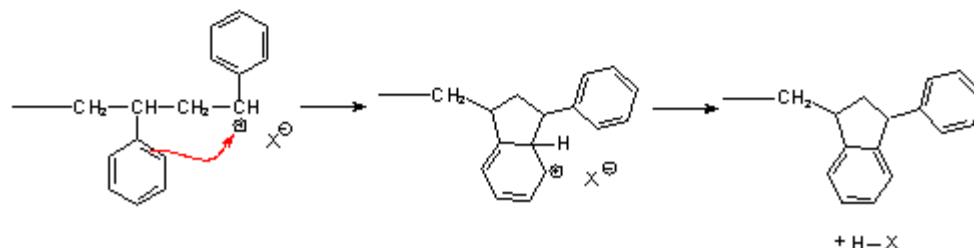
- Chain transfer to monomer.



- Spontaneous ejection of HX:

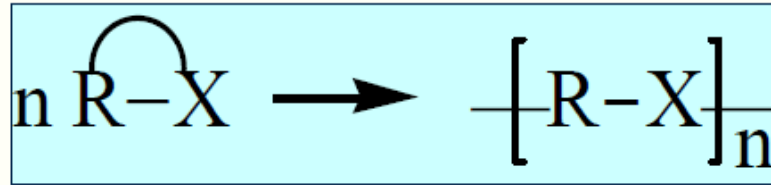


- Backbiting.





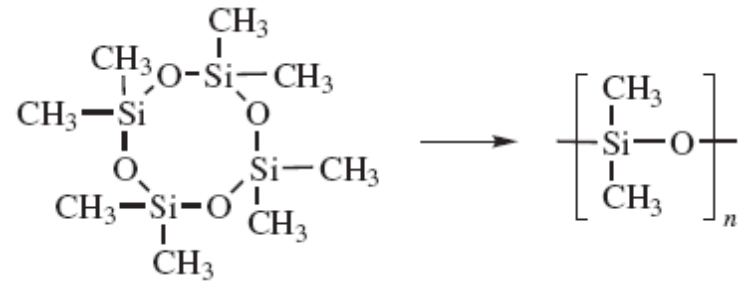
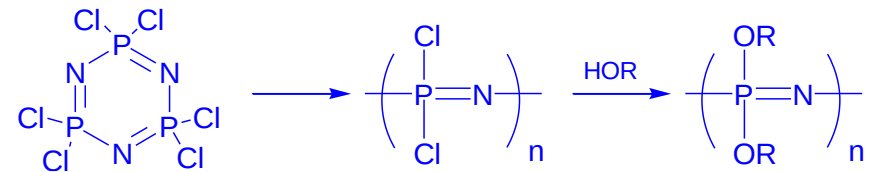
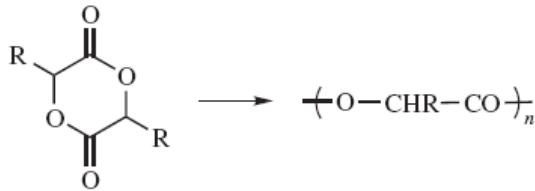
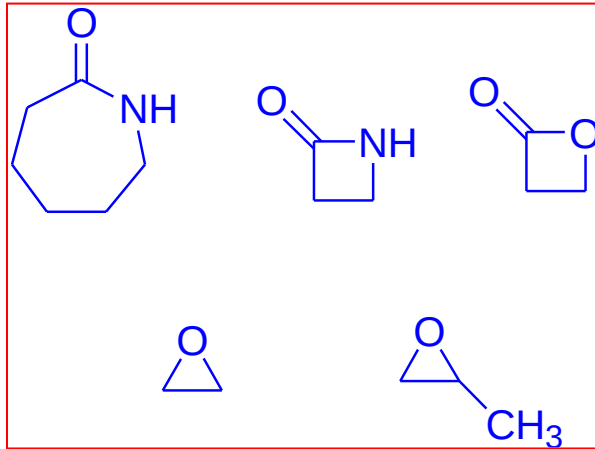
□ Ring-opening polymerization



- Cyclic monomers.
- Driving force: release of ring strain of the monomer molecules
- The reaction resembles chain polymerization in the sense that it proceeds by the addition of monomer to growing chain molecule.
- However, the rate of initiation and subsequent addition reaction are similar.
- Monomers: cyclic ethers, cyclic esters (lactones 内酯), cyclic amides (lactams 内酰胺).

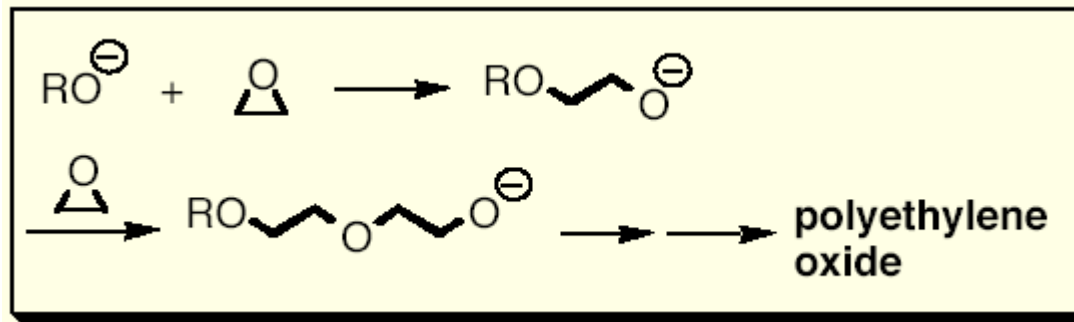


Some examples of cyclic monomers



Biocompatible and biodegradable

polymerization of ethylene oxide initiated by alkoxide





- Unstable ring or the ring which cannot be cyclized easily are very reactive
 - 3,4 and 7-11 membered ring is the most reactive ring.
 - 5,6 membered rings are stable and polymerize slowly, but, it still possible to be polymerized.
 - 3-membered ring is the most easiest to be polymerized.
- It should be noted that some monomers can be polymerized by using both anionic or cationic initiators.
- Ethylene oxide can be polymerized by using acid or base as initiator.

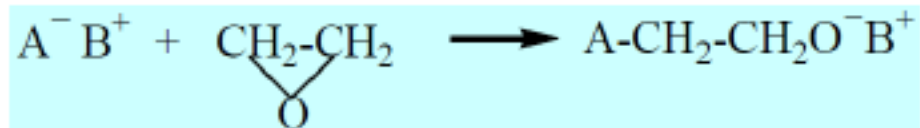


Ring-opening anionic polymerization

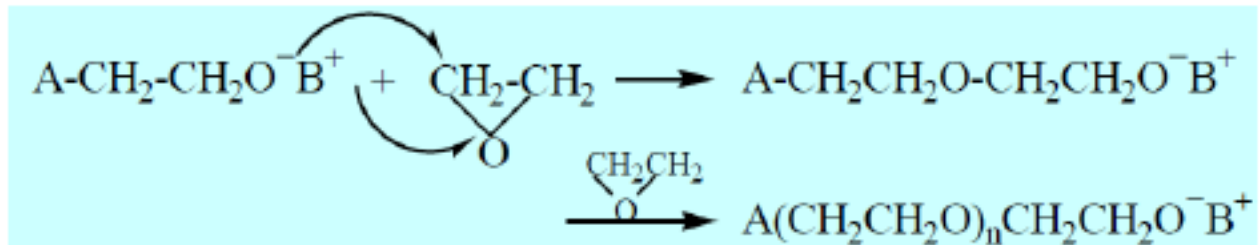
Anionic initiators



Initiation:



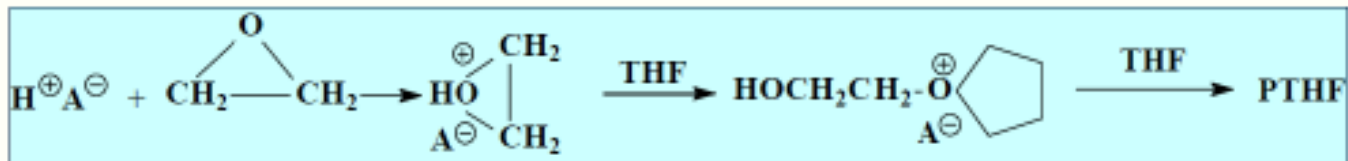
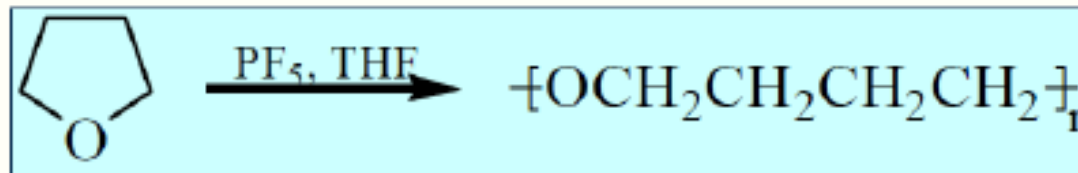
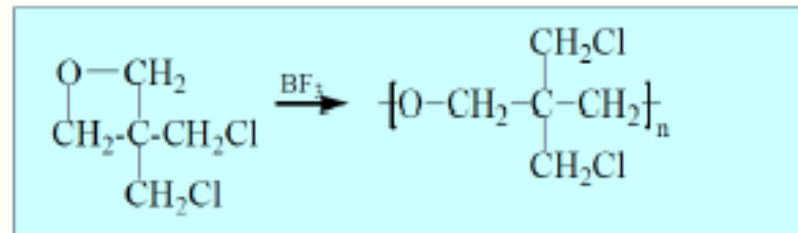
Propagation:





Ring-opening cationic polymerization

Some examples :



□ Coordination Polymerization

- Coordination polymerisation is a form of chain polymerization in which monomer adds to a growing macromolecule through an organometallic active center.
- A polymerization technique offers stereoselective control over the growing polymer chain
- Named after K. Ziegler and G. Natta in the 1950s.



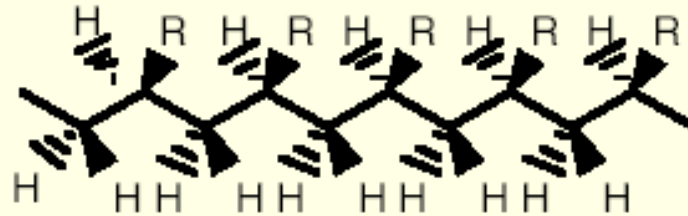
The coordination polymerization is invented by two Italian scientists: Karl Ziegler and Giulio Natta.

They shared the nobel prize in chemistry in 1963.



Polymer Stereochemistry

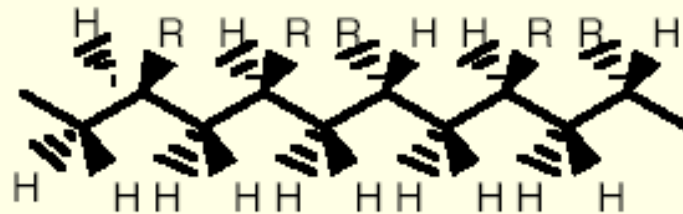
Isotactic



Syndiotactic



Atactic

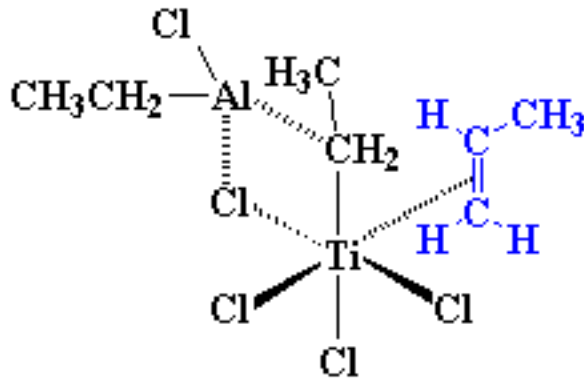


Tacticity and polymerization methods

- The tacticity of a polymer affects the packing between molecules (crystallinity) and thus affects its physical properties (Glass transition, melting temperature, mechanical strength, elasticity etc.)
- Structure Regularity
 - Isotactic > Syndiotactic > Atactic
- Free Radical chain polymerization gives polymers with no control in stereochemistry
- Coordination polymerisation offers isotactic or syndiotactic structures of polymer chains.

Ziegler-Natta Catalysts

- Complexes formed by the interaction of alkyls of metal of groups I-III with halides of transition metals of groups IV-VIII.
- Hundreds of combinations with different reactivity and selectivity.
- Coordination complex-one empty coordination site accommodates the growing polymer chain.



Traditional Ziegler-Natta Initiator

1) Transition-metal component:

Group IV ~ VIII transition-metal components, such as TiCl_3 , TiCl_4 and VCl_4 .

The most widely studied transition-metal: Titanium

α , γ , δ TiCl_3 , TiCl_2 , TiCl_4 , β TiCl_2

2) Group I-III metal component

It is required to obtain high stereoselectivity and high activity.

Such as LiR , MgR_2 , ZnR_2 , AlR_3 , where R is alkyl group or naphthenic group

Aluminum components are by far the most often used :

$\text{Al R}_n\text{X}_{3-n}$, $\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$

The most common Ziegler-Natta Initiator:

TiCl_3 or TiCl_4 & AlR_3 or AlR_2Cl

Effect of components of Ziegler-Natta Initiator (stereoselectivity)

1) Transition-metal component:

Different transition-metal :



Different oxidation states of Titanium:



2) Group I-III metal component

Different metal



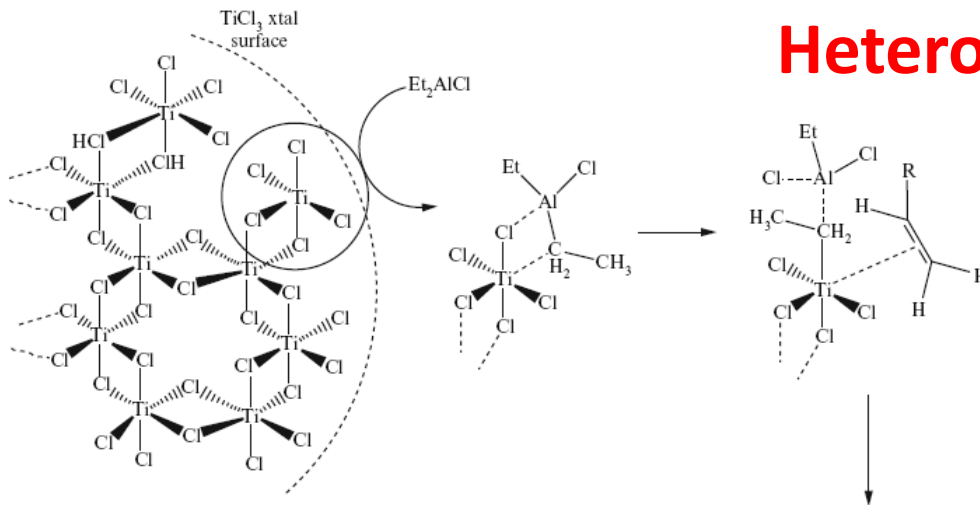
Different halide



Traditional Ziegler–Natta polymerization is an example of heterogeneous catalysis—a dual-phase system where polymer occurs on the surface of the catalyst.

e.g. TiCl_3 or TiCl_4 in association with an aluminum alkyl co-catalyst

Heterogeneous Catalysis

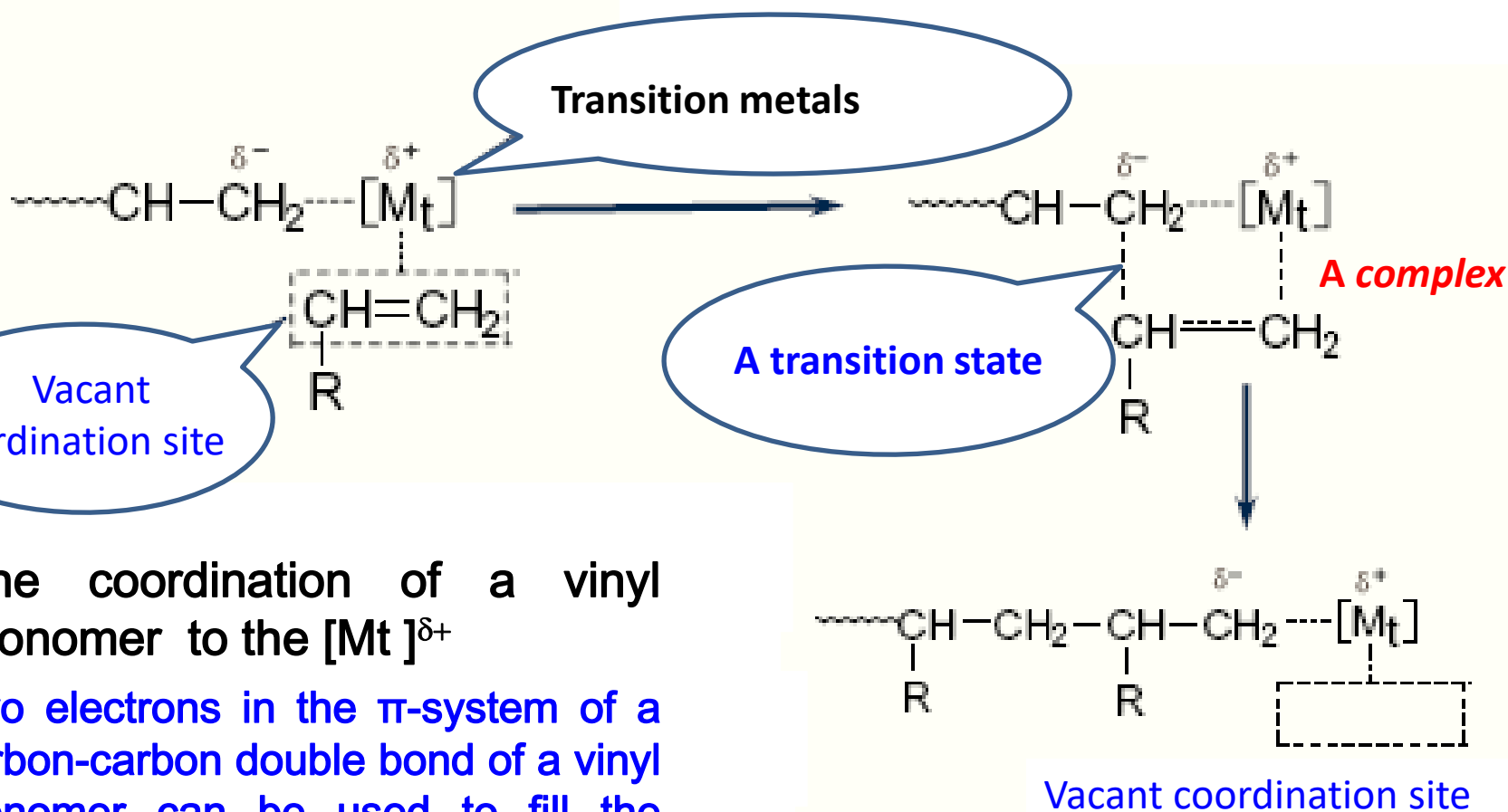


- The aluminum (Lewis acidic) complex acts as an initiator, facilitating monomer coordination through abstraction of a Cl group from the Ti coordination sphere.
- The Lewis acidic Al site also assists in the intramolecular rearrangements that are essential for chain propagation.



Polymerization of a vinyl monomer

Insertion reaction of the monomer to active species



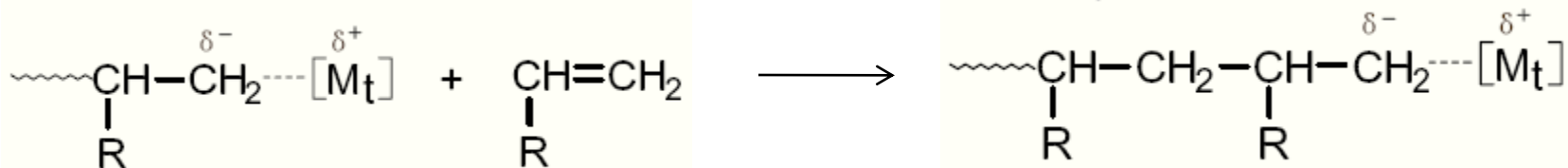


Regioselectivity

Primary versus secondary insertion:

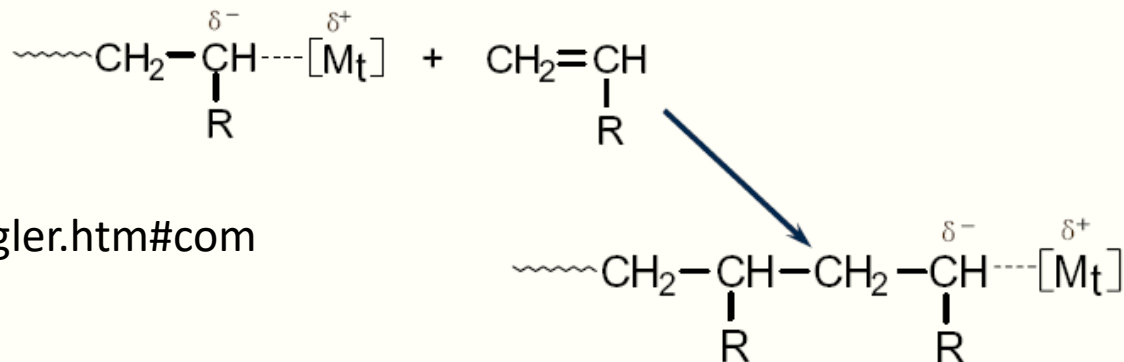
1) Primary insertion: isotactic

The unsubstituted end of the double bond carries the partial negative charge and is attached to the counterion $[M_t]^{\delta+}$



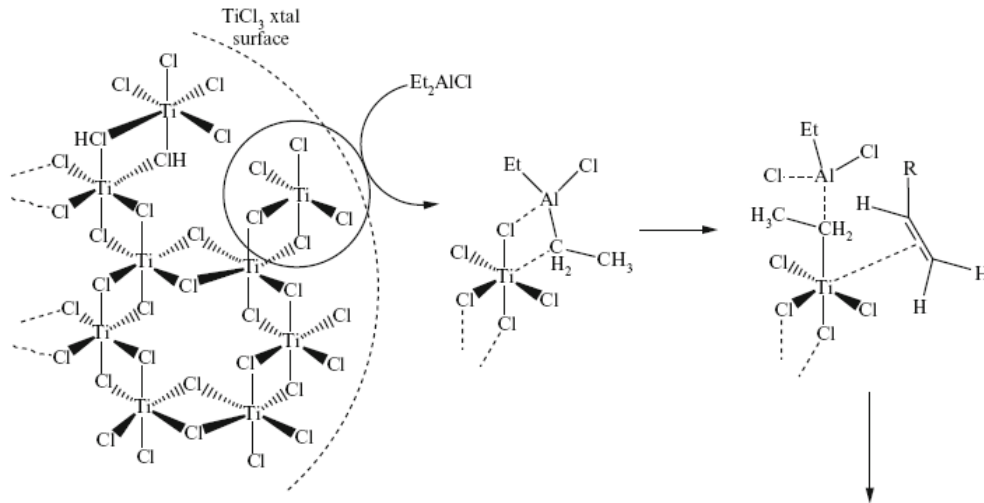
2) Secondary insertion: syndio

The substituted end of the double bond is attached to the counterion $[M_t]^{\delta+}$



<http://pslc.ws/macrog/ziegler.htm#com>

Heterogeneous Catalysis



- Although separation of products from catalyst is easily accomplished within a heterogeneous system, the polydispersity of the product will be relatively high due to multiple reaction sites on the catalyst surface.
- In order to improve the overall selectivity, a variety of single-phase homogeneous catalytic routes have also been developed. Although the polydispersity of the products are much narrower using homogeneous catalysts, the primary limitation is the difficulty in separating products from reactants/catalyst.



Homogeneous Catalysis

- Single-phase homogeneous catalysis are developed to improve the overall stereoselectivity.
- A Kaminsky catalyst is a catalytic system for alkene polymerization discovered in 1980.
- Kaminsky catalysts are based on metallocenes of group 4 transition metals (Ti, Zr, Hf) with methylaluminoxane.
- These catalysts are a type of Ziegler-Natta catalyst, but they are homogeneous and show extremely high activity for polymerization of olefins such as ethylene and propylene.

Metallocene Homogeneous Polymerization Catalysts

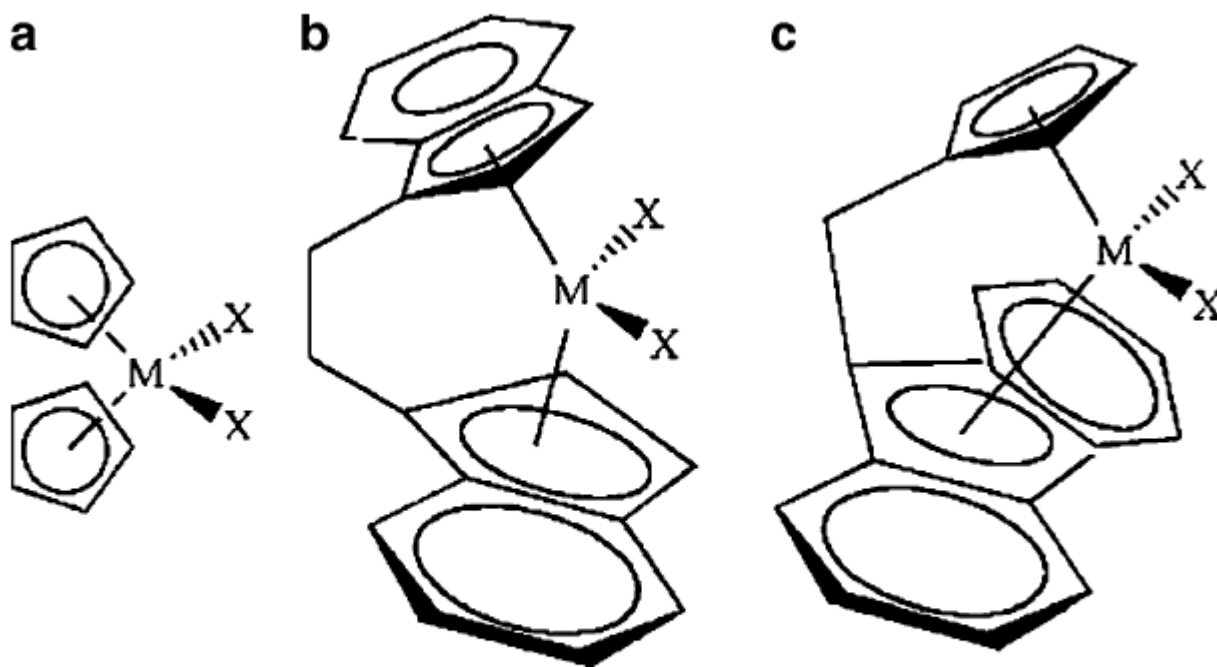


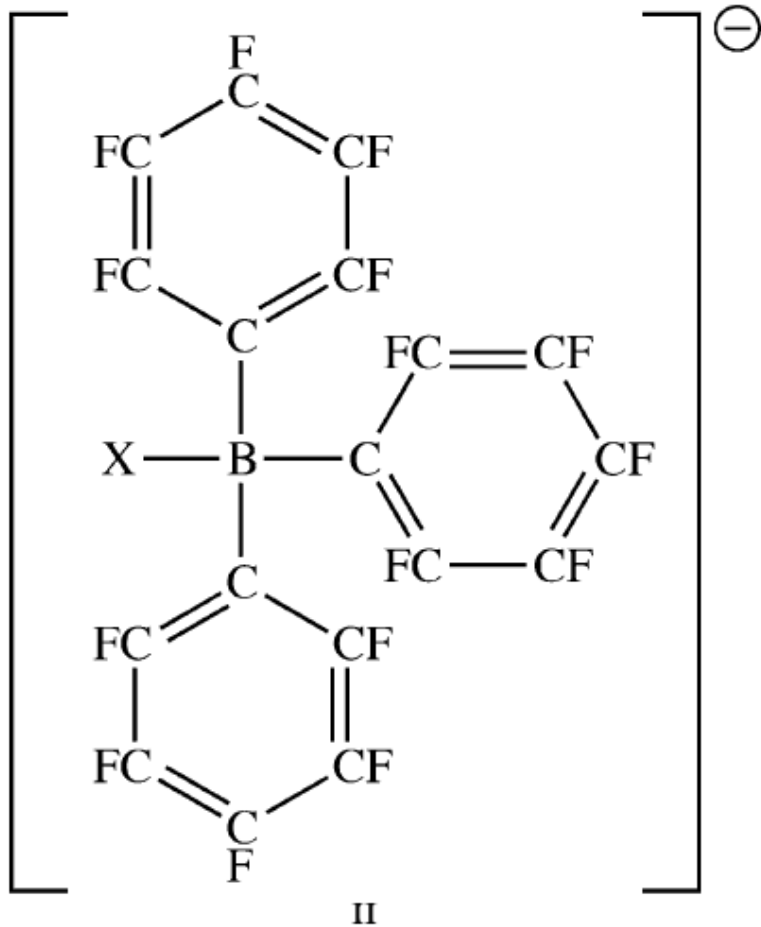
Figure 5.19. Molecular structures of common metallocene homogeneous polymerization catalysts ($M = \text{Zr}, \text{Hf}$, $X = \text{Me}, \text{Cl}$). The cyclopentadienyl ligands may be abbreviated as “cp” in the molecular formula (e.g., $(\text{cp})_2\text{HfMe}_2$).



Co-Catalyst

In order to obtain catalytic activity in the system, a co-catalyst featuring an electronegative center (e.g., B, Al) must also be present in the solution.

- For aluminum co-catalysts, this yields a $[(Cp)_2MX]^+ [(tBu)_2AlX]^-$ cation/anion pair (X=R, Cl).
- This activates the metallocene structure toward polymerization, since the M^+ site more readily accepts electron density from the alkene monomer.



- In order for the ion pair to be active toward polymerization, the anion must be non-coordinating.
- $[BX(C_6F_5)_3]$ (II) is noncoordinating toward the metallocene structure

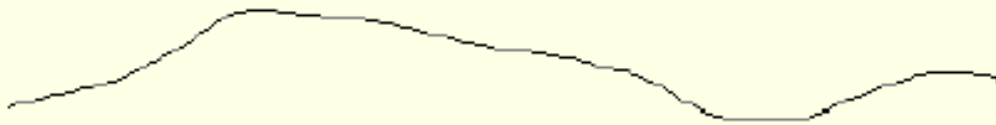


Application of Ziegler–Natta polymerization

- For the polymerization of ethylene, propylene and other olefin monomers with stereo control, isotactic or syndiotactic polymers can be obtained.
- Most catalyst systems are heterogeneous catalysts.
- Some superactive or high-mileage initiator system have very high activity.
 - (50-2000 kg polypropylene/g Ti !)
- PE-much lower degree of branching compared with one obtained by radical polymerization.

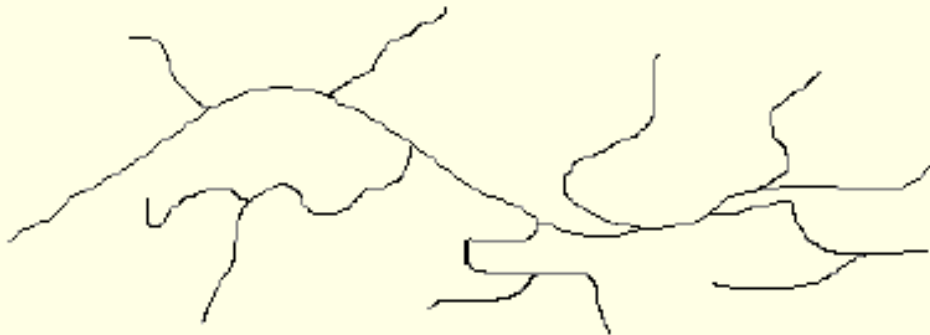
Linear vs. Branched PE

- Some PE have long chains of polyethylene attached to them. This is called branched, or low-density polyethylene, or LDPE.
- When there is no branching, it is called linear polyethylene, or HDPE.
- Linear polyethylene is much stronger than branched polyethylene, but branched polyethylene is cheaper and easier to make.



A molecule of linear polyethylene, or HDPE

**Linear PE or HDPE:
obtained by Ziegler–Natta
polymerization**



A molecule of branched polyethylene, or LDPE

**branched PE or HDPE:
obtained by radical
polymerization;**

Low Density Polyethylene

Produced by free radical polymerization of ethylene gas

- Polymer chain with branching, low crystallinity
- More flexible, melting point $\sim 115\text{ }^{\circ}\text{C}$
- Density $\sim 0.91\text{-}0.94$
- Uses:
 - film and sheet-packaging
 - trash bags
 - household wrap, toys
 - squeeze bottle, kitchenwares
 - wire and cable insulation, and many others



High Density Polyethylene

Synthesized from ethylene using metal as the catalyst

- Linear polymer molecule with no branches, m.p. = 133-138 °C, specific gravity ~ 0.96
- High **crystallinity** (70-90% vs. 40-60%), more opaque than LDPE
- Stiffer, harder, higher tensile strength than LDPE
- Uses
 - containers and lids
 - food bottles, petrol tank
 - motor oil bottles
- HDPE is much stronger than LDPE, but LDPE is cheaper and easier to make.



3. Chain-growth polymerization and Mayo equation

- Free-radical polymerization
- Ionic polymerization
 - (Living) Anionic polymerization
 - Cationic polymerization
 - Ring-open polymerization
- Coordination polymerization: the only chain polymerization can control tacticity
- Living free radical polymerization: Three important features for living polymerization.
 1. Chain termination and chain transfer reactions are absent.
 2. The rate of chain initiation is much larger than the rate of chain propagation, initiators decompose quickly to insure narrow MWD
 3. The polymer chains grow at a more constant rate than seen in traditional chain polymerization and their lengths remain very similar.

5.7 Tradition Synthesis

5.7.1 Step Growth Polymerization

5.7.2 Chain Polymerization

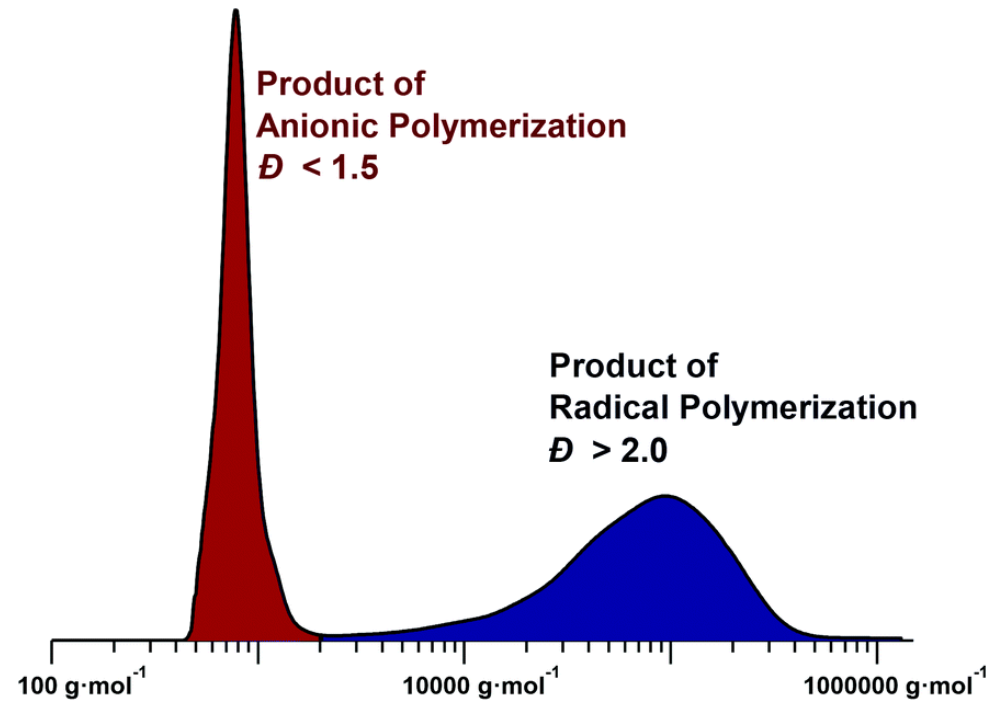
- Free Radical Polymerization
- Ionic polymerization
- Coordination polymerization

5.8 Advanced Synthesis

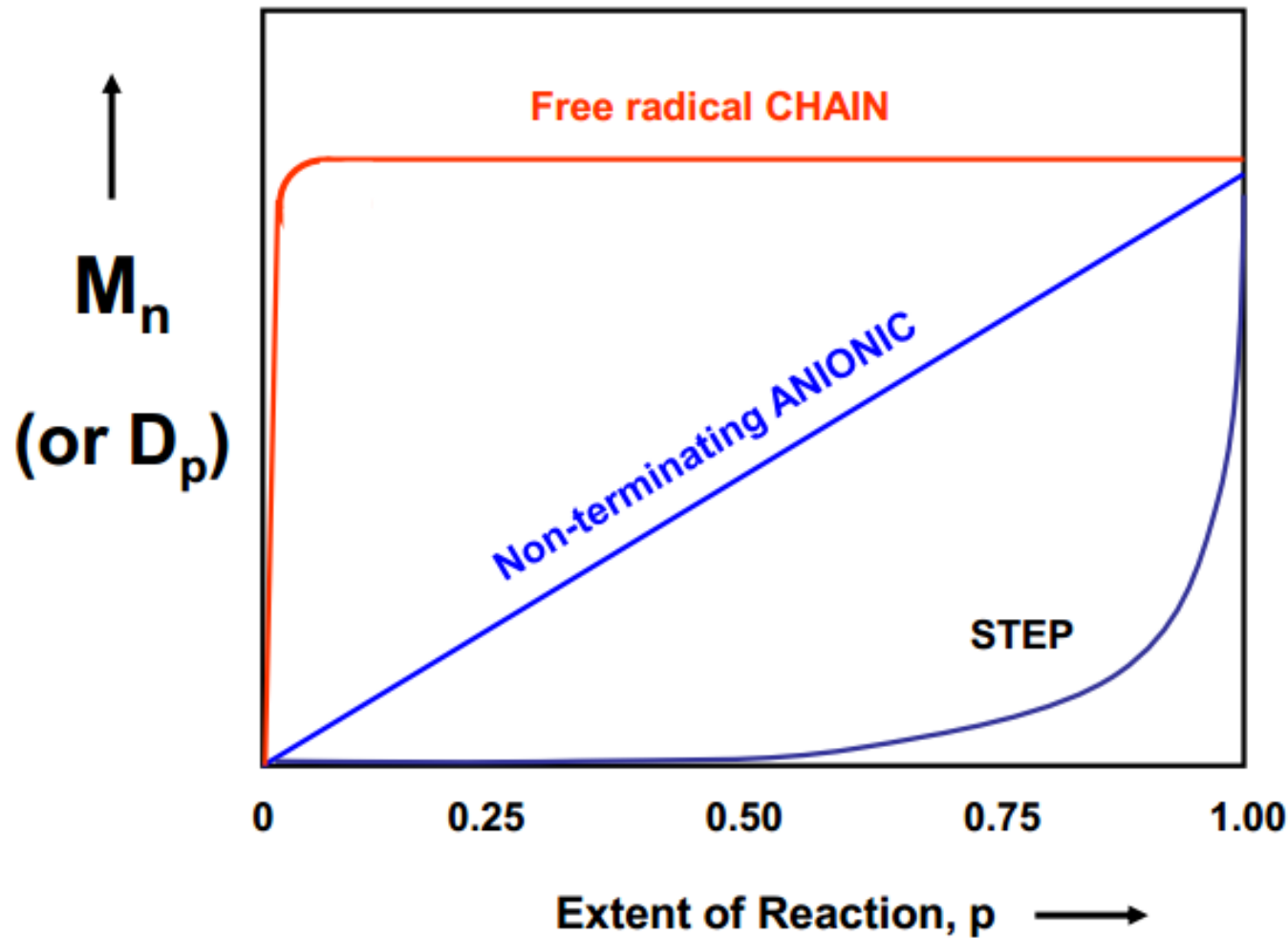
- living free-radical polymerization
- NMP
- ATRP
- RAFT



Free radical polymerization
VS
Anionic polymerization



Ionic polymerizations are fast but are extremely sensitive to small amounts of impurities. Highly polar solvents (water, alcohols, ketones) will react with and destroy or inactivate the initiator.



“Living”

Initiator decomposes quickly, and polymer chains have long life time

Living polymerizations are chain growth polymerizations which proceed in the absence of irreversible chain transfer and termination steps.

Characteristic of Living Polymerizations :

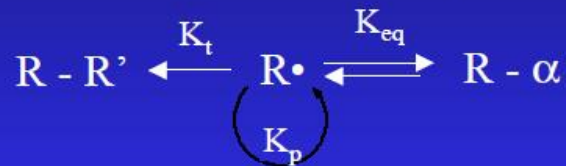
- The reaction proceeds until all monomer is consumed. If more monomer is introduced then the polymerization will continue
- The number average molecular weight, M_n , is a linear function of conversion.
- The number of propagating chains (active centers) is constant and independent of conversion.
- M_n can be controlled by the reaction stoichiometry.
- Sequential monomer addition results in the preparation of block copolymers.
- Resulting polymers will exhibit a narrow molecular weight distribution.



Living Free Radical Polymerization

- Definition: Polymerization in which radicals are not terminated
- Motivation: Superior molecular weight and polydispersity control, ease of block copolymerization, route to unique materials and architecture.
- Approach:

- ◆ Reduce concentration of free radicals to prevent coupling termination.
- ◆ Protect radicals reversibly with an endgroup α

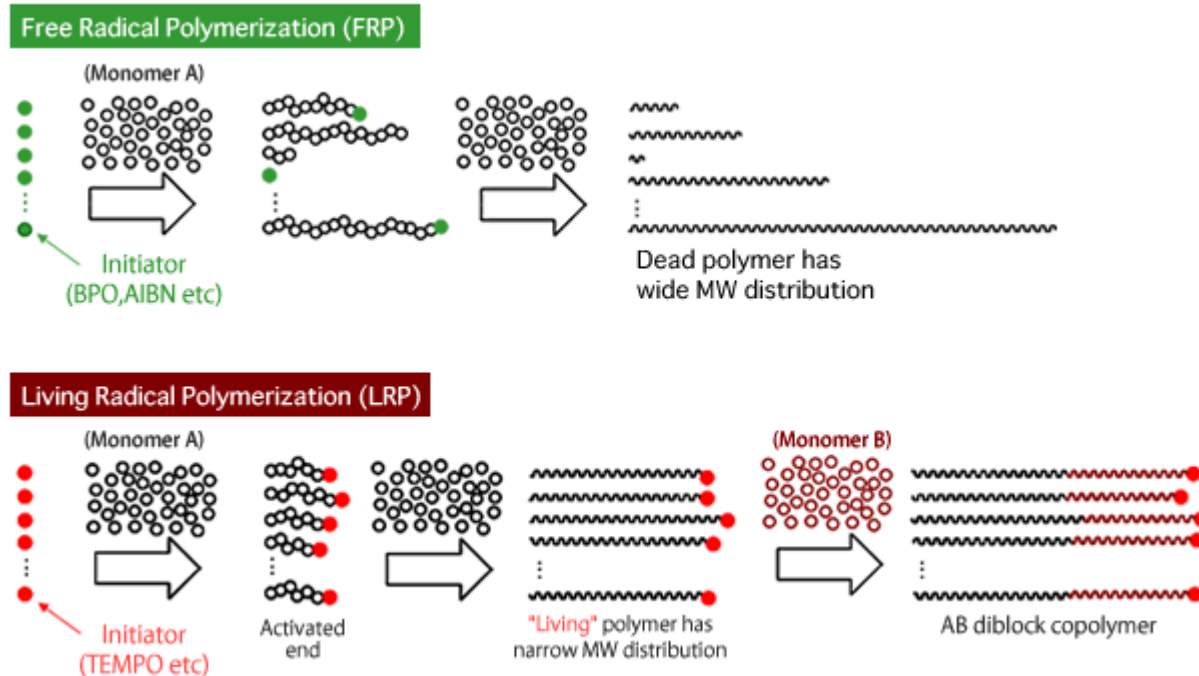


- ◆ Requirements:
 - ◆ $K_{eq} \gg K_t$ ensures living conditions
 - ◆ $K_{eq} > K_p$ ensures low polydispersity and good molecular weight control
- Methods
 - ◆ Atom Transfer Radical Polymerization
 - ◆ Nitroxide-mediated LFRP



5.8 Advanced Synthesis

5.8.1 Living radical polymerization



Features:

1. Chain termination and chain transfer reactions are absent.
2. The rate of chain initiation is much larger than the rate of chain propagation, initiators decompose quickly to insure narrow MWD
3. The polymer chains grow at a more constant rate than seen in traditional chain polymerization and their lengths remain very similar.

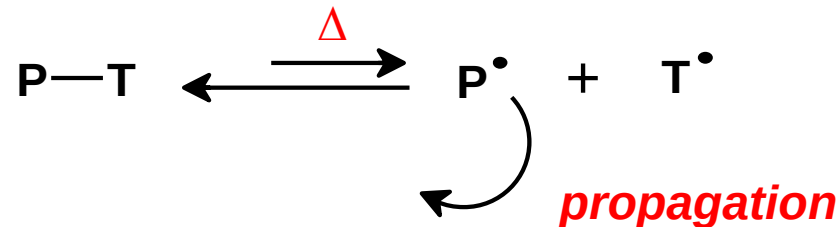


Nitroxide-mediated Controlled/Living Radical Polymerization (NMP)

- A stable free radical (e.g., 1, 2,2,6,6,-tetramethyl-1-piperidinyloxy (TEMPO)) is added to the solution, acting as a radical scavenger.
- The TEMPO radical is exceedingly stable due to the nearby methyl groups that supply electron density and help stabilize the unpaired electron that is delocalized over the N—O bond.
- The coupling of TEMPO with the polymeric radical is reversible, which allows one to control the molecular weight and polydispersity of the resultant polymer by varying the monomer:TEMPO ratio.

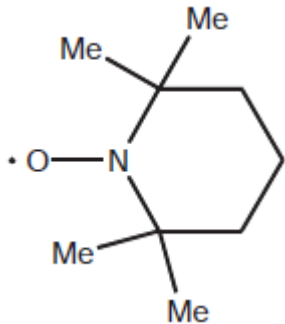


Nitroxide-mediated Controlled/Living Radical Polymerization (NMP)

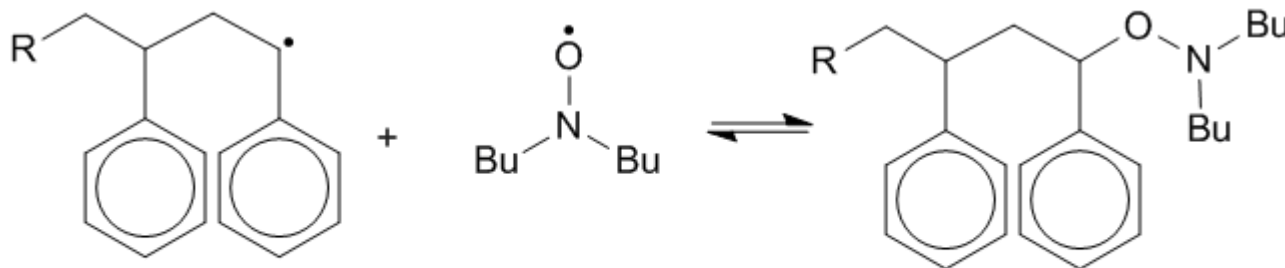


T[•] = Nitroxide

P[•] = Propagating radical



- Addition of a stable nitroxide radical, able to trap the propagating radical in a thermally unstable species.
- The most common nitroxide used as trapping agent is TEMPO (2, 2, 6, 6-tetramethyl-1-piperidinyloxy)

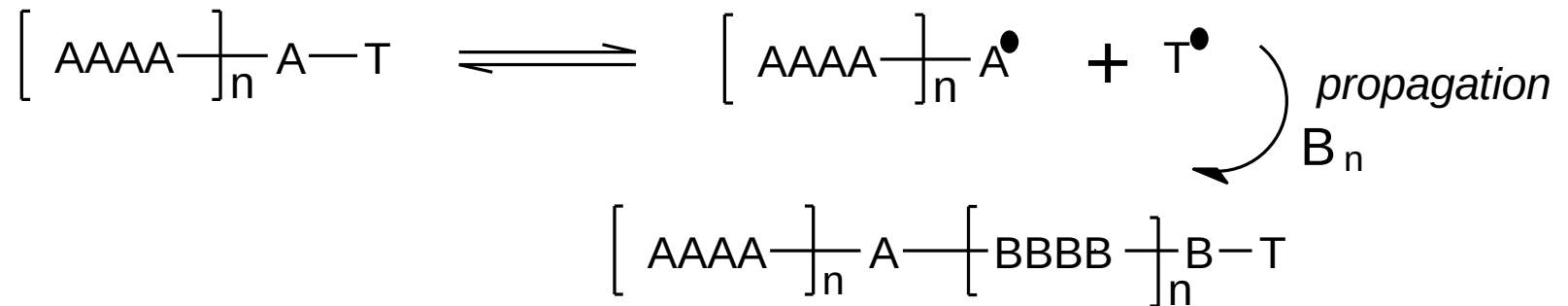


Reversible Termination Reaction

Bu = tert-Butyl

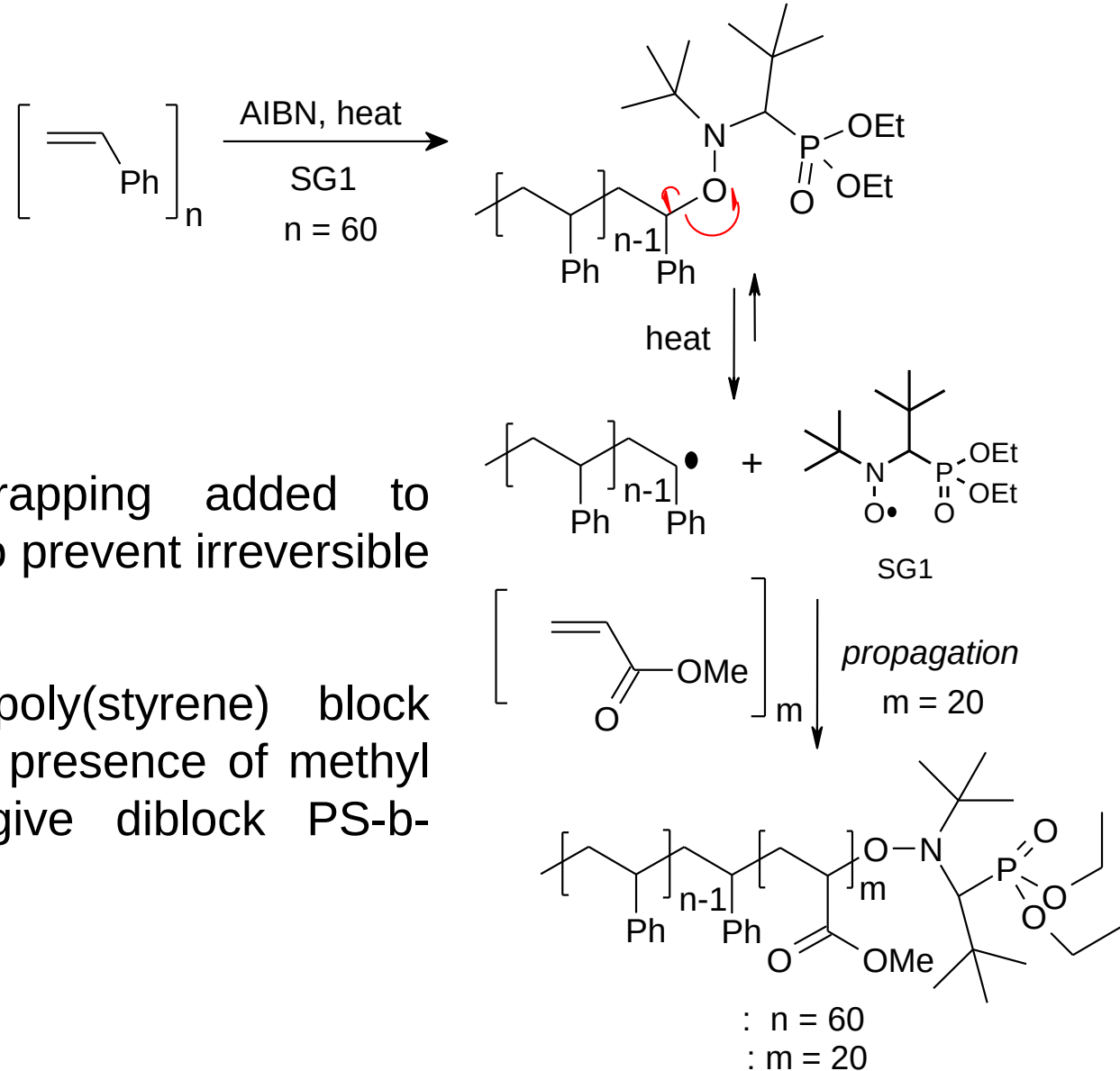
NMP for Block Copolymer Synthesis

Block copolymer synthesis



Features:

1. Molecular weight increases linearly with conversion
2. Narrow molecular weight distributions obtained
3. Polymer chains contain living ends enabling chain extension or block copolymer synthesis



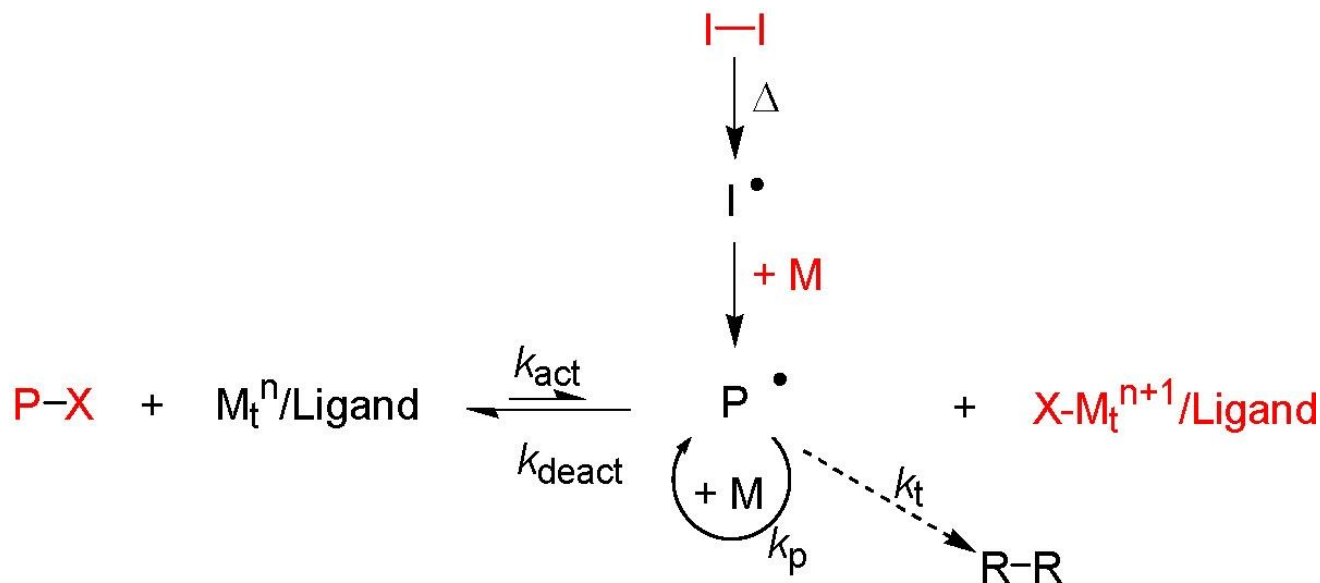
- Reversible trapping added to propagation to prevent irreversible termination
- First living poly(styrene) block heated in the presence of methyl acrylate to give diblock PS-*b*-PMMA



5.8.2 Atom-Transfer Radical Polymerization (ATRP)

ATRP (or transition metal-mediated living radical polymerization) was independently discovered by Mitsuo Sawamoto and by Krzysztof Matyjaszewski in 1995.

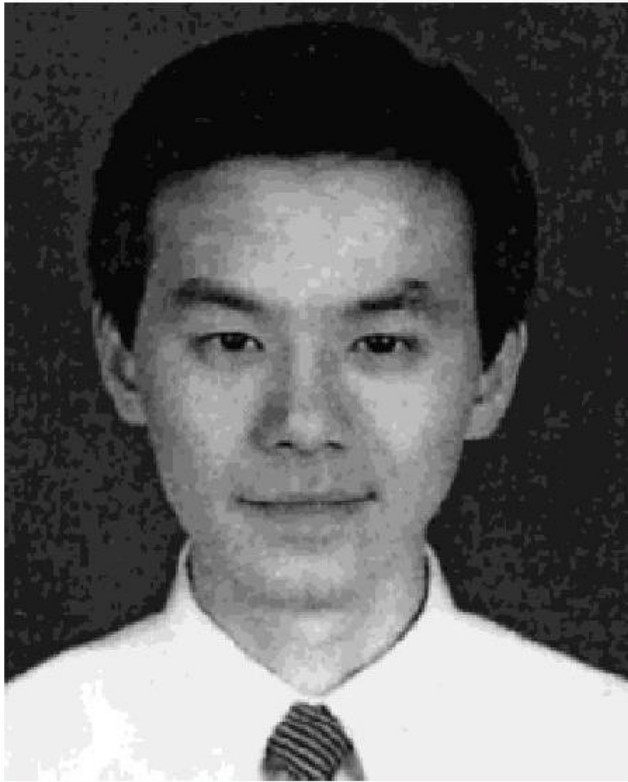
Atom transfer means of forming a carbon-carbon bond through a transition metal catalyst. As the name implies, the atom transfer step is the key step in the reaction responsible for uniform polymer chain growth.





5.8.2 Atom-Transfer Radical Polymerization (ATRP)

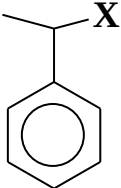
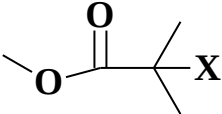
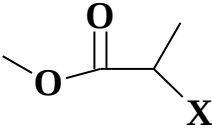
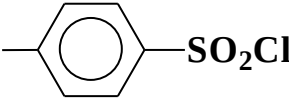
ATRP的发现者



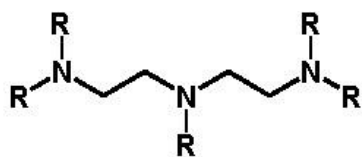
- 1995 年中国旅美博士王锦山博士在卡内基梅隆 (Carnegie Mellon) 大学做博士后研究时首次发现了原子转移自由基聚合 (Atom Transfer Radical Polymerization, 简称 ATRP), 实现了真正意义上的活性自由基聚合, 引起了世界各国高分子学家的极大兴趣。这是聚合史上唯一以中国人为主所发明的聚合方法。



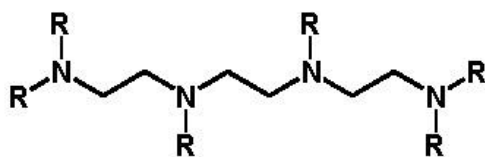
ATRP Systems

Metal	Ligand	Initiator
CuBr, CuCl	Bipyridine	
	Multidentate amine	
RuBr₂	PPh ₃ + Al(OiPr) ₃	
FeBr ₂	PPh ₃	
NiBr ₂	PPh ₃	
PdBr ₂	PPh ₃	
		

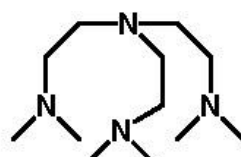
Al(OiPr)₃
三异丙氧基铝



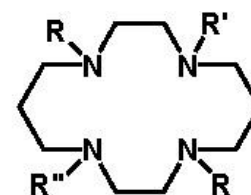
DETA (R = H)
PMDETA (R = Me)



TETA (R = H)
HMTETA (R = Me)



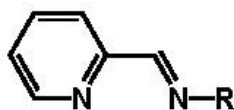
Me₆TREN



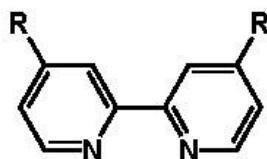
cyclam (R = R' = R'' = H)
Me₄Cyclam (R = R' = R'' = Me)
DMCBCy (R = Me, R' + R'' = CH₂CH₂)



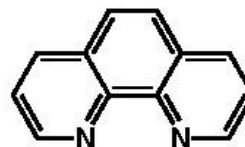
Glim-R



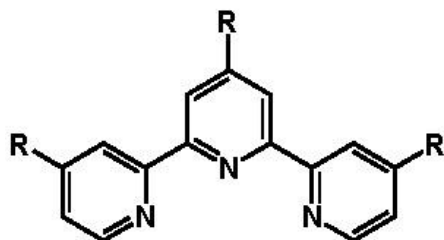
Pylm-R



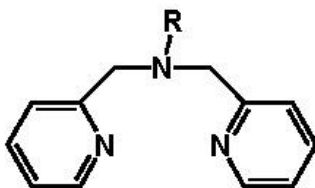
bpy (R = H)
dNbpy (R = CHBu₂)



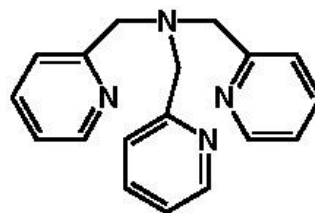
1,10-Phen



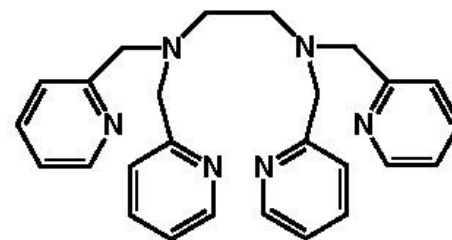
tpy (R = H)
tNtpy (R = CHBu₂)



BPMPra (R = Pr)
BPMA (R = C₈H₁₇)
BPMODA (R = C₁₈H₃₇)



TPMA

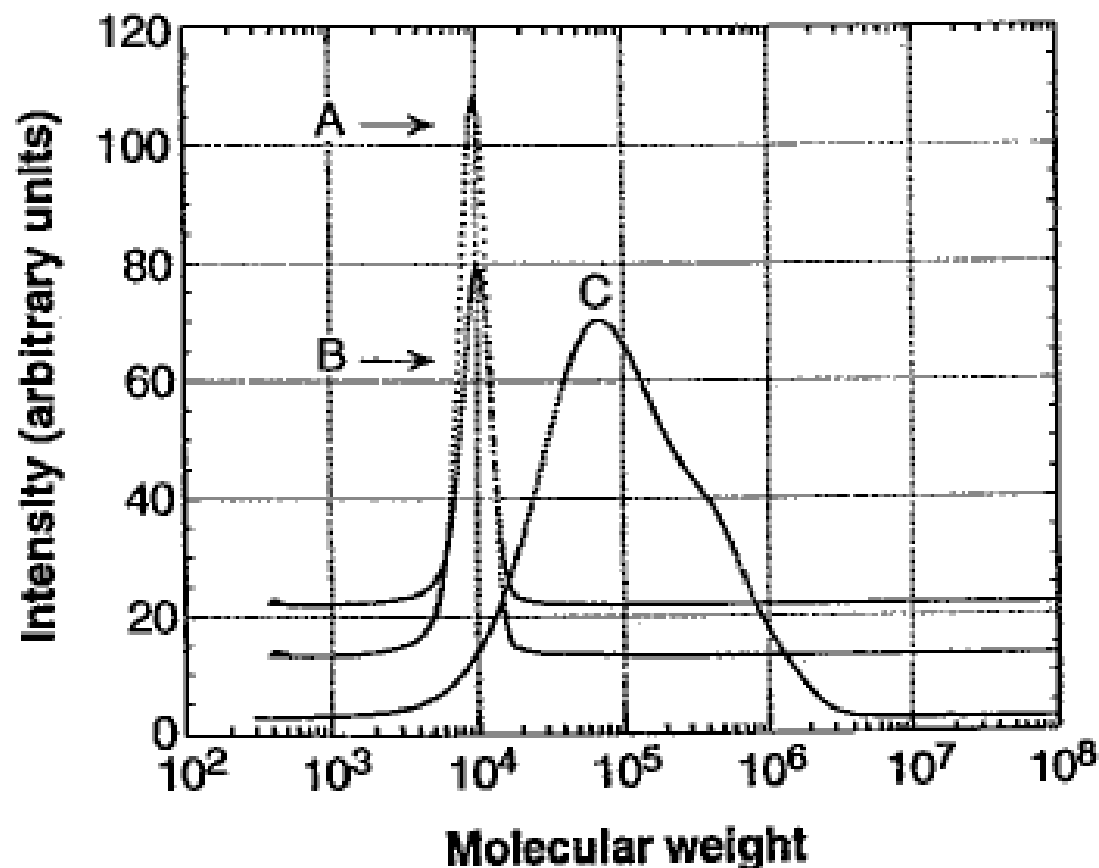


TPEDA



ATRP Applications

Synthesis of polymers with controlled molecular weights



A: PS standard

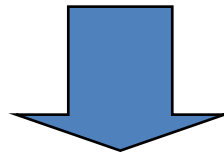
B: PS by ATRP

C: PS by AIBN



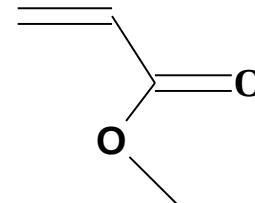
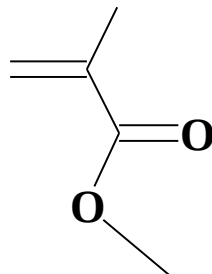
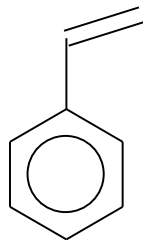
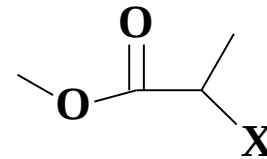
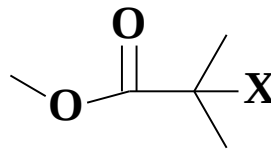
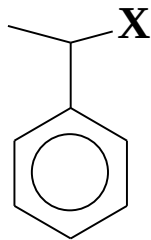
Factors affecting the molecular weight control

$$C-X_{\text{Initiator}} \geq C-X_{\text{polymer}}$$



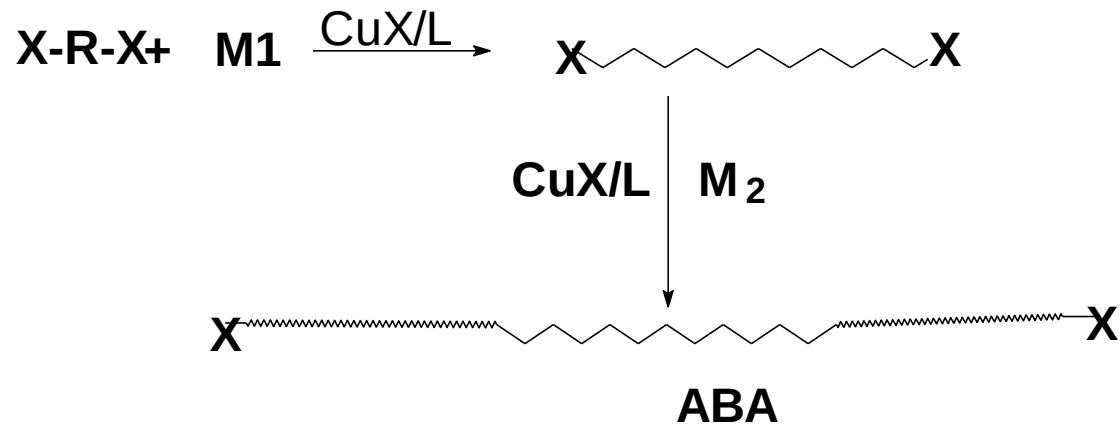
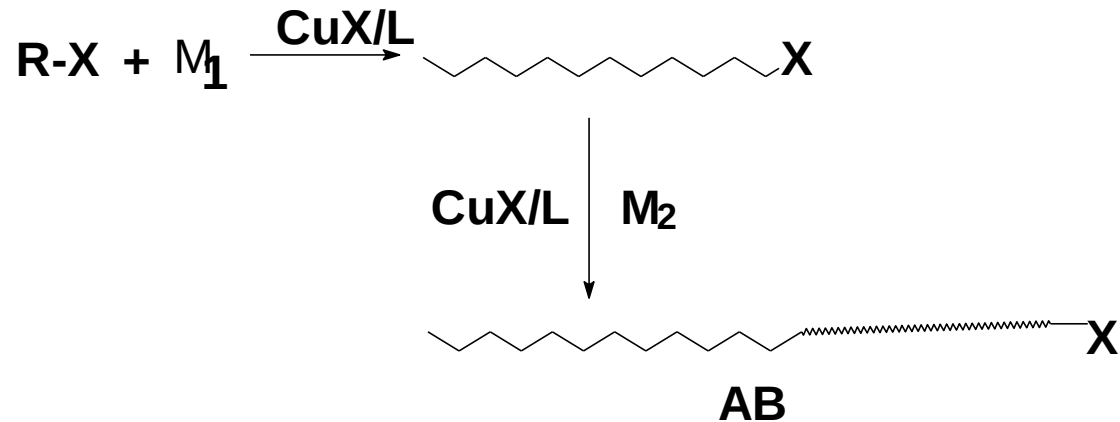
Fast Initiation
Rapid deactivation

Initiator matches Monomer





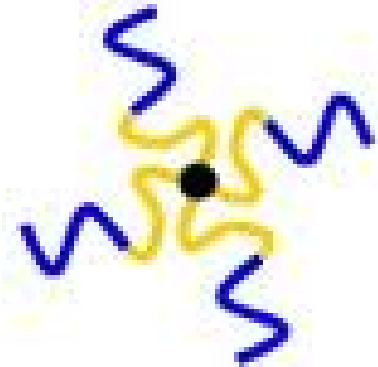
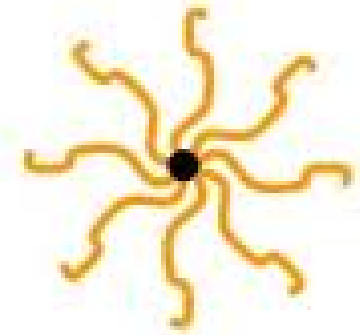
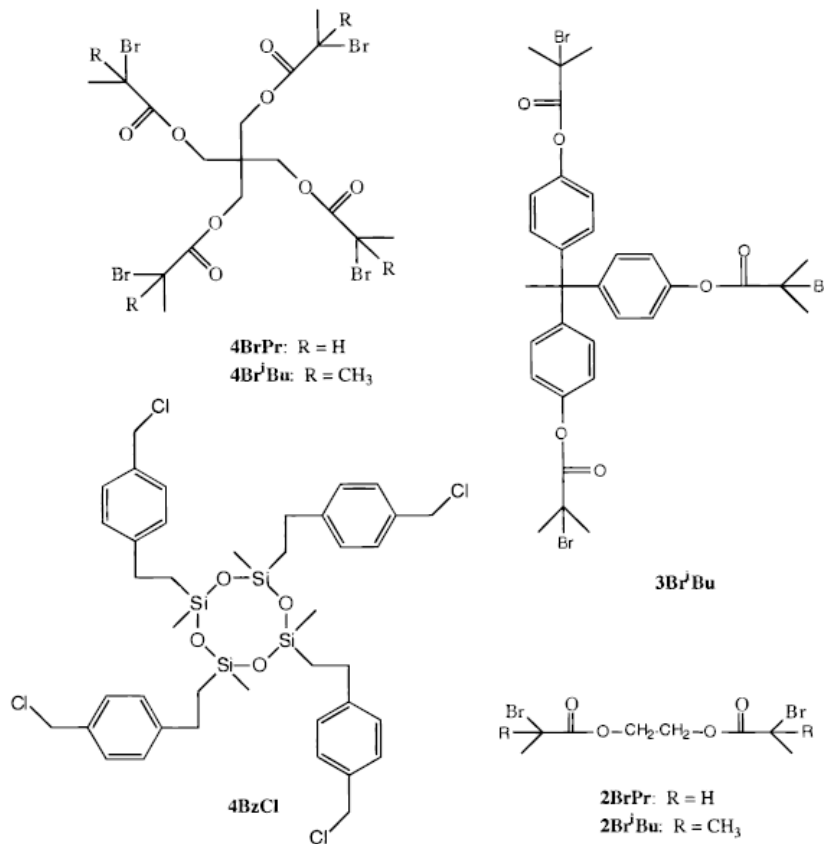
Macroinitiator method



Advantage: Stable macro-initiator, easy operation



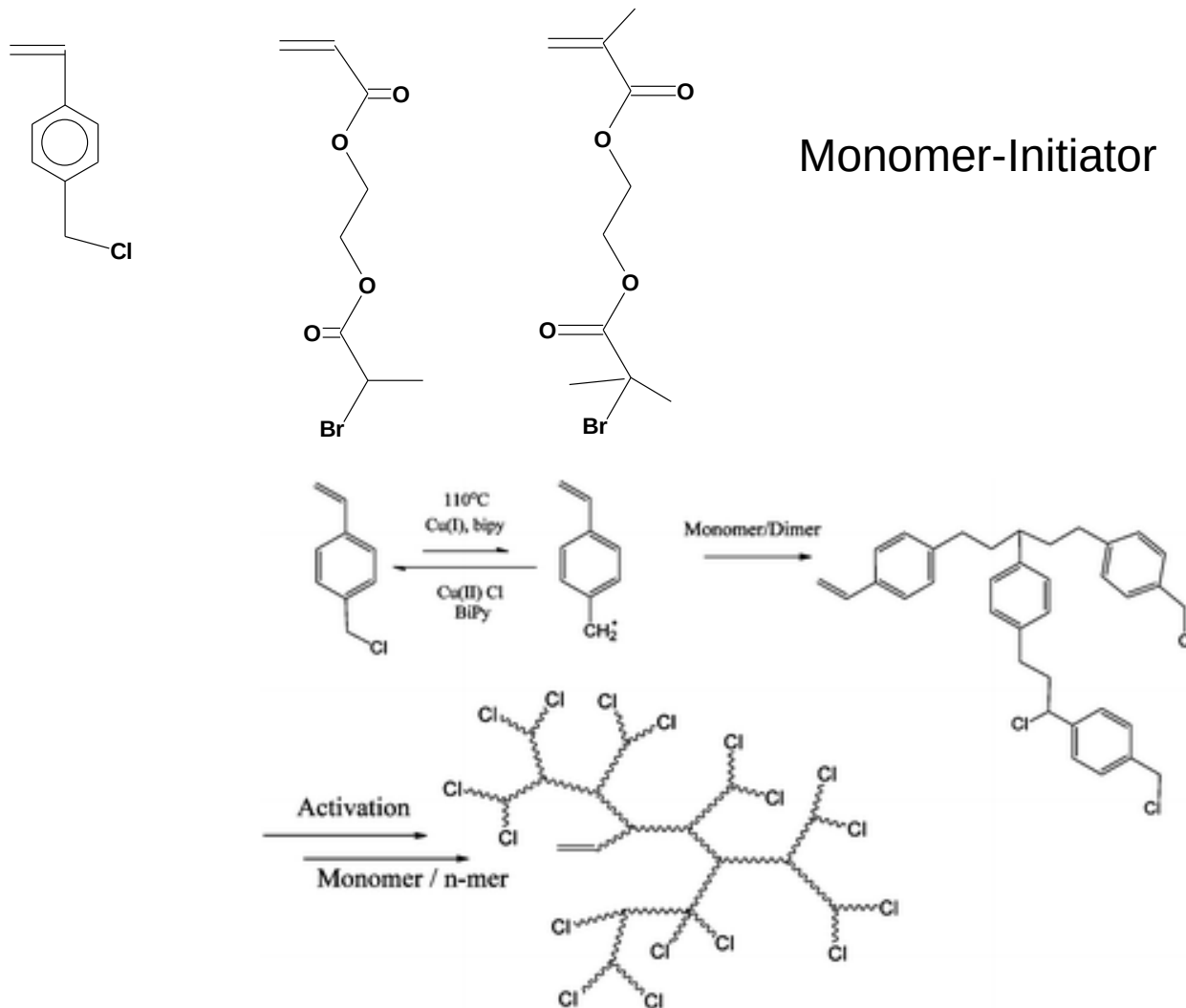
Synthesis of star polymers (polystyrene)

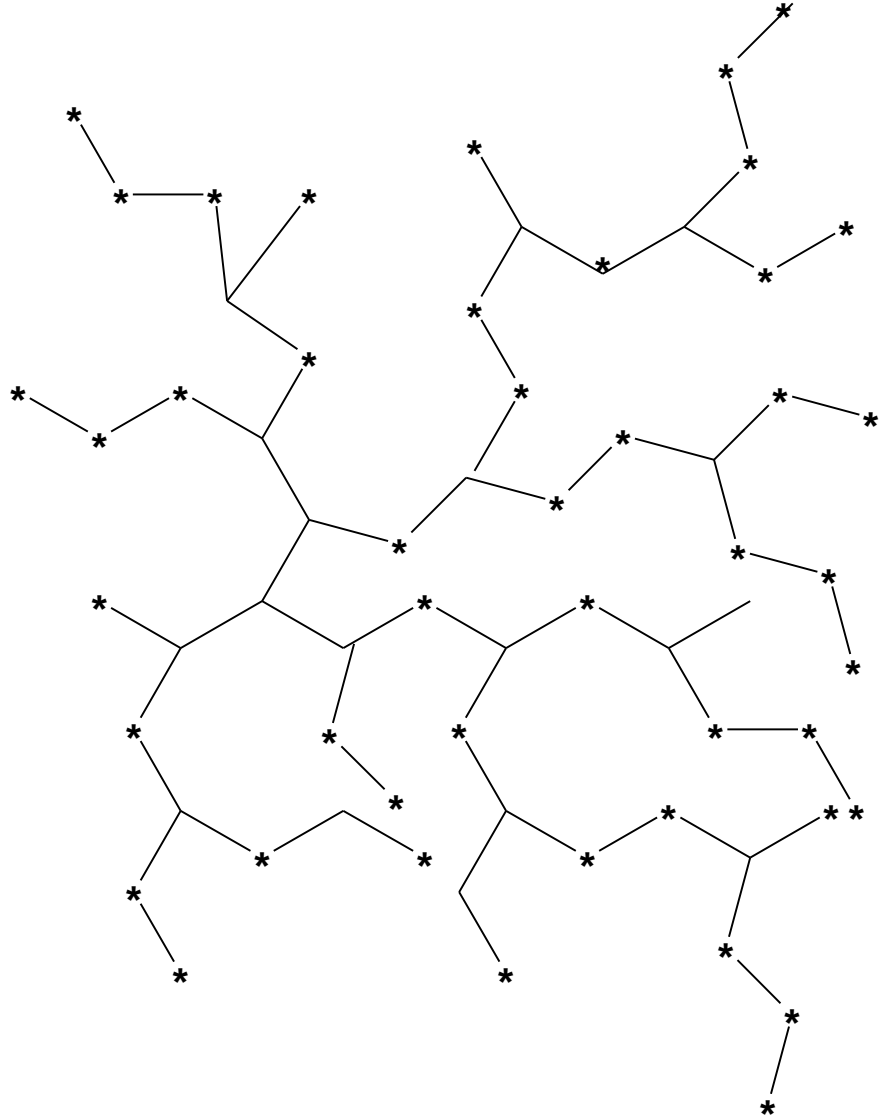
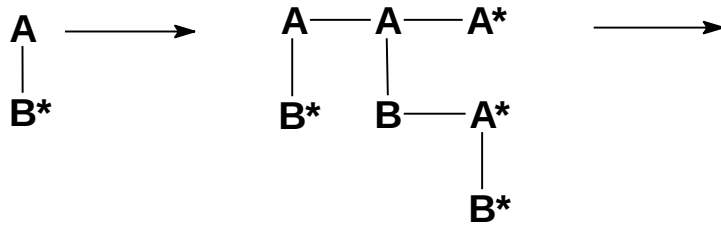


Matyjaszewski, K., Miller, P. J. Pyun, J. Kickelbick, G. Diaamanti,
Macromolecules 1999, 32, 6526



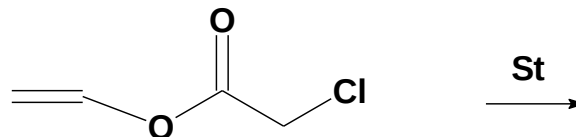
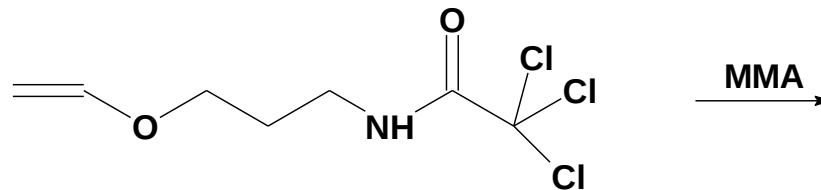
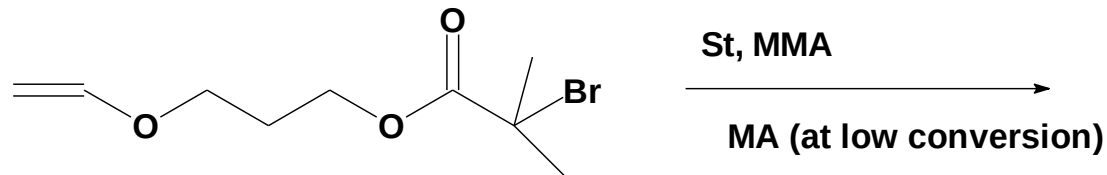
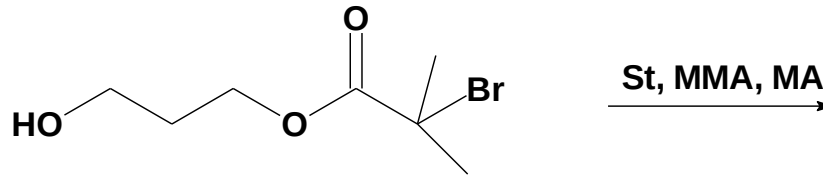
Hyperbranched Polymers







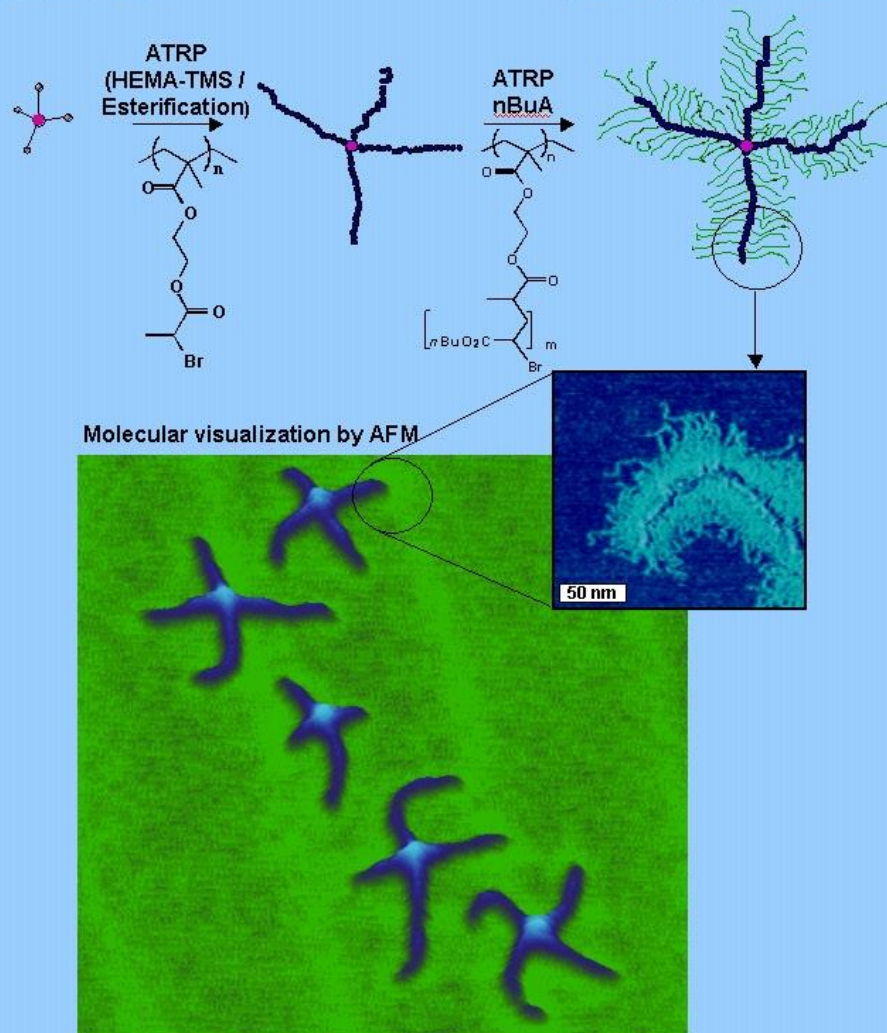
Synthesis of End-functionalized Polymers



Macromolecules

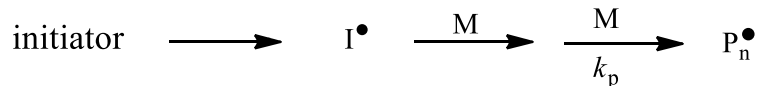
OCTOBER 7, 2003

VOLUME 36, Number 22



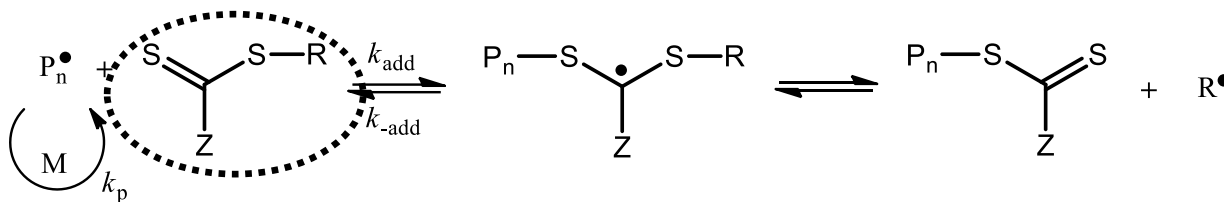
Reversible Addition-Fragmentation Chain Transfer (RAFT) (可逆加成-断裂链转移聚合)

Initiation

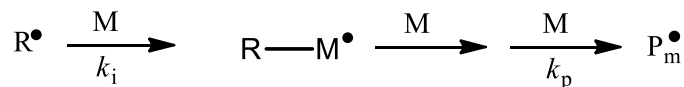


- Normal radical initiators (AIBN, etc.)
- Reversible chain transfer!

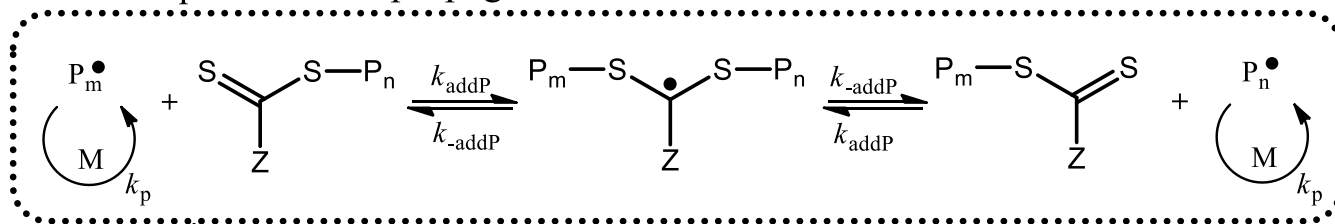
Reversible chain transfer propagation



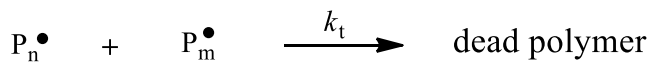
Reinitiation



Chain equilibration propagation



Termination

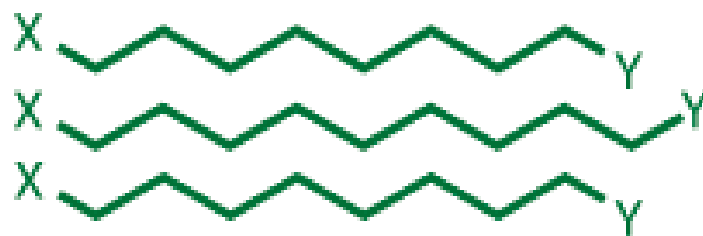


E. Rizzardo, San H. Thang, et al. *Macromolecules* 1998, 31, 5559-5562.

CSIRO Molecular Science, Bag 10, Clayton South, Clayton, Victoria 3169, Australia

Living polymerization

Greater Chain Length Control



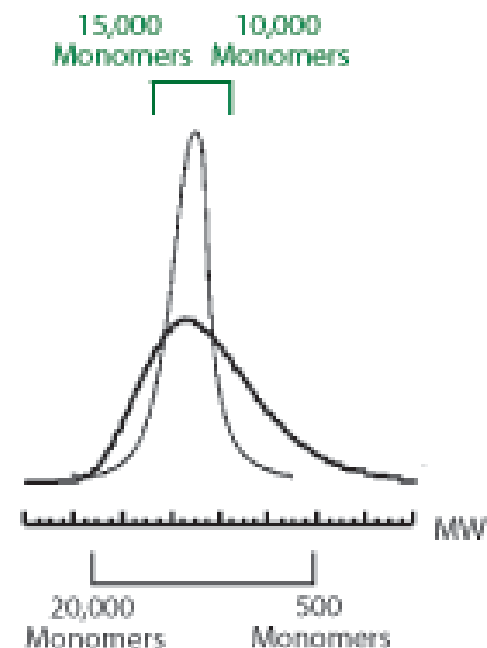
RAFT
Polymerization

Monomer

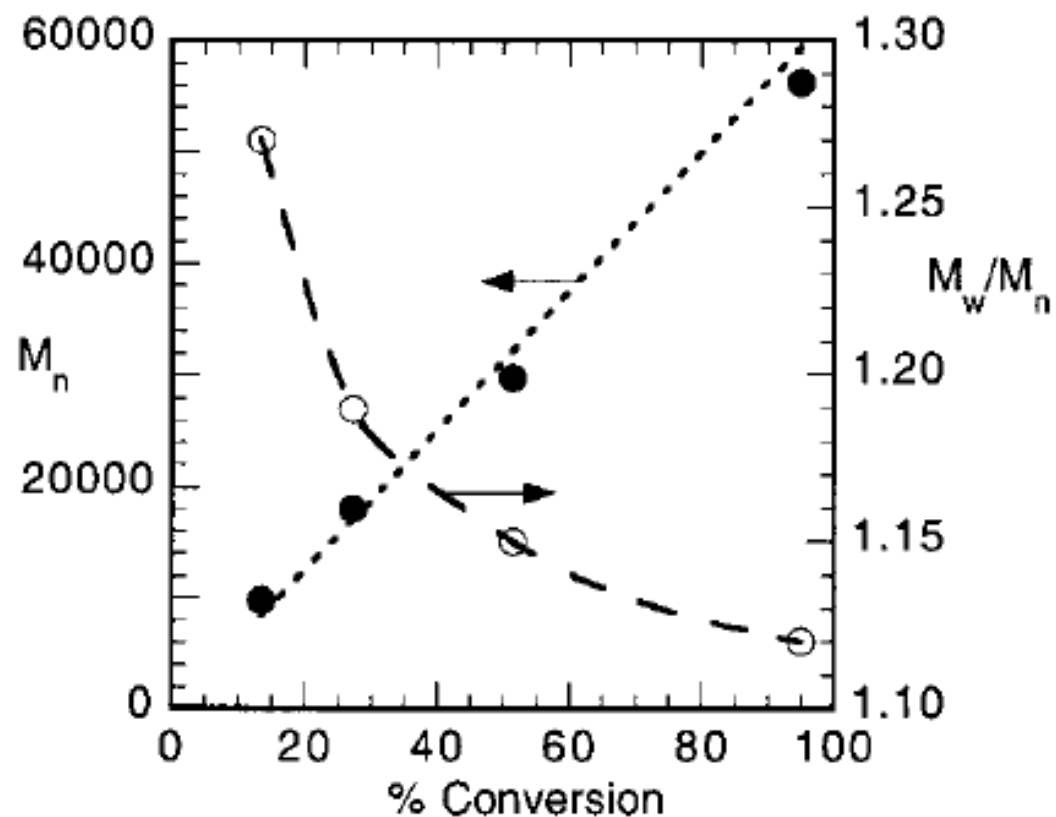
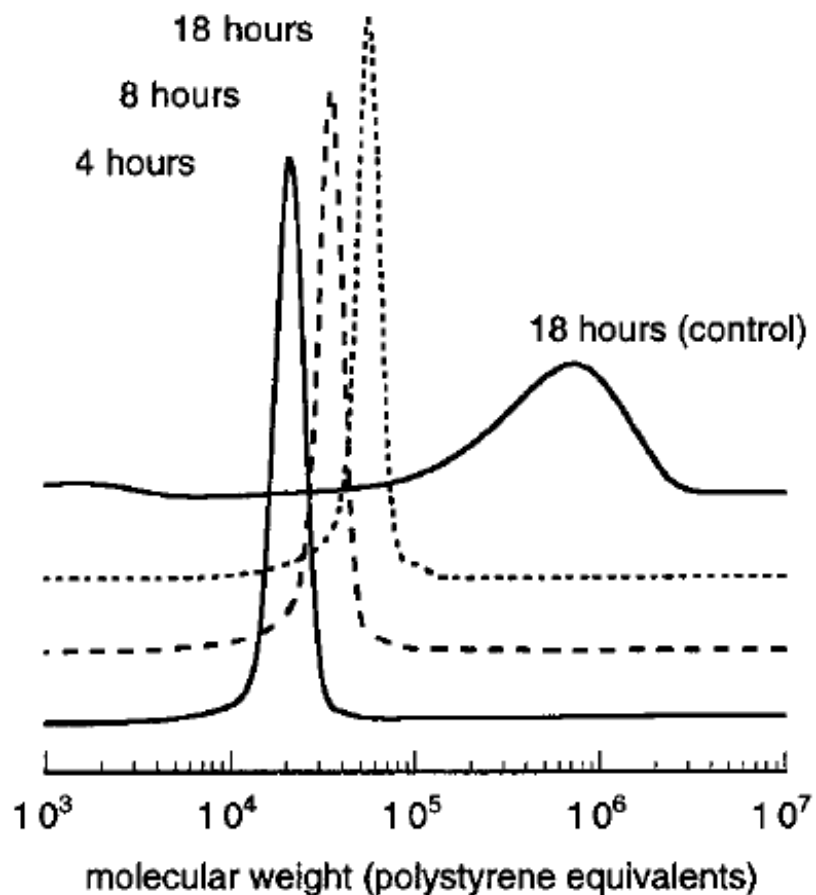
Traditional
Radical
Polymerization



Inconsistent Chain Length



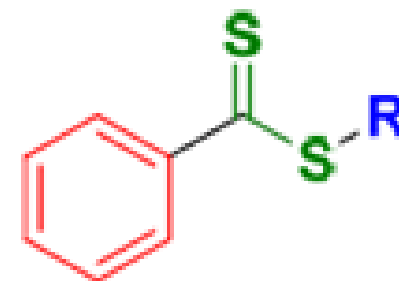
A Typical example of RAFT polymerization



Classes of RAFT Agents: Solubility and reactivity of a RAFT agent depend on the **R** and **Z** groups; as a result, different RAFT agents are more suitable for specific classes of monomers. The main classes of RAFT agents are:

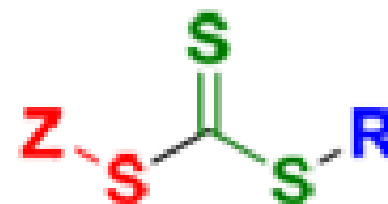
Dithiobenzoates Very high transfer constants

- Prone to hydrolysis
- May cause retardation under high concentrations



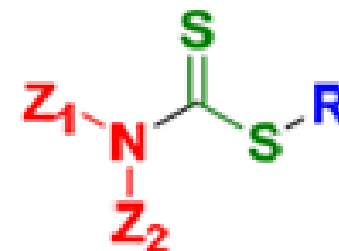
• **Trithiocarbonates** High transfer constants

- More hydrolytically stable (than dithiobenzoates)
- Cause less retardation

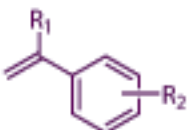
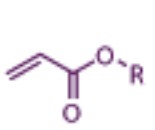
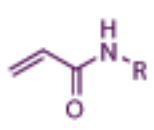
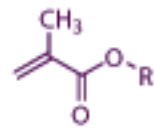
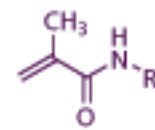
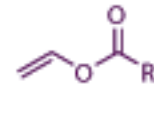
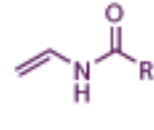
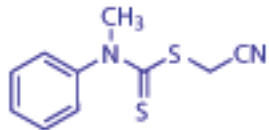
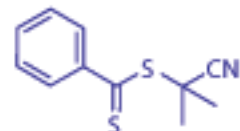
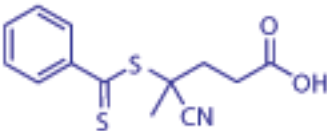
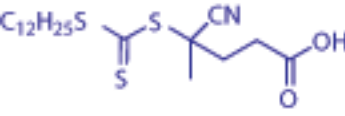
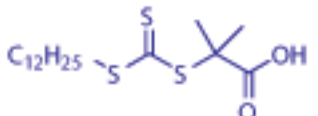


• **Dithiocarbamates** Activity determined by substituents on N

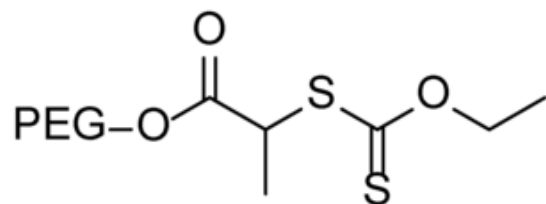
- Effective with electron-rich monomers



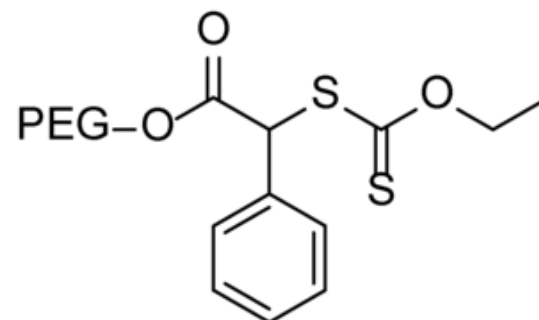
RAFT initiators and corresponding monomers

	 styrenes	 acrylates	 acrylamides	 methacrylates	 methacrylamides	 vinyl esters	 vinyl amides
	—	—	—	—	—	+++	+++
	+++	+++	+++	—	—	—	—
	++	+	—	+++	+++	—	—
	++	+	+	+++	+++	—	—
	+++	++	++	+++	+++	—	—
	+++	++	++	+++	+++	—	—
	+++	+++	+++	+	+	—	—

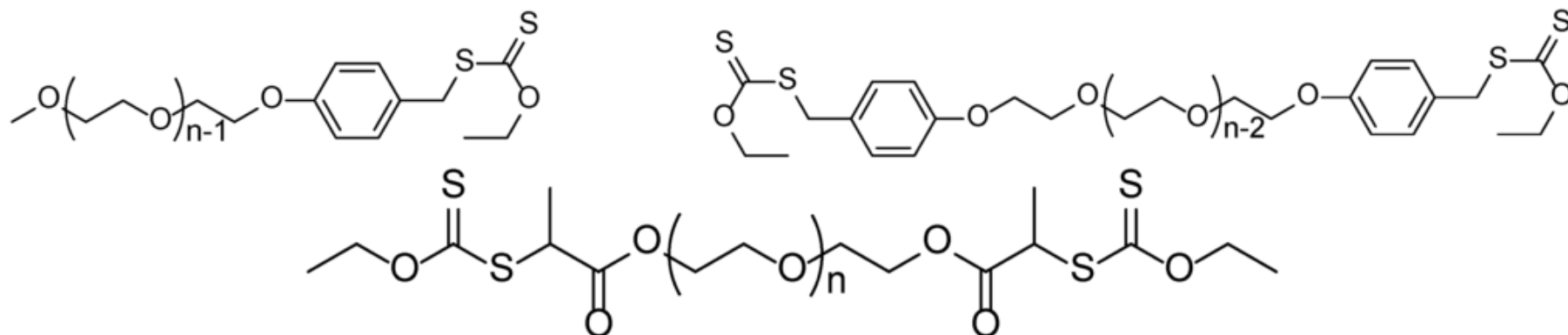
RAFT macro-initiators



PEG-X1

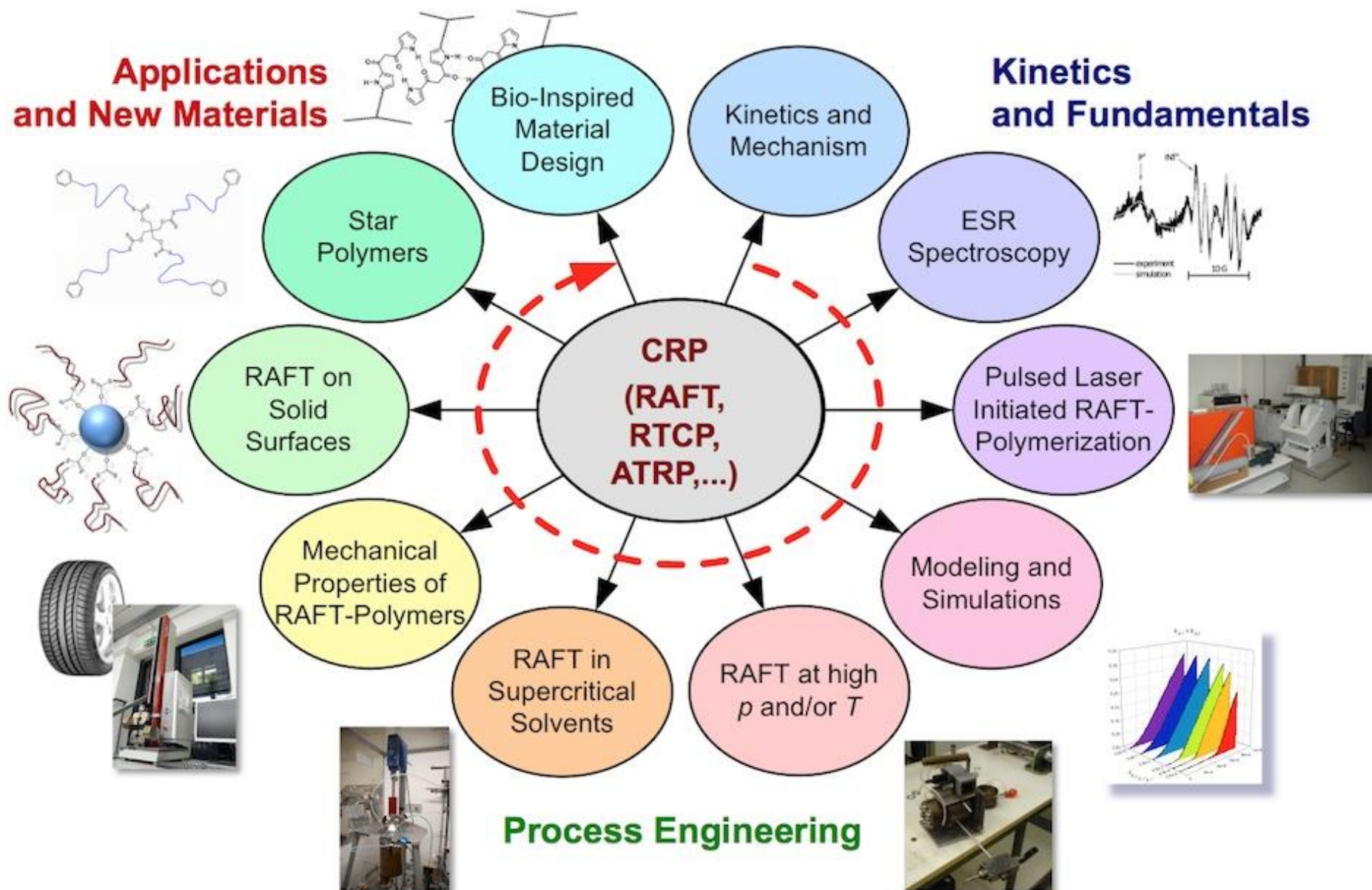


PEG-X2



X-PEG-X

Typical applications of living polymerizations





Summary

5.1 Definition

polymer

monomer

oligomer

degree of polymerization

5.2 Types of polymers and polymerizations

- Carothers classification (from the view of products)
 - condensation polymers
 - Addition polymers
- Flory classification (from the view of mechanism)
 - Step-growth polymerization
 - Chain polymerization
 - Living polymerization



Summary

5.3 Molecular weight

Number-average molecular weight

$$\overline{M}_n = \sum \frac{N_x}{N} M_x = \sum \underline{N_x} M_x$$

Weight-average molecular weight

$$\overline{M}_w = \frac{\sum c_x M_x}{c} = \frac{\sum N_x M_x^2}{\sum N_x M_x}$$

Viscosity-average molecular weight

Polydispersity index (PDI)

$$PDI = M_w / M_n$$



Summary

5.4 Nomenclature of polymer

5.5 Classification

Thermoplastics vs thermosets

Chain structures: linear, cross-linked, branched, dendritic

Monomer composition: homopolymer vs copolymer

Different configuration/conformation

5.6 Characteristics of Polymers



Summary

5.7 Tradition Synthesis

5.7.1 Step Growth Polymerization

5.7.2 Chain Polymerization

Free Radical Polymerization

Ionic polymerization

Coordination polymerization

5.8 Advanced Synthesis

living free-radical polymerization

NMP

ATRP

RAFT



p: 官能团的反应率

$$p = \frac{[M]_0 - [M]}{[M]_0}$$

$[M]_0$ is the initial (at $t = 0$) concentration of the functional groups.

$[M]$ is the concentration of the functional groups at time t .

r: 官能团的化学计量比

$$r = \frac{N_a}{N_b} < 1$$

N_a, N_b : the numbers of A and B functional groups

$\overline{X}_n$: 聚合物的聚合度 $\overline{X}_n$: 每条高分子链所含结构单元数的平均值



Step-growth polymerization

$$\overline{X}_n = \frac{1}{1-p}$$

Carothers' equation

$$f_{\text{avg}} = \frac{\sum N_i f_i}{\sum N_i}$$

Applications of Carothers equations

$$\overline{X}_n = \frac{2}{2 - p f_{\text{avg}}}$$

To calculate the degree of polymerization of reaction system.

If $\overline{X}_n < 0$ in the calculation: system past the gel point at that conversion.

$$p_c = \frac{2}{f_{\text{avg}}}$$

To calculate the critical extent of reaction at the gel point.

p is the extent of reaction of the deficient functional group.



Chain polymerization

$$v = \frac{R_p}{R_i} = \frac{R_p}{R_t}$$

$$\overline{X}_n = \frac{R_p}{(2 - a)(R_t/2)}$$

The fractions of propagating chains, a and $(1 - a)$, respectively, which undergo termination by coupling and disproportionation

$$\frac{1}{\overline{X}_n} = \frac{(2 - a)R_i}{2R_p} + C_M + C_S \frac{[S]}{[M]} + C_I \frac{[I]}{[M]}$$

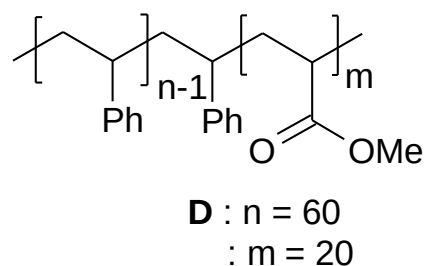
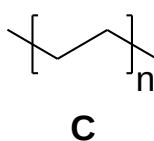
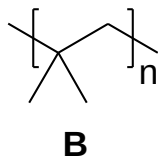
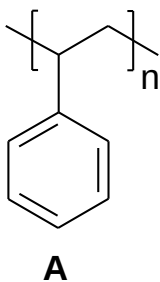
Mayo Equation

chain – transfer constant C

$$C_M = \frac{k_{tr,M}}{k_p} \quad C_S = \frac{k_{tr,S}}{k_p} \quad C_I = \frac{k_{tr,I}}{k_p}$$

Homework 6th

1. What is the advantage/character of living polymerization?
2. If I want to make a polystyrene with a molecular weight of $M_n = 10400$ with an assumed monomer conversion of 80%, what molar ratio between the styrene monomer and its initiator should I use?
3. Using one appropriate monomer for each polymerization classification, discuss the mechanism and kinetics; (a) Step-growth, b) conventional (non-living) chain (addition), c) living chain (addition) polymerizations. In your answer give details of reaction conditions and reagents required.
4. How would you prepare the following polymers? Give reaction conditions, reagents and detailed mechanisms for each polymerization. Name polymers **A-D**.



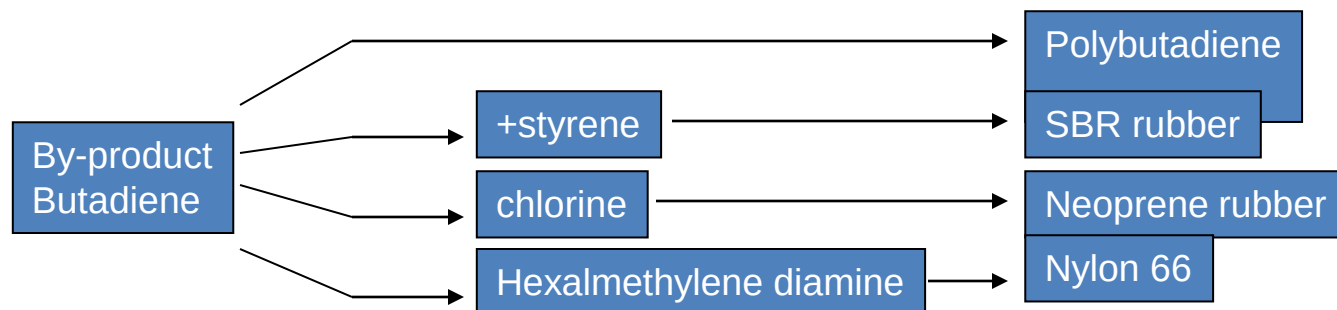
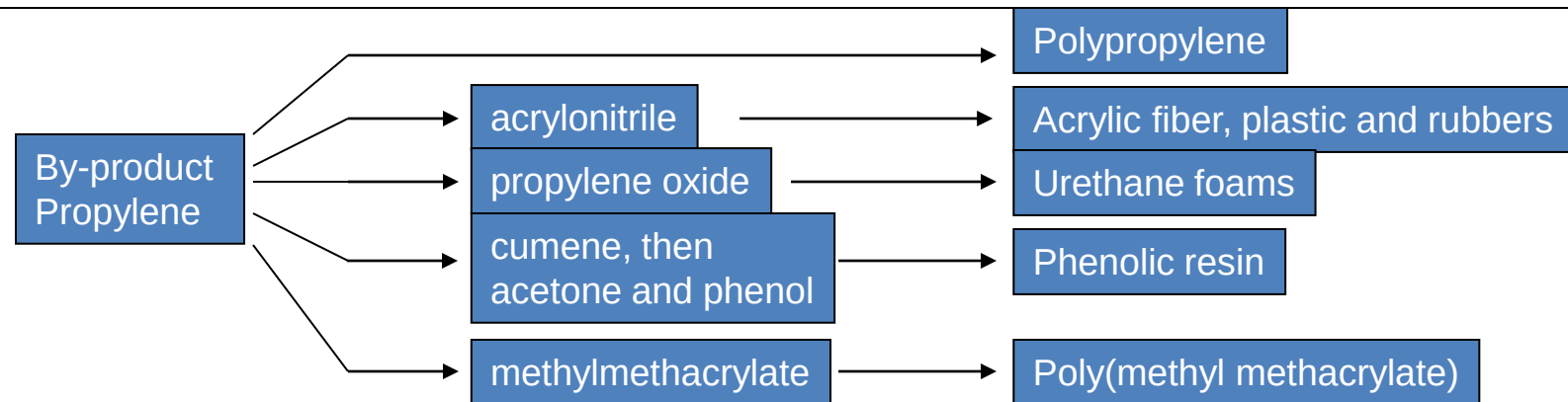
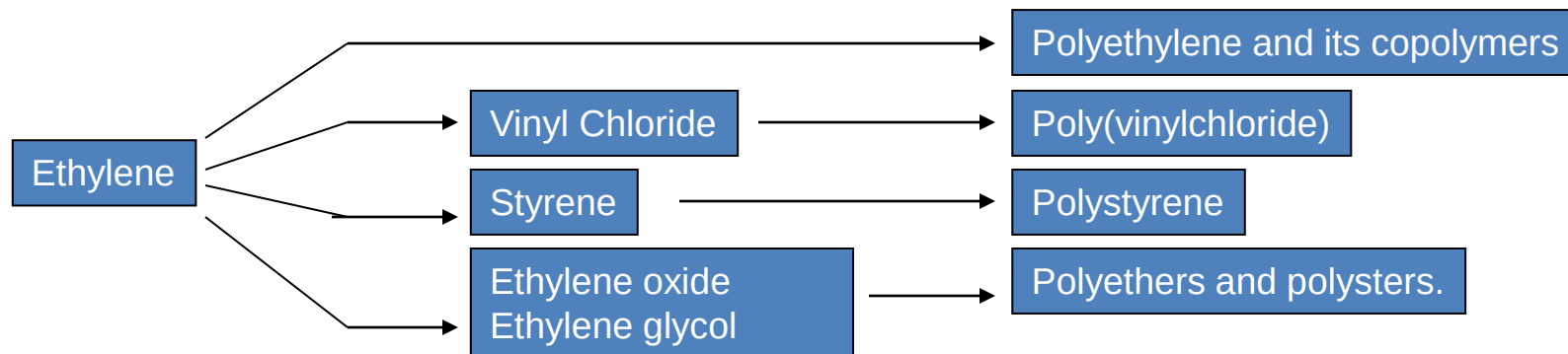
Polymer Everywhere

- Automotive
 - Exterior: bumpers, fenders, paint, tires
 - Interior: dashboard, upholstery, carpeting
 - Engine compartment: intake manifold, radiator
 - Soon to come: H₂ fuel tank, fuel cell membrane
- Aerospace
 - Flight surfaces, canopies & windows, cabin panels, liquid fuel tanks
- Electronics & Entertainment
 - Housing, displays, CD/DVD, circuit boards, capacitors, batteries
- Information Technology
 - Photoresists, dielectrics, photonic devices, displays
- Biomedical Devices
 - Contact lens, heart valves, hip joints, catheters, dental implants, blood vessels, artificial skin
- Civil & Military Infrastructure
 - FRP bridges, pavement, cement, gas & water pipes, paint & coating
- Packaging & Food
 - Containers, films, bubble wrap, disposable utensils, adhesive tape, food additives
- Home
 - Clothing, carpeting, drapery
 - Furniture, appliances, toys
 - Paint, molding, flooring

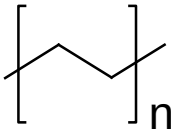
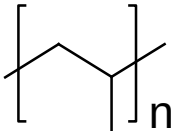
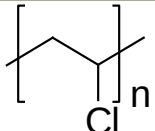
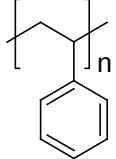
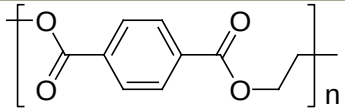
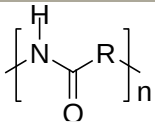
Plastic Resin Codes

標誌* Code*	塑膠種類 Type of plastic	在生產過程中使用塑化劑 Use of plasticisers in production	耐熱程度 Heat resistance	常見的食物接觸產品例子 Common food contact applications
 1 或 1 PET PETE	聚對苯二甲酸乙二醇酯 Polyethylene Terephthalate (PET/PETE)	不需要 Not required	在攝氏80度會變軟 Softens at 80°C	
 2 HDPE	高密度聚乙烯 High Density Polyethylene (HDPE)	不需要 Not required	在攝氏75度會變軟 Softens at 75°C	
 3 或 3 PVC V	聚氯乙烯 Polyvinyl Chloride (PVC/V)	需要 Required	在攝氏80度會變軟 Softens at 80°C	
 4 LDPE	低密度聚乙烯 Low Density Polyethylene (LDPE)	不需要 Not required	在攝氏70度會變軟 Softens at 70°C	
 5 PP	聚丙烯 Polypropylene (PP)	不需要 Not required	在攝氏140度會變軟 Softens at 140°C	
 6 PS	聚苯乙烯 Polystyrene (PS)	不需要 Not required	在攝氏95度會變軟 Softens at 95°C	
 7 OTHER	所有其他塑膠 (例如聚碳酸酯) All other resins (e.g. polycarbonate (PC))	視乎塑膠或合成塑膠而定 (例如PC不需要使用塑化劑) Depends on plastic or combination of plastics used (e.g. not required for PC)	視乎塑膠或合成塑膠而定 (例如PC在攝氏140度會變軟) Depends on plastic or combination of plastics used (e.g. PC softens at 140°C)	視乎塑膠而定 Depend on plastics used  (如PC可用於奶瓶) (e.g. PC can be used for bottles)

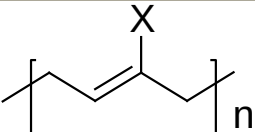
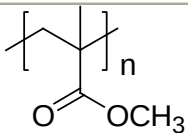
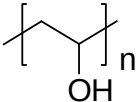
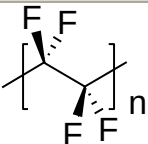
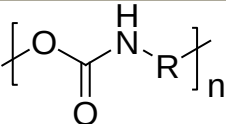
Major Polymers Products Derived from Ethylene and its Derivatives



Chemical Structures and Physical Properties of Major Commercial Plastics

Plastic	Chemical Structure	Material Type at RT	Physical Properties
Polyethylene (HDPE, LDPE, LLDPE)		Crystalline; T_m 110-140 °C T_g -120 °C	Strong; tough; excellent resistance to chemicals; easily melt processable
Polypropylene		Crystalline; T_m 140-165 °C T_g -20 °C	Strong; tough; resistance to heat; barrier to moisture and chemicals
Polyvinylchloride		Amorphous rubber; T_g -70 °C	Clear; easily blendable; excellent resistance to oil and grease
Polystyrene		Amorphous glass; T_g 100 °C	Clear; easily melt and foam processable
Polyesters (polyethylene terephthalate)		Crystalline; T_m 225 °C	Clear; strong; tough; good barrier to gases and moisture
Polyamides (nylon)		Crystalline; T_m 230 °C	Strong; resistance to heat and chemicals; easily spun into fibers

Chemical Structures and Common Uses of other Plastics:

Plastic	Chemical Structure	Material Type at RT	Physical Properties
Polydienes Poly(butadiene, $x=H$; polyisoprene, $X=CH_3$)		Amorphous rubber; T_g: $-70\text{ }^{\circ}\text{C}$	Impact modifier for glassy polymers; when crosslinked tires, gaskets, hoses
Polymethyl methacrylate		Amorphous glass; T_g: $105\text{ }^{\circ}\text{C}$	Optical fibers; home furnishings; lenses (Plexiglas)
Polyvinylalcohol		Amorphous glass; T_g $80\text{ }^{\circ}\text{C}$	Water soluble, so additives as thickening agents
Polytetrafluoroethylene		Crystalline; T_m $370\text{ }^{\circ}\text{C}$, T_g $125\text{ }^{\circ}\text{C}$	Insulation; gasket; heat and chemical resistive piping; cookware (Teflon)
Polyurethanes		Crosslinked solid; T_g $-50\text{ }^{\circ}\text{C}$	Thermoset; coatings; bowling

Homework 2nd

- Give an example to show the relationship among monomer, repeating unit, and DPI
- List a few examples for the configuration affecting polymer properties
- Calculates the number-average and weight-average molecular weight according to the following data

Calculates the number-average and weight-average molecular weight

Components	W_i	M_i
A	0.05	13000
B	0.09	20000
C	0.20	34000
D	0.18	45000
E	0.21	70000
F	0.13	112000
G	0.10	132000
H	0.04	156000

Average molecular weight

Assumption: the total mass of the polymer is 100 g

M_i	$\#i$	$\#i \cdot M_i$	W_i / g	$N_i = W_i / M_i$	$N_i / \sum N_i$	$N_i \cdot M_i / \sum N_i$
13000						
20000						
34000						
45000						
70000						
112000						
132000						
156000						

$$\bar{M}_n = \frac{\sum_i N_i M_i}{\sum_i N_i} = \sum_i \frac{N_i M_i}{\sum_i N_i} =$$

$$\bar{M}_w = \frac{\sum_i W_i M_i}{\sum_i W_i} = \sum_i \frac{W_i M_i}{\sum_i W_i} =$$

Homework 3rd

1. Tell the difference among step-growth polymerization, chain polymerization, condensation polymerization, and addition polymerization.
2. Examine the polymeric materials in your immediate vicinity and determine whether they were formed via step-growth polymerization or chain polymerization
3. To increase the M_w in the process of polyester production, the byproduct (water) is required to be removed as much as possible from the polymerization system. Why? Can you explain it in your own words or show a example with simulation data.
- 4 Describe and draw the structure of the polyester obtained in each of the following polymerizations:
 - a. $\text{HO}_2\text{C R CO}_2\text{H} + \text{HO R}' \text{ OH}$
 - b. $\text{HO}_2\text{C R CO}_2\text{H} + \text{HO R}'' \text{ OH}$
OH
 - c. $\text{HO}_2\text{C R CO}_2\text{H} + \text{HO R}'' \text{ OH} + \text{HO R}' \text{ OH}$
OH

Will the structure of the polymer produced in each case depend on the relative amounts of the reactants? If so, describe the differences.

- 5 Calculate the extent of reaction at which gelation occurs for the following mixtures:
- Phthalic anhydride and glycerol in stoichiometric amounts
 - Phthalic anhydride and glycerol in the molar ratio 1.500:0.980
 - Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.990:0.002
 - Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.500:0.700

Compare the gel points calculated from Carothers equation with those using the Flory statistical approach. Describe the effect of unequal functional groups reactivity (e.g. for the hydroxyl groups in glycerol) on the extent of reaction at the gel point.

Homework 4th

1. Explain the effects of the initiator concentration on the degree of polymerization (free-radical route) of styrene.
 2. Show by chemical equations the polymerization of acrylonitrile initiated by the thermal decomposition of cumyl hydroperoxide.
 3. Design a method to measure the chain transfer constant C_M for polymerization of styrene. Point out the simple experimental conditions and the calculation method.
- Hint : * the chain transfer to initiator (C_I) is 0 if using AIBN as initiator

Disproportion termination

$$\overline{(X_n)}_0 = v$$

$$R_p = -\frac{d[M]}{dt}$$

$$\frac{1}{\overline{X_n}} = \frac{2k_t R_p}{k_p^2 [M]^2} + C_M + C_I \frac{[I]}{[M]} + C_S \frac{[S]}{[M]}$$

此题为选做题，可做可不做。有余力的同学建议做

Homework 5th:

- 1) What is the difference among free-radical polymerization, anionic /cationic polymerization, and coordination polymerization? (Please take examples to explain)
- 2) What are the mechanism for syndiotactic and isotactic placements in propene polymerization?
- 3) Teach yourselves about these advanced polymerization, and you are welcome to show your slides in next class:

Living polymerization

- NMP polymerization
- ATRP polymerization
- RAFT polymerization



p: 官能团的反应率

$$p = \frac{[M]_0 - [M]}{[M]_0}$$

$[M]_0$ is the initial (at $t = 0$) concentration of the functional groups.

$[M]$ is the concentration of the functional groups at time t .

$\overline{X}_n$: 聚合物的聚合度

$$\overline{X}_n = \frac{N_0}{N} = \frac{[M]_0}{[M]}$$

N : unreacted carboxyl groups at time t

N = the total number of molecules present in the system = $[M]$

r: 官能团的化学计量比

$$r = \frac{N_a}{N_b} < 1$$

N_a, N_b : the numbers of A and B functional groups